

USER MANUAL

Boom Bap Producer Pads

Version 1.498.0 · VST3, CLAP and Standalone for Windows

A pad sampler and sequencer built for boom bap: pads, a step sequencer with groove, a synth, a bass line that slides, two turntables, stems and a mixer with buses — all in one instrument.







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distributed by EdensFrequency

edensfrequency.github.io/boom-bap-producer-pads

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CHAPTER 01

Welcome

Boom Bap Producer Pads is a beat-making instrument for your computer. Load sounds onto pads, play them, program beats and bass lines, play and write parts for a built-in synth, sample records on two built-in turntables, and mix it all. It runs inside music software (as a VST3 or CLAP plugin) or on its own as a standalone app, on Windows.

These docs are written for musicians, not programmers.

Words you'll see

- **DAW**: your music software (FL Studio, Ableton Live, Reason, Bitwig...).
- **Plugin (VST3 / CLAP)**: a program that runs *inside* a DAW. VST3 and CLAP are two plug shapes; your DAW supports at least one of them.
- **Standalone**: the plugin running as its own app, with no DAW needed.
- **Sample**: a recorded sound, like a drum hit, a vocal phrase, or a slice of a record.
- **Pad**: one of the squares in the grid. Each holds a sample and plays it when hit.
- **Bank**: a whole set of pads with their patterns, like a separate kit. There are four (A-D).
- **Chop**: cutting a sample into pieces, each on its own pad.
- **Pattern / step sequencer**: a loop of 16 steps where you switch on the moments a pad should play.
- **BPM**: beats per minute, the tempo. **Key**: the musical key a sound is in.
- **MIDI**: the language music gear uses to talk to computers. A **MIDI controller** is a keyboard or pad controller that sends it.
- **Preset**: your whole setup saved under a name, to reload anywhere.
- **Stems**: separate parts of a sound, like its low end, its hits vs its sustained notes, or a record's vocals, drums and bass.
- **Synth**: an instrument that makes its sound from scratch (no sample), played from a keyboard.
- **Bus**: a channel several sounds go through together, so one effect or fader treats them all.

CHAPTER 02

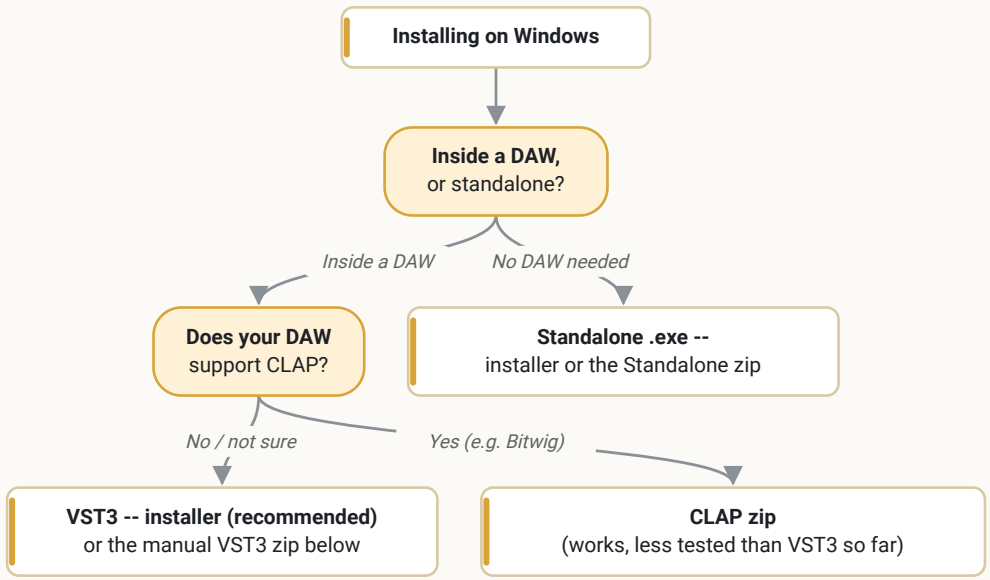
Installing

Windows only: VST3, CLAP and a standalone app.

Requirements

- A VST3 host: Reason, FL Studio, or Ableton Live confirmed working. Others that support VST3 should work too but haven't been tested yet. **Not** required for the standalone app below — it runs entirely on its own.
- A CLAP build is also available (`Boom Bap Producer Pads.clap.zip` on the Releases page) for hosts that support the CLAP format (e.g. Bitwig) — less tested than the VST3 build so far, but the same plugin either way.

Which path do I want?



Recommended: the installer

Download `BoomBapProducerPadsSetup-X.Y.Z.exe` from the project's Releases page and run it. The wizard installs both the VST3 (into the shared system VST3 folder every host scans, or a folder you choose) and the standalone app, and silently installs the WebView2 Runtime if it isn't already on your system (needed for the DISCOVER tab's YouTube embed — most Windows 11 machines already have it). It never touches your presets, DISCOVER crate, or settings, even when reinstalling over an existing version.

A **Check for Updates** shortcut is added to the Start Menu alongside the app — it checks for a newer release and, if one exists, downloads and runs the same installer wizard.

The sections below are the manual alternative — useful if you want full control over exactly what gets copied where.

As a VST3 plugin (inside a DAW), manually

1. Get `Boom Bap Producer Pads.vst3` — download the latest `Boom Bap Producer Pads.vst3.zip` from the project's Releases page and unzip it.
2. Copy the whole `Boom Bap Producer Pads.vst3` folder into your system's VST3 folder: `C:\Program Files\Common Files\VST3\`.
3. Rescan plugins in your DAW:
   - **FL Studio**: Options → Manage Plugins → Find/rescan plugins. Look for "Boom Bap Producer Pads" under Generators (it's an instrument).
   - **Ableton Live**: Preferences → Plug-Ins → make sure VST3 is enabled, then rescan (or refresh the browser's Plug-Ins list).
   - **Reason**: Preferences → Advanced/VST → enable VST3 plugins and rescan, then add it as an instrument device.
4. If you already had a previous version loaded in an open project, remove and re-add the plugin instance (or restart the DAW) — some hosts keep the old binary mapped in memory even after a rescan.

As a standalone app (no DAW needed)

Download `Boom Bap Producer Pads-Standalone.zip` from the Releases page (or build the project yourself, which also produces `Boom Bap Producer Pads.exe`) and run the `.exe` directly — no install step, no host to configure. On first launch, open its audio settings to pick an output device (and a MIDI input device if you want to play it from a controller). See usage/saving-and-standalone.md for the one behavioural difference from running inside a DAW (the step sequencer needs its own "Seq Play" toggle, since there's no host transport to follow).

Troubleshooting

- **Plugin doesn't appear after rescanning**: confirm the folder is really at `C:\Program Files\Common Files\VST3\Boom Bap Producer Pads.vst3\` and not nested inside itself (e.g. `...\Boom Bap Producer Pads.vst3\Boom Bap Producer Pads.vst3\...`) — some file managers/scripts can nest a folder copy if the destination already exists. Delete the whole thing and copy fresh if in doubt.
- **Old version still showing**: the plugin binary can be locked by a running DAW that already has it loaded. Close the DAW (or remove that plugin instance) before replacing the file, then reopen.
- **Standalone app has no sound**: open its audio settings (usually a small icon/menu in the window) and confirm the right output device is selected — it doesn't inherit any system default automatically the first time.

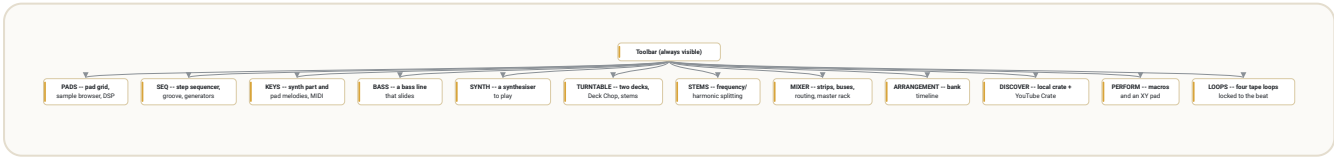
CHAPTER 03

A quick tour

The plugin has twelve tabs, switched via the buttons in the toolbar strip (top-left, next to the preset controls): **PADS**, **STEMS**, **TURNTABLE**, **KEYS**, **SEQ**, **DISCOVER**, **ARRANGE**, **BASS**, **SYNTH**, **MIXER**, **PERFORM** and **LOOPS**. The preset bar and output meter stay visible on every tab, and a **< Back** button next to them walks back through your last few tab switches. The banner shows the installed version number (e.g. `v1.123.0`) directly under the plugin name — check it before reporting a bug, since the fix you're looking for might already be in a newer build.



The PADS tab — the default view



Your first beat, in six steps

If you've never used a pad sampler before, this is the shortest path from nothing to a playing beat:

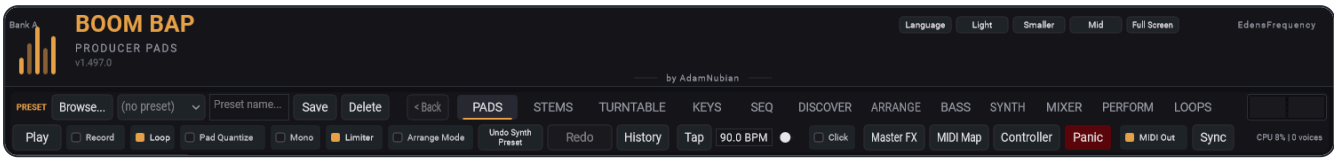
1. **Load sounds.** On the PADS tab, point the browser at a folder of drum sounds, or drag files onto pads. See Pads.
2. **Play them.** Click the pads, or hit them on a MIDI controller.

3. **Make a pattern.** On the SEQ tab, select the kick pad and light up some of the 16 steps, then do the same for the snare and hats. Or let **Generate...** write one for you. See SEQ tab.
4. **Press play in your DAW.** The pattern plays in time with your song (running standalone, use **Seq Play**).
5. **Shape and balance.** Tweak each pad on the PADS tab, then balance everything on the MIXER tab. See DSP controls and MIXER.
6. **Save it.** Save your DAW project, or save a preset to reuse the kit anywhere. See saving-and-standalone.md.

Want to sample a record instead? Put it on the TURNTABLE, press Play, and hit **Chop** on the beat. See Deck Chop.

CHAPTER 04

The toolbar



The banner and both toolbar rows

Two rows below the banner. Row 1 holds the preset bar, the **< Back** button, and the tab buttons; row 2 holds transport, pattern-editing tools, and everything that used to spill off a single-row toolbar as the plugin grew. Both rows are visible on every tab.

Row 1 — presets, back, tabs

- **PRESET** dropdown — pick a saved preset to load it immediately.
- **Click the PRESET label** to choose what loading takes: **Everything**, **Sounds only** (the preset's samples and settings, keeping your patterns) or **Patterns only** (its steps, banks, bass line and synth part, keeping your sounds). The label then reads PRESET, SOUNDS or PATTERNS, and the sound browser loads kits the same way. Undo puts back what was there.
- **Preset name** field + **Save** — type a name and click Save to store the current full kit as a preset: every pad's sample, all its DSP settings (filter, envelope, pitch, bitcrush, mute/solo, choke group, color, rename), all 4 banks (samples and patterns together, see Step sequencer), swing, and the turntable's own settings. Saving over an existing name overwrites it. While a preset loads or saves, **Loading preset...** or **Saving preset...** shows over the plugin; it goes when it's done (or counts down the samples still loading).
- **Browse...** (left of the preset list) — the sound browser: your kits (these presets) and synth sounds in one list. **Search** names, categories and tags; narrow it by kind, category or tag, or show only your **favourites** (click a star). Give the selected sound a **Category** and **Tags** ("dusty, warm, vinyl..."), then **Load** it (or double-click); a synth sound can go **To Bass** instead. **Undo** puts back what was there. The browser keeps its tags in its own file and never changes your preset files, and it opens where you left off.
- **Delete** — removes whichever preset is currently selected in the dropdown.
- **< Back** — returns to whichever tab you were on before your last few tab switches, walking further back each time you click it again (it doesn't create a new "forward" entry of its own, so it can't bounce you between the same two tabs).
- **PADS / STEMS / TURNTABLE / KEYS / SEQ / DISCOVER / ARRANGE / BASS / SYNTH / MIXER / PERFORM / LOOPS** — switch tabs. The active tab is highlighted in orange.
- **Meter** (far right) — live stereo output level. Turns red when clipping.

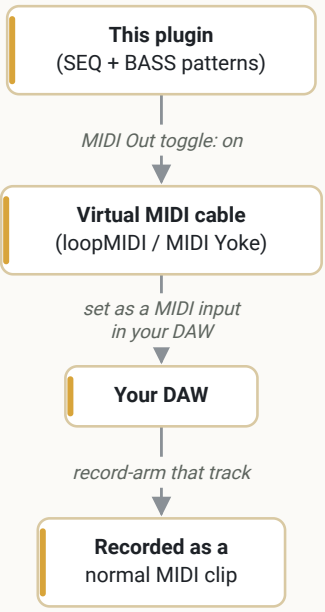
Row 2 — transport, editing, tools

- **Play/Stop/REC** — the global transport button. Inside a real DAW this just reflects the host's own play/record state (a plugin generally can't command its host's transport) — the button text and colour follow along but

clicking it has no effect there. Running standalone, it's a genuine Play/Stop for the whole plugin.

- **Record** — arms recording (the same armed state the PADS tab's own Record toggle shows); right-click for MIDI Learn like most toggles here. A hit goes on the step nearest where you played it. On the PADS tab, **Rec...** sets how: **Quantize strength** (100% puts hits right on the step; less keeps some of your timing, 0% all of it), **Overdub** (add to what's there) or **Replace** (the first hit on a pad clears that pad's steps), and a **Count-in when Play starts** (Standalone). Hold **Erase** and hit pads to clear their steps as the playhead passes.
- **Loop** — toggles looped playback of whatever's currently previewing (the KEYS tab's pattern preview, for example) — not a host parameter, just a plain preference.
- **Pad Quantize** — when on, hitting a pad live (MIDI or a click) waits for the next beat instead of firing right away. Handy for performing along with the sequencer without your hits drifting off the grid. Sequencer steps aren't affected — they're already on the grid.
- **Mono** — when on, triggering any pad stops every other currently-sounding pad, kit-wide. Different from a pad's own Choke Group (right-click a pad to set one): Choke Group only cuts other pads sharing the same numbered group, Mono affects every pad at once with no grouping needed. A real parameter, so it's automatable and MIDI-learnable like anything else. Off by default.
- **Limiter** — a brick-wall limiter on the final master output, to catch accidental clipping when several pads, the bass, the synth and the decks all peak at once. On by default; it's in Export WAV too. **Right-click** it for its **Ceiling** (how loud the output can get: -0.1, -0.3, -1 or -2 dB; -0.3 to start) and **Release** (how fast it lets go: Fast 10 ms, Medium 50 ms, Slow 200 ms -- slower is smoother, faster is louder).
- **Arrangement Mode** — turns on the ARRANGEMENT tab's section timeline; while it's on, the SEQ tab's own bank buttons are disabled since the timeline drives bank switches instead.
- **Undo / Redo** — steps back (and forward) through the edits that change a lot at once: chopping and slicing (and moving or deleting a chop), Deck Chop, every **Generate...**, Randomize, Humanize, Euclid, Ghost notes, Flip, Clear Pattern, Clear Pad, Trim Silence and Import Kit — up to 20 levels. (The KEYS tab's synth part has its own undo.) Ctrl+Z / Ctrl+Y (Ctrl+Shift+Z also works for Redo) work from any tab. **History** opens a popup listing every step in the undo/redo stack by name, so you can see what you're about to step through instead of guessing.
- **Tap** — tap along to set the standalone tempo (standalone only; hidden inside a real DAW, where the host's own tempo already applies). The **BPM** field next to it does the same job by direct entry — type a number or drag it.
- **Click** — a metronome, standalone only, same visibility rule as Tap/BPM above (a real DAW already has its own click).
- **Master FX** — opens a small popup with 3 effect slots applied to your final mix, regardless of which tab is showing. Pick an effect type per slot. Reverb, Delay, Saturation, and Formant Shift slots also show real knobs (Size/Damping/Mix, Time/Feedback/Mix, Drive, or Shift) — adjustable, automatable, and MIDI-learnable like any other knob in the plugin. The same popup is also reachable from the MIXER tab's Master strip, via its own **FX** button.
- **MIDI Map** — opens a popup listing every active MIDI Learn CC mapping in the plugin in one place, so you can review or clear a binding without hunting down the control it's attached to. See MIDI Learn.
- **Controller** - opens the DJ controller window: teach the plugin your DJ controller's jog wheels, buttons, faders and pads so they play the decks and pads, and save it by name. See Using a DJ controller.
- **Panic** — immediately silences every playing pad, the bass and the SYNTH. For a stuck note during live use, not something you'd need routinely. (The decks keep playing: they aren't notes.)
- **MIDI Out** — on by default. While on, the pattern actually playing on the PADS grid and the BASS lane is also sent out as real, sample-accurate MIDI note-on/off (not just triggering this plugin's own audio) — useful if you

want to capture what the plugin is generating as an editable MIDI clip. Getting it to actually land in your DAW's own piano roll generally needs a virtual MIDI cable (e.g. loopMIDI or MIDI Yoke): route this plugin's MIDI output to the virtual port, then set that same port as a MIDI input in your DAW and record it — that one-time setup happens in your DAW/OS, not in this plugin. If you just want a one-off pattern as a file instead, the SEQ and KEYS tabs both have their own MIDI export/drag options that don't need any of this.



- **Sync** -- MIDI clock, lit while it's on.
  - **Send MIDI clock**: a clock for whatever's playing (your DAW's transport, or the standalone's Play) on the plugin's MIDI output, with Start, Stop and the song position, so a drum machine, another app or FL Studio can follow.
  - **Follow MIDI clock** (the standalone): the tempo, Play, Stop and position follow a clock coming in on any MIDI input you've turned on in its audio / MIDI settings. The BPM shows the tempo it hears. As a plugin in a DAW it already follows the DAW, so this is greyed there.
  - **Syncing with FL Studio...** shows the setup. You need a MIDI cable between the two programs -- a virtual one such as loopMIDI (free) or a real interface. For FL Studio to lead, turn on master sync (MIDI clock) for that cable's output in FL's MIDI settings, turn the cable on as an input here, and choose Follow MIDI clock. For this to lead, pick the cable as this standalone's MIDI output, choose Send MIDI clock, and set FL Studio to sync to external MIDI clock from that input.
  - Ableton Link isn't built in yet.
- **CPU / voice count** (far right) — a live readout of processor load and how many voices are currently sounding.

Banner strip (top, next to the logo)

- **Light / Dark** — switches the whole plugin between the dark theme and a light one (easier on the eyes in a bright room or in daylight). Every open window of the plugin switches at once, and it remembers your choice next time. The button shows the theme you'd switch *to*. The record on the turntable stays black either way.
- **Language** — the plugin's language. Pick one from the list (it's used the next time you open the plugin's window, and remembered), open the Languages folder, or make a template for translating it into yours. See

Languages, and translating the plugin.

- **Smaller / Mid / Full Screen** — one-click window sizing. Full Screen fits your actual display, whatever size that is. The window is also still freely resizable by dragging any edge or corner, and always keeps its proportions while you do.
- A small readout next to the size buttons shows **Bank X** at all times, and — while Arrangement Mode is on — the current section's name and its position in the timeline (e.g. "Bank A - Chorus (2/5)"), so you always know where you are without switching to the ARRANGEMENT tab to check.

Scrolling over any knob (DSP panel, Turntable, or this toolbar's own BPM/ Swing/Volume/Pan sliders) nudges its value by a small, precise step — handy for fine adjustments without needing to drag.

Presets are stored per-user at `%APPDATA%/Boom Bap Producer Pads/Presets/*.bbpreset` — they're independent of any DAW project, so they carry across projects and sessions. Saving a preset also copies every sample it uses (across all 4 banks) into a `<preset name>_data/` folder alongside it, so the preset keeps working even if you later delete or move the original sample files — a common workflow once a one-shot from a "used samples" folder has done its job.

DAW project saves (not named presets) still reference your samples by their original file path, same as always.

CHAPTER 05

Pads and the sample browser



The PADS tab

This is the main PADS view: sample browser on the left, the 4x4 pad grid in the middle, and the waveform/DSP column on the right, with the performance controls along the bottom and the keyboard under everything.

Sample browser

Top-left panel.

- **Choose Folder...** — point the browser at your own sample library. Remembered across sessions.
- **BPM, key and length** show at the right of each file DISCOVER has analysed (its library -- see the DISCOVER tab).
- **Search** box — filters the visible tree by filename (case-insensitive substring match). Folders you already had open keep showing what they last scanned until you collapse and reopen them; newly opened folders always reflect the current search.
- **Preview** — when ticked, clicking a file plays it. The small slider next to it sets how loud previews play (DISCOVER's too); it's saved with the project.
- **Fit** (next to Recent) — when ticked, a loop you put on a pad plays at your song's tempo (time-stretched, so the pitch holds) and in its key (the one set on KEYS). One-shots stay as they are.

- **Double-click a file** to load it onto whichever pad is currently selected.
- **Drag a file onto a pad** to load it there (Windows Explorer and your DAW's own browser work too -- see below).

Dropping files anywhere

Drag audio or video from Windows Explorer (or your DAW's browser) onto:

- **a pad** — loads it onto that pad;
- **the Sample Editor** — loads it onto the pad it's showing;
- **the deck** (TURNTABLE) or the **bass lane** — loads it there;
- **the DISCOVER crate** — adds it (a folder adds its audio and video);
- **anywhere else** — a menu asks what to do with it: load it onto the selected pad, onto the next empty pad (several files: onto empty pads in order), send it to the deck, add it to the DISCOVER crate, or **match the project tempo to it** (sets the tempo from the file's detected BPM).

The pad grid

16 pads. By default they're laid out bottom-left to top-right (pad 1 bottom-left, pad 16 top-right), matching how many hardware pad controllers number their physical pads — so the pad you see selected is the pad you actually hit. Click **Hardware Layout** in the toolbar to switch back to plain top-left-to-bottom-right reading order. Either way, the underlying MIDI note mapping is unchanged (notes 36–51 always trigger pads 1–16 in the same order) — this only affects which grid cell each pad's box is drawn in.

Each pad plays one sample, one-shot (it always plays from the start of its region; triggering it again while it's still playing cuts the previous hit and restarts — different pads never cut each other off, unless **Mono** mode is on — see The toolbar).

- **Chop lanes** (right-click a sliced pad > **Chop lanes...**) -- paint a value for each chop straight over the waveform. The window shows the sample the pad was sliced from, each chop as a column (with its pad number); pick a lane and drag across the columns. Letting go plays the last chop you painted.

Lane	What a chop gets
Volume / **Pan**	Its level and place.
Pitch	Up or down, -12 to +12 semitones.
Reverse	Top half plays it backwards.
Stutter	Its start played 2, 3, 4, 6 or 8 times, in the chop's own length.
Filter	The cutoff (top = open), with the filter picked next to the lane: Low-Pass, Low-Pass 24, Band-Pass, High-Pass, Notch or Comb.
Delay / **Reverb**	How much goes to the shared delay (a dotted 1/8 at the tempo, bouncing left and right) or reverb.

Random fills the lane with musical values (pitch mostly on 0, 5, 7 and 12), **Repeat** copies the chops up to the last one you painted over the rest (paint 1-4, then Repeat), **Clear** puts the lane back to normal. Each stroke or button is one **Undo**.

Each chop is a pad, so what you paint is that pad's own setting: it plays wherever the chop does -- hit, SEQ, the keyboard, exports -- and shows on the pad's own controls. (A lock on a SEQ step only changes that one step.) Stutter suits one-shots; a looping pad ignores it.

- **Drum synth** (right-click a pad > **Drum synth...**) -- make a drum from scratch instead of loading one: **Kick**, **808**, **Snare**, **Closed hat**, **Open hat**, **Clap**, **Rim**, **Tom** or **Cowbell**, shaped by five knobs:

Knob	What it does
Tune	Up or down a semitone at a time (a kick at +12 is an octave higher).
Decay	How long it rings -- a tight hat or an open one, a short kick or a long 808.
Tone	Darker or brighter: the click on a kick, the sizzle on a snare or hat.
Punch	The attack: how far a kick's pitch drops, a snare's snap, how spread a clap's slaps are.
Drive	Saturation: louder, rounder, then gritty.

  Every change plays the new drum. On an empty pad it starts with a kick; on a pad that already has a sound, the first change replaces it (one **Undo** brings the old sound back). The drum is saved as a WAV in **Music > Boom Bap Captures > Drum Synth**, so it plays like any sample -- in SEQ, with the crush and stretch modes, in exports and kit exports -- and opening **Drum synth...** again shows its knobs where you left them.

- **Drag a pad into your DAW.** Drag a loaded pad (without Shift) onto a track and its sound -- the trimmed region -- lands there as a WAV named after the pad ("Pad 3 - Kick.wav"), kept in **Music > Boom Bap Drag Out**. Shift+drag swaps two pads instead.
- **< and >: try the next sample.** Hover a loaded pad and arrows appear on its left and right edges. **>** loads the next sound in the same folder as the pad's sample, **<** the one before (in the order Explorer lists them, going round at the end). The pad keeps its settings -- pitch, filter, envelope, effects -- so you hear each kick through the same chain. If nothing is playing, the new sound plays once; while a beat plays, you hear it in the pattern instead. The same two are in the pad's right-click menu.
- While a pad plays, a **gold line** moves across its waveform to show how far through the sound it is.
- **Full Level** (next to Key Pads) — every pad you play hits at full velocity, however hard you hit it. The steps of a pattern keep their own.
- **Velocity curve** (right-click a pad) — how that pad answers how hard you play: **Linear**, **Soft** (gentle hits come through louder) or **Hard** (only real hits get the top).
- **16 Levels from this pad** (right-click a pad) — pads 1-16 all play that one sound: **By velocity** (pad 1 the softest, pad 16 full) or **By tune** (pad 1 as it is, a semitone up each pad). **Off** in the same menu.
- **Pad 1's MIDI note** (right-click a pad) — the note pad 1 answers to, the others following: C0, C1 (36, the usual), C2 or C3. MIDI Out and the pattern's MIDI export use the same notes.
- **DAW output** (right-click a pad) — which of the plugin's 16 outputs the pad goes to in your DAW: **Its own** (pads 1-16 have one each; the others go to the main mix) or **Out 1**-**Out 16**. Several pads can share one. The output has to be switched on in your DAW's routing.
- **Pressure (aftertouch)** (right-click a pad) — on a controller that sends it, pressing into a held pad changes **Its volume** or opens **Its filter**. Until you press, the pad sounds as it always does.
- **MIDI channel** (right-click a pad) — **Any** (as usual) or one channel, and then the pad answers its note only there.

- **Key Pads** (next to Hardware Layout) — play the pads from your computer keyboard. The keys are laid out like the pads on screen:

Hardware Layout on	Hardware Layout off
`1 2 3 4` = pads 13-16	`1 2 3 4` = pads 1-4
`Q W E R` = pads 9-12	`Q W E R` = pads 5-8
`A S D F` = pads 5-8	`A S D F` = pads 9-12
`Z X C V` = pads 1-4	`Z X C V` = pads 13-16

  Hold **Shift** to play softer. On a bigger grid the keys cover the first 16 pads' corner of it. It works on every tab except TURNTABLE (whose letters chop the record), so you can play pads while you're on SEQ, KEYS, SYNTH and the rest. While Key Pads is on, **E** plays a pad instead of resizing the Sample Editor (use its Expand button), and typing in a text box still types. It's remembered for next time.

- **Keyboard and screen readers. Tab** moves through the controls; on a pad, **Space** or **Return** plays it and **Shift+F10** opens its menu. Screen readers (Narrator, NVDA) read each pad as "Pad 3: <its sample>", and each knob, fader and dropdown by its name ("Cutoff", "Pad 3 volume", "Grid size").
- **Click an empty pad** — opens a file chooser to load a sample onto it.
- **Click a loaded pad** — selects it (highlights gold) and triggers it (flashes, and glows orange while the sample is still playing).
- **Quick Stop** — a small stop icon appears in the top-left corner of any pad that's currently playing (it takes over the Choke Group letter badge's spot, which comes back once playback stops). Click it to cut the pad off right away instead of waiting for a long sample to finish.
- **Drag an audio file onto any pad** — loads it directly, from Windows Explorer or your DAW's own sample browser if it supports OS-level drag-and-drop.
- **Drag a loaded pad's sample out** (past a small threshold) — exports its current region to a temp WAV and starts a real file drag: drop it onto another pad to copy the sample there, or drop it into your DAW's own browser/playlist to pull it out of the plugin entirely.
- **Right-click any pad** — Load Sample… / Clear / Rename…, **Send to the deck** (to chop it again on the TURNTABLE tab, as it is or as it sounds -- see Chop a pad again), plus Choke Group, **Layers** (see below), Color, and three **Insert FX** submenus (Insert FX 1/2/3 — up to 3 effects in series on that pad). Each slot can hold Chorus, Flanger, Phaser, Transient Designer, Harmonic Exciter, Stereo Doubler, Reverb, Delay, one of 3 Saturation flavors (Tube, Transformer, Console), or Formant Shift, a **Ring Mod** (Freq and Mix: bell-like, metallic tones), a **Compressor** (Threshold, Ratio, Makeup), a **Lo-fi Box** (Bits, Rate, Mix), a **Tremolo** (in time with the song: Rate 1/2-1/32, Depth, Shape), a **Tape Stop** (Stop past halfway slows the sound to a halt over Time -- map Stop to a macro or MIDI button to throw it), a **Gate** (Threshold, Release, Range: the room off a snare, the hiss between hits), an **Era** (one knob from newer to older -- fewer bits, a lower rate, the top and bottom rolled off, wow, hiss and crackle -- and a Mix), a **Scratch** (the sound winds back and forth like a record under a hand, in time with the song -- Depth, Length 1/16 to 4 bars, Mix; it follows the pad's drawn Shape on the Mod page if you've drawn one), and under them **Bypass** (the effect off, its settings kept), **Move earlier / later in the chain** (and, in the pad menu, **Insert chains**: save a pad's three effects under a name, or put a saved set on this pad or every selected one) (and **Inserts in parallel**: every effect works on the pad's dry sound and they're mixed together, instead of one feeding the next) (the slots run 1, 2, 3; moving one keeps its settings) and **Parallel compression (preset)** -- the Compressor squashing hard, mixed in at half, for thicker drums that still hit -- a **Shape Filter** (a filter that opens and shuts in time with the song along a shape you draw), or an **EQ** (with **Edit**

EQ..., see the MIXER tab) ("None" is the default — sounds exactly like it always did). Every slot but EQ also gets an "Edit Knobs…" option in that same submenu for adjusting their parameters (Size/Damping/Mix, Time/Feedback/Mix, Drive, or Shift).

- **Copy pad** / **Paste pad here** (right-click menu) — copy one pad, then right-click another and paste **All of it**, the **Sample only**, the **Settings only** (its sound: level, filter, envelope, tune, crush, inserts, modulation) or the **Pattern only** (its steps). Mute, solo, colour and layers stay as they were. A paste takes what the copied pad holds at that moment, and Undo takes it back.
- **Name and colour pads by sound** (right-click menu) — listens to every loaded pad and colours it for what it sounds like: kick red, snare orange, hats yellow, perc green, loop blue, bass purple. A pad you haven't named gets the name too ("Kick", "Open Hat"...). It's a guess from the sound -- recolour or rename any it gets wrong; Undo takes it all back.
- **Tune its root note to C** (right-click menu, on a loaded pad) — finds the note the sound plays at (an 808 in G, say) and sets the pad's Tune and Fine so it plays as a C. Playing the pad chromatically from KEYS is then in tune. A sound with no clear note (a hat, noise) is left alone. Undo takes it back.
- **Key Shift Pad** (right-click menu, on a loaded pad) — spreads that pad's sample across your other empty pads, each one a semitone higher than the last. Turns one chop into a playable chromatic instrument across the grid, without needing the KEYS tab.
- **Layers** (right-click menu) — makes this pad play one of several pads each time it's hit. That's how you get drums that don't sound machine-gunned, and hits that change character with how hard you play. Load the variations onto pads next to each other, right-click the first one and pick:
  - **Velocity layers:** soft hits play the first pad, harder hits the next ones, the hardest hits the last one. Load them softest first.
  - **Round robin:** each hit plays the next pad in turn.
  - **Random:** each hit plays one of them at random, never the same one twice in a row.
  - **Stack (all at once):** each hit plays all of them together, each at its own level -- a kick with a sub under it, a chord from single notes, or a sample's stems (STEMS > **To pad layers** builds one).

  Then choose how many pads (this pad and up to 7 after it). If you've batch-selected pads (Ctrl+click), you can also build the group from those. The pad you right-clicked shows **VEL**, **RR**, **RND** or **STK** and the count in its corner; the others show **in N** (the pad that plays them). They all stay normal pads with their own chop, envelope and effects, and you can still play or program them on their own. Program the pattern on the first pad and the variations happen by themselves, in the bounce too. **Off** undoes it. A pad can only be in one group.

- **Key zones** (right-click > Layers) -- several pads become **one instrument across the keyboard**. Each pad holds a note of the instrument (a piano's C3, C4, C5...), named in its file name ("Piano_C3.wav", "Rhodes F#4.wav"). Load them on pads side by side, right-click the first and pick **Key zones > Pads 1-4** (up to 16 pads). Then select that first pad and play the keyboard, the KEYS piano roll or MIDI channel 2: every note plays the nearest recorded note, tuned from there -- so a C sounds like the C sample and an F like the nearest one, not one sample stretched across five octaves. The pad shows **KEYS**. If a pad's file has no note in its name, it says which.
  - Quickest way in: **Kit... > Import > Folder as one instrument** puts a whole folder of named notes onto empty pads and sets this up for you.
  - The SYNTH tab's **To Pads... > One per octave** makes one too.
  - Notes in names count middle C as **C4** (60). A pack that calls it C3 just sits an octave higher -- play it an octave down.

  - **Articulations** (right-click the first pad > Layers > Articulations): load each articulation of an instrument (sustain, staccato, pizzicato...) as its own key zones, then on the first one pick which of them plays on keys **C#1**, **D1** and **D#1** (**C1** is itself). Press those keys while you play to switch.

- **Layer the synth under it** (right-click a pad) -- the SYNTH's sound goes on an empty pad and is stacked with this one, so every hit plays your sample and the synth together. Balance them with the two pads' Volume, or put both on one PERFORM macro.
- **Blend the layers (morph)** (right-click > Layers, on a Velocity layers or Key zones pad) -- instead of jumping from one layer to the next, the layers **crossfade**: in between two of them you hear both, mixed so the level stays even. The pad's corner tag gets a **~**. Pick what moves the blend:
  - **Blend by velocity** -- how hard you hit sets the mix: soft hits the first pad, hard hits the last, and everything in between a smooth mix of the two nearest. No more sudden jump at a velocity boundary.
  - **Blend by the mod wheel (CC1)** or **by expression (CC11)** -- the mix moves **while the note holds**: push the wheel up and a held string or horn note grows from soft to loud. Wheel down plays the first pad, up the last; expression starts at full.
  - **Blend by an LFO** -- the mix sweeps back and forth by itself from each hit. **Mix / centre** sets where it sweeps around, **LFO rate** how fast (0.25-8 Hz) and **LFO depth** how far.
  - **Blend at a fixed mix** -- two sounds stacked at the **Mix** you pick (0% the first, 100% the last): a kick with a little of another kick's click, two drum-synth pads blended into one.
  - **Switch (no blend)** goes back to plain layers.

  In **key zones**, pads whose files are on the **same note** become that note's soft-to-loud layers (load "Strings_C4_soft" and "Strings_C4_loud" and give both C4): the blend crossfades between them. A note with just one sample gets a level instead -- the wheel takes it from quiet (-18 dB) up to full. Export WAV plays the blend as it sounds live (the wheel and expression stay where they are when you export). The blend is saved with the project and the kit.

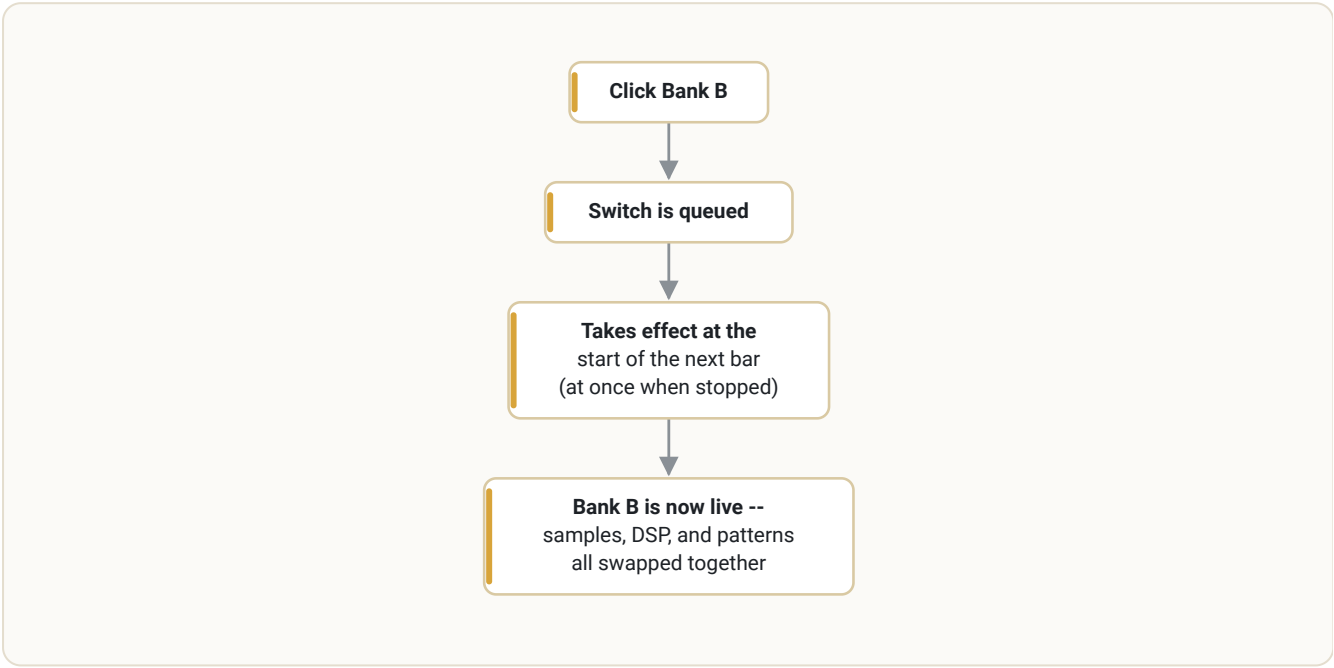
- **Favorite** (right-click menu) — protects this pad's step pattern from Randomize and Flip the Sample (see Step sequencer) — shown with a small gold star.
- **Locked** (right-click menu) — Kit shuffle leaves this pad alone; a small padlock shows beside the star.
- **Draw a waveform...** (right-click menu) — draw one cycle of a wave with the mouse (or start from Sine, Triangle, Saw, Square or Random; **Smooth** rounds it off), then **Put on pad**: it becomes a looping sound at C4, in tune from the keys.
- **Sample FM** (right-click menu) — a sine wobbles the sample's speed very fast, for metallic, bell-like or growling tones. Pick how much (Light to Wild) and the sine's speed against the note (x0.5 to x7); it follows the note as you play up the keys. Exports have it too.
- Supported file types: `.wav`, `.aif`/`.aiff`, `.flac`, `.ogg`, `.mp3`, `.m4a`/`.aac`, `.wma` — and videos (`.mp4`, `.mov`, `.mkv`, `.avi`, `.wmv`, `.webm`, `.m4v`, `.3gp`, `.mpg`), which load their soundtrack. See DISCOVER → using videos for what Windows can decode and the 20-minute limit.

Pads also respond to MIDI: notes 36–51 (C1 upward) trigger pads 1–16 with real velocity sensitivity from your MIDI controller/keyboard. Clicking a pad in the UI always triggers at a fixed velocity.

Bank A / B / C / D buttons (right under the grid, next to "Editing pad N" — also available on the SEQ tab) switch between 4 complete kits — 4 banks x 16 pads = 64 addressable pad slots in total. Switching banks swaps *everything*: every pad's sample, every DSP setting, and every pattern, all together, so each bank is a genuinely

separate kit rather than just an alternate pattern for the same 16 samples. A bank switch is bar-quantized — while the song is playing it takes effect at the start of the next bar, not the instant you click, so it never chops a pattern off mid-phrase. While it's stopped, it switches straight away.

Right-click a bank button on the SEQ tab to **Copy Into Bank X** (this bank's pads and pattern over that one's) or **Swap With Bank X** (the two trade places -- pads, patterns, all 16 pattern slots and grid sizes). Undo takes a swap back.



Grid size dropdown (next to the Bank buttons) — each bank can be its own size: 4x4 (16 pads, the default), 5x5 (25), 6x6 (36), 7x7 (49), or 8x8 (64). Picking a bigger size adds more pads to the grid immediately — the pads themselves get smaller to fit the same space, the plugin window doesn't grow. This applies to whichever bank is currently active and takes effect right away, unlike Bank switching itself (no bar-quantize wait — there's no audio content to keep in sync, just how many pads are addressable). Every other bank keeps its own independently-set size, so Bank A can stay a compact 4x4 while Bank B runs a full 8x8. The SEQ/KEYS/DISCOVER tabs and MIDI note range all follow whichever size the active bank is currently set to; see Output routing for the one thing that's capped at 16 regardless of grid size.

Kit... (just left of the Grid size dropdown) exports the active bank as a SoundFont, SFZ, DecentSampler preset, sliced WAV or a whole FL Studio pack, and imports kits from those formats or a folder of samples. See Kit import & export. **Kit... > Shuffle the unlocked pads** gives each loaded pad in the bank a new sound like its own, picked from the sounds DISCOVER has analysed (a kick stays a kick). Lock the pads you want to keep first.

CHAPTER 06

Kit import and export

The **Kit...** button on the PADS tab (just left of the grid size, under the pads) moves a whole bank's kit in and out of the plugin: take your kit into FL Studio, another sampler or a DAW, or bring a kit made elsewhere in, with each sound on its own pad.

Everything works on the **bank you're looking at** (A, B, C or D).

Export kit

Every loaded pad goes out exactly as it plays: trimmed to its region, reversed if Reverse is on, at its Normalize level. Pad 1 is always MIDI note 36, pad 2 is 37, and so on.

Choose	You get	Load it in
SoundFont (.sf2)	One file with every pad on its own key	DirectWave, FL Studio's Fruity SoundFont Player, Kontakt, most samplers
SFZ	`Kit.sfz` plus one WAV per pad, in a new folder	Any SFZ player (sforzando, TX16Wx, Sitala...)
DecentSampler preset	A `.dspreset` and a `Samples` folder, in a new folder	DecentSampler (free)
Sliced WAV with markers	Every pad back to back in one WAV, with a slice marker at the start of each	FL Studio's Slicex or Fruity Slicer - each pad becomes a slice
FL Studio pack	A folder with all of it: `One-shots` (a WAV per pad, ready for FPC), the SoundFont, the sliced WAV, the bank's pattern as MIDI, and a short how-to	FL Studio
Ableton Live Drum Rack	A `.adg` and a `Samples` folder: a Simpler per pad on its key, with volume, pan, tune, envelope and choke group	Ableton Live 12 (drag the .adg onto a track)
Drum program (.xpm)	A `.xpm` with its WAVs beside it: each pad on its key, with volume, pan, tune, envelope and mute group	Samplers that load .xpm drum programs
Bitwig / Reason	A `Samples` folder, each WAV named by its key (`036 C1 - 01 Kick.wav`), and a how-to	Bitwig's Drum Machine (drop them all on C1), Reason's Kong (one per pad)

What carries over: volume, pan, tuning (Tune, Fine and Speed), the envelope (attack, decay, sustain, release), choke groups, and Loop. The SFZ export also keeps Reverse, Loop and **Play To End** as settings.

Layers (a pad's right-click **Layers** groups) carry over too: the group's sounds all go on the first pad's key. SFZ and DecentSampler keep velocity layers, round robin and random. SoundFonts can only split by velocity, so a round-robin or random group goes out as each pad on its own key. Filter, bitcrush and insert effects stay in this plugin -

these formats have nothing to hold them.

Exports never overwrite: a folder that already exists gets a number, like `Boom Bap Kit A (SFZ) (2)`. The message afterwards has a **Show in Explorer** button.

Why not FPC, DirectWave or Kontakt files directly?

Those programs save their own kits in private formats that aren't published (Kontakt's are encrypted), so no other program can write them reliably. But every one of them **opens** the formats above: FPC takes the one-shot WAVs, DirectWave and Kontakt load the SoundFont, Slicex loads the sliced WAV.

Import kit

Import **replaces the pads of the bank you're looking at**: the bank is cleared, each pad is reset to its default settings, and the kit's sounds go on. **Undo** puts the previous kit back, settings and all.

Choose	What happens
SoundFont (.sf2)	Every sound in the SoundFont's first preset gets a pad. The sounds are saved as WAV files in `Documents\Boom Bap Producer Pads\Imported Kits` (pads always play from a file).
SFZ	Every region gets a pad, with its volume, pan, tuning, envelope, choke group, reverse and loop.
DecentSampler preset	Every sample gets a pad, with the same settings, and the group's name as the pad's name.
WAV with slice markers	Each slice becomes a pad playing that part of the file - a WAV saved by FL Studio's Edison, one exported from here, or any WAV with markers.
Folder of samples	Every audio file in the folder, in name order (`2 Snare` before `10 Clap`). Subfolders aren't included.
Folder as one instrument	A folder of one instrument's notes, each named with its note (`Piano_C3.wav`, `Piano_F#3.wav`...), goes onto empty pads as **key zones**: the first pad plays the whole instrument across the keyboard, each note from the nearest sample (see Key zones). Up to 16 notes -- with more, an even spread from the lowest to the highest is kept. Where a note has several files (velocity or round-robin takes), the first by name is used.

Where each sound lands:

- A sound mapped to **one key** goes on that key's pad (key 36 = pad 1). That's how drum kits line up with the right pads.
- Sounds that cover **a range of keys** (a multisampled piano or strings) fill the empty pads from the lowest range to the highest.
- If there are more sounds than pads, the rest are skipped and the message says how many. Pick a bigger **Grid Size** (up to 8x8) and import again.

Good to know:

- A SoundFont with several presets (a whole General MIDI set) imports only the first one.
- **Velocity layers and round robin come in as Layers.** Samples that share a key in the kit each get a pad, and the pad on that key plays them as a group: by velocity, in turn, or at random, as the kit says (see Layers).

Samples that share a key but play *together* in the source (a stack) just get their own pads.

- SoundFonts have no "play to the end" setting, so a Play To End pad exported to SF2 comes back with a release as long as the sound, which plays the same way.
- Some commercial libraries encrypt their samples. Those can't be read, and the message names a missing sample so you can tell.

CHAPTER 07

The Sample Editor

Top-right panel of the PADS tab — shows the waveform of whichever pad is currently selected.



The Sample Editor, expanded

- **Expand** / **Full Screen** — make the waveform bigger: Expand takes over the sound controls' space, Full Screen the whole tab. **E** steps through Normal, Expanded and Full Screen; **Escape** goes back to Normal.
- **Trim handles** (the two vertical orange bars) — drag to set the region the pad actually plays. Everything outside the region is dimmed.
- **Click inside the waveform** — drops a cyan cut-point marker.
- **Chop** — splits the current region at every cut-point marker you've placed, and assigns the resulting slices to this pad and the pads after it (up to the last pad of the grid — slices never wrap back around).
- **Clear Cuts** — removes any pending cut-point markers without chopping.

Slice

The row under Chop cuts the region for you. Pick a **mode**, set its amount with the slider, and press **Slice**: slice 1 stays on this pad, the rest go on the pads after it. **Undo** puts it back.

Mode	The slider sets	What it does
Equal	how many slices (2-64)	Cuts the region into equal pieces.
Transients	sensitivity	Cuts at every hit. Low sensitivity keeps only the big hits (kick, snare); high catches the quiet ones too (ghost notes, hats).
Beat grid	slice length: 1 bar, 1/2 bar, 1 beat, 1/8, 1/16	Cuts on the beat, from the sample's detected tempo, lined up with its first hit. Each line settles onto a hit that's close by, so a played loop still cuts on its real hits.
Find chops	how many chops (2-32)	Picks strong hits that start a phrase (on the beat, especially the one) and makes a chop from each to the next, at most two bars long. The quick way to pull the best parts out of a loop.
Random	how many slices	Cuts at random points (nudged onto nearby hits). Press again for a different set.

- If there are more slices than pads left, the rest are skipped and a message by the Slice button says how many. Pick a bigger Grid Size, or slice from an earlier pad.
- **Beat grid** needs a tempo. If none was detected in the sample, it estimates one from the region, and failing that uses your project's tempo (the message says so).
- Cuts are placed a hair before each hit so the attack is never shaved off, and every slice fades in and out over a millisecond or so at the points where it was cut — no clicks, even cutting through a held note. A pad playing a whole sample keeps its instant attack.
- To reverse one slice, select its pad and turn on **Reverse** in the sound controls — each slice is its own pad.
- **More** has two extras:
  - **Slice into the next empty bank:** the same slices, but onto pads 1 onward of the next bank that has no sounds and no pattern. The plugin switches to that bank, and your current kit isn't touched. While the song is playing, the switch (and the slices) happen at the start of the next bar. Undo takes the slices back.
  - **Save slices as MIDI...:** saves this pad's slices as a MIDI file that plays them in their original order and timing. Drag it into your DAW and the loop plays back from the pads, ready to rearrange. Or skip the dialog: drag the small dotted grip beside **More** straight onto a track.
- **Quantize** toggle — when on, placing a cut point or dragging a trim handle snaps to the nearest beat-grid line (shown as faint vertical lines once a tempo is detected) instead of the exact pixel you clicked. With no detected tempo, it snaps to the nearest transient instead.
- **Alt-click** inside the region to **hear the slice** there — from the cut point before your click to the one after it (or the region's ends). It places no cut, so you can check every slice before you Chop.
- **Spectrum** toggle — shows the sample as a spectrogram: time left to right, low frequencies at the bottom, louder brighter. Handy for finding where a hat or a vocal sits before you chop. It follows the zoom.
- **Zero X** toggle — when on, cut points and trim handles land on the nearest zero crossing (within 5 ms), so a slice's edges don't click. With Quantize on too, they snap to the beat or transient first, then to the crossing beside it.
- **Process...** — edits for the sound itself:
  - **Trim silence** trims leading and trailing silence (below roughly -48dBFS) from the current region automatically. Only scans within the region you've already set, so running it again after a manual trim narrows further rather than re-scanning material you already cut away.

  - **Fade in** (over the region's first tenth, at most a second) and **Fade out** (over its last quarter, at most two seconds; silent after the region).
  - **Gain +3 dB / -3 dB**, **Remove DC offset**, and **Crop to the region** (keeps only what's between the trim handles).
  - **Loop seam** shows where a looping pad wraps round: the region's end on the left, its start on the right, zoomed in (**+** / **-**). A clean loop's waveform runs straight across the middle line. Drag the left side to move the end, the right side to move the start (**Zero X** keeps them on zero crossings), or press **Smooth seam** to move the end a little to where it joins up. **Hear** plays the region. The Loop page's **X-Fade** smooths whatever's left.
  - **Warp the hits to the grid** lines up a loose loop: each hit is pinned to its nearest 1/16 (the region counted as whole bars at the song's tempo), and Time-Stretch stretches each part between them on its own, so the hits land on the grid and nothing else moves. It turns Time-Stretch on. The markers show as lines at the hits, with a tick where each now plays. **Alt+Shift-click** the waveform to add a marker (on the same grid) or take one away; **Clear the warp markers** takes them all off. They're saved with the pad.

  Each of these saves the edited sound as a new WAV in the "Boom Bap Pad Bounces" folder in your Music folder and puts it on the pad, so your project keeps it. The region and the pad's name stay. Undo puts the sound from before back.

- **BPM / Key** readout — automatically detected when you load a sample (shows `--` for either if nothing could be estimated — a one-shot hit with no discernible tempo, for example). **Sync** sets this pad's Speed so its detected BPM matches your host's current tempo, one click.
- **Freq** toggle — switches the waveform from the plain gold colour to a tint by frequency content (bass-heavy = red, treble-heavy = blue), so you can spot where the low end and the brighter transients sit at a glance.
- **Zoom slider** — 0 (fully zoomed out) to fully zoomed in.
- **Ctrl+scroll** on the waveform — zooms in/out anchored to wherever your cursor is, like a code editor. Plain scroll (no ctrl) pans left/right once you're zoomed in.

Recording into a pad

Rec (bottom left of the Sample Editor) records your audio input straight onto the selected pad: a vocal, a guitar, a record playing on a real turntable, anything coming into your computer.

1. **Turn the input on** (once):
   - **Standalone:** choose your microphone or audio interface as the input in the audio settings. If there's no input yet, Rec offers to open them; you can also right-click Rec. The little meter beside Rec moves when sound is coming in.
   - **In a DAW:** send a track, or your interface's input, to Boom Bap Producer Pads' **sidechain input**. Every DAW does this a little differently; look for "sidechain" in the plugin's routing. The input stays off until you do, so nothing changes for projects that don't use it.
2. **Press Rec.** It says *Waiting for sound...* and starts recording the moment sound comes in, keeping a few milliseconds from just before so the attack isn't cut off. Right-click Rec and pick **Start straight away** to record from the moment you press it instead.
3. **Press Stop** when you're done (or **Cancel** while it's still waiting). Recordings stop by themselves after 2 minutes. Or right-click Rec and pick **Record 1 / 2 / 4 / 8 bars, then stop**: the take is exactly that long at the song's tempo, counted from the first sound when it waits for one.

- **Count-in:** right-click Rec and pick **Count in 1 bar** or **2 bars**. You hear a click on each beat (high on the one), then it records, without waiting for sound. While the song is playing (in your DAW, FL Studio included, or here) the take starts exactly on a bar line, so it lines up with the grid.
- **Loop takes:** with a length in bars, tick **Loop takes** in the same menu. Rec then keeps going, pass after pass, until you press Stop: each pass is its own take, saved and put on the pad as it finishes (so you hear the newest), and an unfinished last pass is dropped. **Put a take from the last loop on the pad** brings back an earlier one. With a count-in, every pass starts on a bar line.

The take is saved as a WAV in your **Music\Boom Bap Recordings** folder (named after the pad and the time) and loaded onto the pad it was armed on, so your project, presets and kits keep it like any other sample. Right-click Rec → **Open the recordings folder** to find them. Right-click Rec → **Grade** prints what Rec and Grab record through one of the vintage converters (12-bit, 8-bit and so on) as it's saved, or keeps it **Clean** -- the grit is in the file, so it goes everywhere the sample does.

Right-click Rec → **Auto-sample an outside synth** plays notes out of the plugin's MIDI out and records what comes back into its input: **one note** (C4), **one per octave** (C2 to C6) or **every 4 semitones**. Each note is held for 1.5 seconds and given 1.5 seconds to ring out; then the takes are trimmed and put on empty pads side by side as **key zones** -- select the first and play it from the keys. Route the synth first: its MIDI in from the plugin's MIDI out (MIDI Out on), its sound into the plugin's input (the Standalone's audio settings, or your DAW's routing). What's coming in is never played through the plugin, so you won't hear it twice (or get feedback) while recording.

Note Map: the notes in a sample

Notes (in the editor's tools) turns on the **Note Map**: a grid over the waveform showing which notes are sounding through the sample, C at the bottom rising to B at the top -- brighter where a note is stronger. Use it to find the notes in a loop, then play them back on the pad or send them to the synth.

- **Pick notes**: click or drag across cells to select them (a drag that starts on a selected cell deselects). **Select Above** picks every cell over the threshold slider beside it -- higher keeps only the strongest note at each moment.
- **Octave** shifts every picked note up or down.
- **Preview** plays the picked notes in order on this pad (at a fixed pace, not your tempo); **Loop** repeats them.
- **Commit** writes the picked notes onto this pad's pattern, a melody of the pad itself (see the pad melody on the KEYS tab).
- **To synth part** puts the picked notes into the KEYS synth part instead -- each held to the next -- so the synth can double or replay the sample's melody.
- **Chords** reads the sample's chords (one a beat, across its bar) and writes them into the synth part as major and minor triads.
- **To synth part** and **Chords** replace the synth part's notes; Undo on the KEYS tab brings them back.

Grab: the last few seconds, onto a pad

Grab (next to Rec) is always listening. Press it and the last 30 seconds -- of whatever the plugin has been playing -- go onto the selected pad, the quiet bits at either end trimmed off. Nothing to arm, nothing to miss: play around until something sounds good, then Grab it.

- It's **resampling** in one click: jam a beat, Grab it, chop it on the pad or send it to the deck (pad menu), and build on it.

- **Right-click Grab** for the length (the last **10**, **20** or **30** seconds, or the last **1**, **2**, **4** or **8 bars** -- cut exactly on the bar lines, ready to loop; that needs the song to have played, and works after Stop too), what it keeps -- **what's playing** (the default) or **the audio input** (set up as for Rec, above) -- and **Open the captures folder**.
- Grabs are saved as WAVs in **Music\Boom Bap Captures** and loaded onto the pad like any sample, so your project keeps them.
- Switching what it keeps starts the listening afresh.
- **Straight into your DAW:** drag the small dotted grip beside Grab onto a track. The last seconds are saved to the captures folder the same way and dropped in your DAW, without touching any pad.

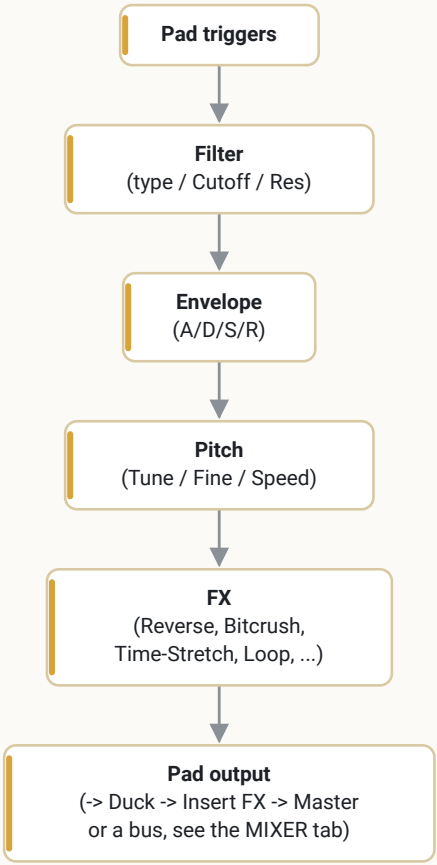
CHAPTER 08

A pad's sound controls

Below the sample editor on the PADS tab — the same per-pad chain, always following whichever pad is selected. Every knob/toggle on the Sound page is a real, host-automatable parameter and supports MIDI Learn. Scrolling over any knob nudges its value by a small, precise step.



The pad's sound controls



The buttons along the top of the panel pick what it shows: **Sound** (the controls below), **Mod** (the Mod page: LFO, pitch envelope, velocity response, loop shaping and Voices) or **Grain** (the Grain page: granular mode). When a pad is in granular mode its button reads **Grain ***, so you can tell from any page.

Faders (top of the panel) switches every knob on the Sound page to a vertical fader instead — same controls, same values, same MIDI Learn bindings, just a different shape (bottom-to-top strips instead of dials), if that's a layout you find quicker to read or automate by ear. **Expand** grows this panel by temporarily shrinking the sample editor above it, for more room to work the strips precisely. Both are purely visual — nothing about the sound or the underlying parameters changes.

- **Filter** — a type picker, Cutoff (20 Hz–20 kHz) and Res(onance) (0.1–10), reset fresh on every trigger. The types:
  - **Low-Pass** (the default) keeps the lows and removes the highs: turn Cutoff down to make a sound darker or more muffled.
  - **High-Pass** removes the lows: thin out a sample so it sits above the kick and bass.
  - **Band-Pass** keeps a band around Cutoff: the "telephone" or "radio" sound. Res makes the band narrower.
  - **Notch** removes a narrow band around Cutoff and keeps the rest.
  - **Low-Pass 24**, **High-Pass 24** and **Band-Pass 24** do the same as their plain versions, twice as steeply -- a cleaner cut.
  - **Comb** rings at Cutoff's note and its harmonics; Res sets how much.
  - **Ladder** is a warm 24 dB low-pass that sings when you turn Res up; **Diode** is its squelchier, 18 dB cousin. **Vowel** makes the sound talk: sweep Cutoff from bottom to top for "a", "e", "i", "o", "u"; Res sharpens it.

They're the SYNTH's own filters.

Switching away from Low-Pass while Cutoff is fully open (20 kHz) moves Cutoff somewhere audible for you (200 Hz for High-Pass, 1 kHz for the others), since a fully open high-pass would be silent.

- **Hold** and **Curve** (in the envelope row) — Hold keeps the sound at full level for a while after the attack (up to 2 seconds) before the decay starts. Curve bends every stage of the envelope: 0 is straight lines (as it always was); turn it up and each stage moves fast, then settles -- snappy and punchy; turn it down and each starts slow.
- **Shape, Comp and Tone** (QUICK, beside Pitch) — three quick moves for the pad: **Shape** up makes each hit snappier, down softens it; **Comp** squeezes the sound together and brings it back up; **Tone** down is darker, up brighter. They come after its insert effects, and exports have them too. Double-click one to zero it.
- **Envelope** — Attack, Decay, Sustain, Release (0–5 seconds, Sustain is a 0–1 level). This is position-based, not held-note-based: since pads are one-shot, Attack/Decay ramp in from the start of the region and Release ramps out over the last R seconds *before* the region's natural end.
- **Pitch** — Tune (±24 semitones), Fine (±100 cents), Speed (0.5x–2.0x). The **x2** and **/2** buttons beside Speed double or halve it in one click, which takes it an octave up or down. By default Speed is a simple resample-based stretch, so it changes pitch too — turn on **Time-Stretch** (in the FX row) to decouple them: Speed then changes duration only, while Tune/Fine keep controlling pitch on their own. Time-Stretch runs in the background and takes a moment to catch up after you change Speed or first turn it on — playback keeps working normally with the old behavior in the meantime. The menu beside Time-Stretch picks **how** it stretches:

Stretch mode	What it sounds like
Clean	Smooth joins -- the stretch tries not to be heard.
Tight	Old samplers' stretch: short slices repeated or skipped, joined as they fall. A light warble; drums stay punchy.
Loose	Longer slices: an audible stutter and wobble, good on chords and loops.
Buzz	Very short slices: a robotic buzz on voices and sustained notes -- the classic jungle stretch. Push Speed further for more.
Tonal	Long, smooth slices: held notes and chords stay steady.
Texture	Tiny blurred grains: a soft smear for pads, drones and noise beds (not drums).

Tight, Loose and Buzz are the vintage modes; picking one turns Time-Stretch on. They leave the pitch where it was to within a few hertz (that small drift is the warble). Time-Stretch and its mode are kept with the project, the pad's bank and a saved kit. **Key Snap** (in the FX row) rounds Tune to notes that are actually in the project's key (KEYS > Key and Scale) -- or, with the scale on None, the pad's detected key -- instead of any raw semitone; the pad's detected key needs to have been detected first (see the BPM/Key readout in the sample editor).

- **FX** — Reverse (plays the region backward), Bitcrush (Bits: 2–16, Rate Div: 1–32, a sample-rate-reduction/decimation effect for lo-fi grit -- see **Crush modes** below), Time-Stretch and Key Snap (see Pitch above), **Loop** (repeats the region instead of stopping — best used with a held MIDI note or the on-screen keyboard, since releasing the note is what stops it), and **Play To End** (ignores note-off, always plays the full region regardless of how short the trigger was).

- **Crush modes** -- the menu beside Bits and Rate Div. Old samplers sounded the way they did for more than their bit depth: each ran at its own sample rate, and some filtered the sound before sampling it (warm, dark) while others didn't (sharp highs folding down into strange ringing tones -- the "dirty" sound). Each mode is one of those converters, with its own rate, bits and filters:

Mode	What it does to the sound
Raw	The Bits and Rate Div knobs, as before.
Gritty 12-bit	26 kHz with no filter going in: highs fold down into grit. The classic dusty drum sound.
Warm 12-bit	40 kHz, filtered going in and out: round, a little dark, no harshness.
Crunchy 8-bit	22 kHz, fewer steps and folding highs: crunchy, buzzy.
Lo-fi 8-bit	11 kHz: dull and small, like an old radio or a games console.
Soft 16-bit	32 kHz: clean, with the top end gently rolled off.

  Picking a mode turns Bitcrush on, and greys out Bits and Rate Div (the mode sets them). It works the same at any project sample rate, plays the same in Export WAV, goes with the pad when you swap two pads, and is a "Pad N Crush Mode" parameter your DAW can automate.

  AA (anti-alias, under the mode) overrides the converter's input filter: **auto** (as the mode has it), always **on** (smoother and darker, even on Gritty and Crunchy) or always **off** (the bright, ringing aliasing, even on Warm and Lo-fi).

- **Normalize** — scans the pad's current region and boosts it so its peak hits 0dBFS. **Reset** removes that boost. Non-destructive; recomputed automatically if you re-chop or re-trim.

The Mod page



The Mod page: Life and the four Vary amounts along the bottom

Press **Mod** at the top of the panel. These controls make a pad's sound move and respond to how hard you play. Like everything else in this panel they follow the selected pad, and they're saved with your project, in bank kits and in presets. They're also copied by A/B compare and apply to every batch-selected pad. Unlike the Sound

page, they aren't DAW parameters, so they can't be automated or MIDI-learned -- except the LFO's **Depth** (see below). **Reset Mod** turns all of them back off for the pad (the filter type and Voices stay).

- **LFO**: a slow wobble. Pick what it moves:
  - **Pitch** for vibrato. Low Depth is a gentle vibrato; high Depth is a siren, up to an octave either way.
  - **Filter** for a "wah" sweep, up to 4 octaves either way around Cutoff.
  - **Volume** for tremolo. Full Depth dips all the way to silence.
  - **Pan** to swing the sound left and right.

  Shape sets how it moves: Sine (smooth), Triangle, Square (on/off, softened so it doesn't click), Saw Down, or Random (a new level each cycle). **Rate** is in cycles per second. Turn on **Sync** to lock it to the tempo instead, from once a bar down to every 1/16 note. The LFO starts from the beginning on every hit, so a pad sounds the same each time you play it (Random included).

  Depth can be automated in your DAW. Each pad has a parameter called **Pad N LFO Depth** (Pad 1 LFO Depth, Pad 2 LFO Depth...): draw automation for it to open a filter sweep over a build-up, or bring in vibrato on the last bar. The Depth knob follows the automation while it plays. It's also saved with the pad everywhere the other Mod settings are.
- **Shape**: a movement you draw, as a second LFO, a step pattern or an envelope. Pick what it moves (**Pitch**, **Filter**, **Volume** or **Pan**), how far (minus turns it upside down), and press **Draw...**: click to add points, drag them, drag a line to bend it, or switch to steps; set its length (1/16 to 4 bars, in time with the song) and whether it **loops** or plays once from each hit and holds its end. A volume shape in eighths is an instant chop; a pitch shape that drops once is a tape stop.
- **Filter Envelope**: **Amount** moves the filter's cutoff at the moment of the hit (up to 4 octaves up, or down), and **Sweep** is how long it takes to come back to the Cutoff knob. Up with a short Sweep gives a pluck or a "pew"; up with a long one, a slow opening. It works with every filter type.
- **Pitch Envelope**: **Amount** bends the pitch at the moment of the hit (up to 24 semitones either way), and it glides back to the pad's own pitch over the **Glide** time. A positive Amount with a short Glide (50-150 ms) gives a kick or tom a punchy "thump"; a long Glide gives a tape-start or riser feel.
- **Velocity**: how the pad responds to how hard you hit it. **Filter** makes softer hits darker, up to 4 octaves of Cutoff at the softest. **Start** makes softer hits start later into the sound (up to 50 ms), skipping some of the attack. Both do nothing on full-velocity hits.
- **Loop** (needs **Loop** turned on on the Sound page): **Ping-Pong** plays the loop forward, then backward, then forward again. **X-Fade** (up to 500 ms) smooths the loop point: the end of the loop fades into its start, so a loop chopped at an awkward spot doesn't click or thump each time round. X-Fade works on forward loops (not Reverse or Ping-Pong).
- **Voices**: how many hits of this pad can sound at once.
  - **1 - cut** (the default): a new hit stops the one before it, with a tiny fade so it doesn't click.
  - **2 to 8**: earlier hits keep ringing underneath the new one. Use this for long cymbals or 808s that shouldn't choke themselves, rolls that overlap, or playing chords on the KEYS tab from one pad. When a pad runs out of voices, its oldest hit fades out.

  Choke groups, Mono mode and the stop button still cut every voice. The pad shows as playing while any of its hits still sounds.
- **Width** and **Mono below** (beside Voices), **Phase** and **Offset** (in the Loop row, on the right):
  - **Width**: 0% mono, 100% as it is (double-click), 200% twice as wide.

  - **Mono below**: everything under this frequency is made mono, so the pad's low end stays centred (under 40 Hz is off).
  - **Phase**: flips the pad's polarity. Two layered kicks or snares can partly cancel each other and sound thin; if they do, flip one.
  - **Offset**: plays the pad's sound up to 10 ms later -- slide one layer until the two hit together and sound fullest. Unlike SEQ's Nudge, it moves the sound however the pad is played.

- **Release**: a pad that plays when you **let go** of a note on this one -- the thump of a piano key coming up, a string's release, a breath at the end of a vocal note. Load that sound on another pad and pick it here. The menu beside it says how it fades: the longer the note was held, the quieter the release (half as loud every 0.5, 1, 2 or 4 seconds, or **No fade**), and it follows how hard the note was hit.
  - It plays when a note **ends**: notes from your DAW's piano roll or a MIDI controller, the on-screen keyboard, and SEQ steps given a **length** (the note is let go where the length runs out). A step with no length just plays its sample out, so there's no release. Export WAV plays the releases too.
  - With **key zones**, give each zone pad its own release, or just the zones you want.

- **Follow**: this pad's **Volume** or **Filter** moves with another sound's level -- another pad, the bass, the synth or a deck.
  - **Volume, +**: the pad only comes through while the other sound plays -- a gate. Put a long pad or a noise riser on it and follow the kick for a chopped, pumping texture.
  - **Volume, -**: the pad dips while the other sound plays -- a duck (like Duck on the MIXER, but from any one sound).
  - **Filter, +**: the filter opens as the other sound gets louder -- hats that brighten with the bass. **-** closes it. Up to four octaves.
  - It reacts within a few milliseconds and isn't in **Export WAV** (an export renders each pad on its own, without the others to listen to).

- **Vary**: every hit comes out a little different, so a hat played on every 16th or a snare roll doesn't sound like a machine gun. **Life**, the first bar on the row, does it all with one knob. At 100% each hit gets up to half a semitone of pitch, 30% pan, an octave of filter and 12 ms of start; 20-40% sounds like a real player. The four thin sliders beside it set each amount on its own, and Life then shows the average. Each is 0 (off) by default; drag one, double-click to reset:
  - **st** -- random pitch, each hit up to this many semitones higher or lower. 0.1-0.3 is a live drummer; a whole semitone is a wobbly tape.
  - **% pan** -- each hit up to this far left or right of the pad's Pan.
  - **oct** -- the filter cutoff, each hit up to this many octaves higher or lower (hear it with the cutoff below fully open).
  - **ms** -- each hit starts up to this much later into the sound, softening or blurring the attack.

  It isn't truly random each time: the 1st hit of a pad after you press Play always gets the same change, the 2nd hit its own, and so on. So a beat sounds the same every time you play it from the start, and **Export WAV** sounds exactly like it. (A pad hit live while the beat plays counts as one of its hits too.) With **Layers**, the hit's change is shared by the layer that plays -- and in a blend by every layer, so they stay in tune with each other.

Snapshots and macros

Beside **Store A** and the A/B button:

- **Snaps**: four sound snapshots for the pad. **Store this pad's sound** in 1-4, **Recall** it on the pad, or **Recall it on every pad that has one** -- a whole kit's sound at once. A snapshot is the pad's sound (not its sample). From your DAW: the **Sound Snapshot** parameter (Hold, 1-4; automate it in FL Studio) or MIDI program changes 0-3 recall one on every pad that has it. Snapshots are saved with the project. Under the snapshots: **Randomise: a nudge** / **a lot** shakes up the pad's filter, envelope, fine tune, crush and quick knobs (never its sample, level, pan or tune), and **Init this pad's sound** puts it back to how a new pad starts. Undo brings either back.
- **Macros**: two knobs for the pad. **Moves...** under each picks what it turns -- any Sound control, or a Mod or Grain knob -- each from where it is now toward its far end. Right-click a knob to MIDI Learn it.

The mod matrix, MIDI Learn and formants



The Matrix: Steps on the volume, the LFO on the filter, and the Steps bars

- **Matrix...** (Mod page, on the Shape row): four slots, each a source -- **LFO**, **Shape**, **Velocity**, **Random** (a new value each hit), **Key** (the note played), **Mod wheel**, **Pressure** -- moving the **Pitch** (12 semitones at 100%), **Filter** (4 octaves), **Volume** or **Pan** by its amount. They add to what the LFO and shape already do.
- **Steps** is a source you draw. Under the four slots, drag the 16 bars up or down (right-click one to clear it). Then pick how many steps it uses before it goes round again, each step's length (1/32 to a bar, at the song's tempo) and **Smooth** (0% jumps, more glides into the next step). It restarts with every hit. Send it to Volume for a trance gate, to Filter for a moving wah, or to Pitch for a little melody on one hit.
- **Steps 2** is a second one. Click **Steps** or **Steps 2** above the bars to draw either, and pick either as a slot's source, so one pad can have a gate on its volume and a different pattern on its filter.
- **Song** (next to Smooth): off, the steps start again with every hit. On, they follow the song's beats instead, so a hit on beat 3 starts on the step that belongs to beat 3, and a pad played in different places of the bar moves differently each time. With the song stopped, Song falls back to starting with the hit.
- **Right-click any Mod or Grain knob** for MIDI Learn. While a project macro is assigning, the right-click adds the knob to the macro instead. (In FL Studio, automate a macro to automate these.)
- **Formants** (Mod page): keeps the voice's character when the pad is pitched up or down -- no chipmunk. It makes the pad sound about 23 ms late while on, so use it for melodies and vocals rather than drums.

The Grain page

Press **Grain** at the top of the panel and turn on **Granular**. Instead of playing its sample from start to end, the pad now plays a *cloud* of tiny overlapping pieces of it ("grains"), each a few milliseconds to half a second long and faded in and out. Use it to turn a vocal, a chord or a sustained note into a pad, a texture or a drone, or to freeze a

single moment of a sound and hold it.

Like the Mod page, these settings follow the selected pad, are saved with your project, in bank kits and presets, and are copied by A/B compare and batch selection. They aren't DAW parameters. **Reset Grain** puts the knobs (and the shape) back to their defaults and leaves Granular on or off.

Shape (beside the Granular switch) sets how each grain rises and falls:

- **Smooth** (the default): fades in and out -- soft, blended clouds.
- **Flat**: full level between short fades, so each grain keeps more of the sample's own attack and body.
- **Sharp**: starts at once and dies away, like a plucked or struck note -- rhythmic, glitchy clouds, especially at low Density.
- **Swell**: the reverse: rises and stops -- a backwards, breathing texture.

The level stays about the same whichever shape you pick.

- **Cloud**
  - **Size**: how long each grain is (10-500 ms). Short grains sound buzzy and blurred; long grains sound more like the sample itself.
  - **Density**: how many grains start each second (1-100). Low is sparse and stuttery, high is a smooth wash. The level stays about the same as you turn it up.
  - **Position**: where in the sample the cloud starts, from 0% (the start) to 100% (the end). With Reverse on, it counts from the end.
  - **Scan**: how fast the cloud moves through the sample. **100%** moves at the sample's own speed, **50%** at half speed (a slow stretch), **0** *freezes* it on one moment, and below 0 moves backward. Double-click the knob to go back to 0. When the cloud reaches the end of the sample it carries on from the start.
- **Spread**
  - **Scan: free / 1 / 2 / 4 / 8 bars** (top row): free lets Scan set the speed; a number of bars makes the cloud cross the whole sample in exactly that many bars at the song's tempo, so a long texture lines up with your beat.
  - **Jitter**: scatters each grain's start a little around the cloud's position, so the cloud sounds less mechanical. High Jitter mixes moments from all over the sample.
  - **Pitch**: each grain plays up to this many semitones above or below the pad's pitch, at random. A few semitones gives a chorus-like shimmer; 12 gives a cloud of notes.
  - **Stereo**: how far grains spread left and right.

How long a granular hit lasts: as long as the pad would normally play (its region at its pitch and speed), with the Sound page's envelope shaping it. For a sound that keeps going for as long as you hold the note, turn on **Loop** on the Sound page. The pad's Tune, Fine and Speed set the grains' pitch, and its filter, envelope, LFO and FX all still apply. A pitch LFO or the Pitch Envelope bends every sounding grain as it plays, so vibrato and pitch drops work on a cloud the way they do on the plain sample. **Keys place** (beside Scan sync): play the pad from the keyboard (or MIDI channel 2) and each note picks where the cloud sits -- an octave below C3 is the start, an octave above the end -- instead of changing its pitch. Granular mode also sounds the same in **Export WAV** and **Export Full Song**. It replaces the loop shaping options (Ping-Pong and X-Fade) while it's on.

Output routing

Pads 1-16 each have their own dedicated stereo output pair, so you can send an individual pad to its own channel in your DAW's mixer instead of the shared master output — useful for processing the kick separately from the rest of the kit, for example. This is off by default; enable it from your DAW's own multi-output routing UI for this plugin instance (exactly how varies by DAW — look for "add output" or similar on the plugin's mixer channel). Once a pad's own output is enabled, that pad's audio goes *only* to its own channel, not also into the main mix.

If a bank's grid size is set larger than 4x4, pads 17 and up always play through the main mix — dedicated per-pad outputs are capped at 16 regardless of grid size, since a real stereo output bus per pad adds up fast on the host side (tested up to a noticeable routing-dialog slowdown well before 64).

CHAPTER 09

SEQ: the step sequencer



The SEQ tab

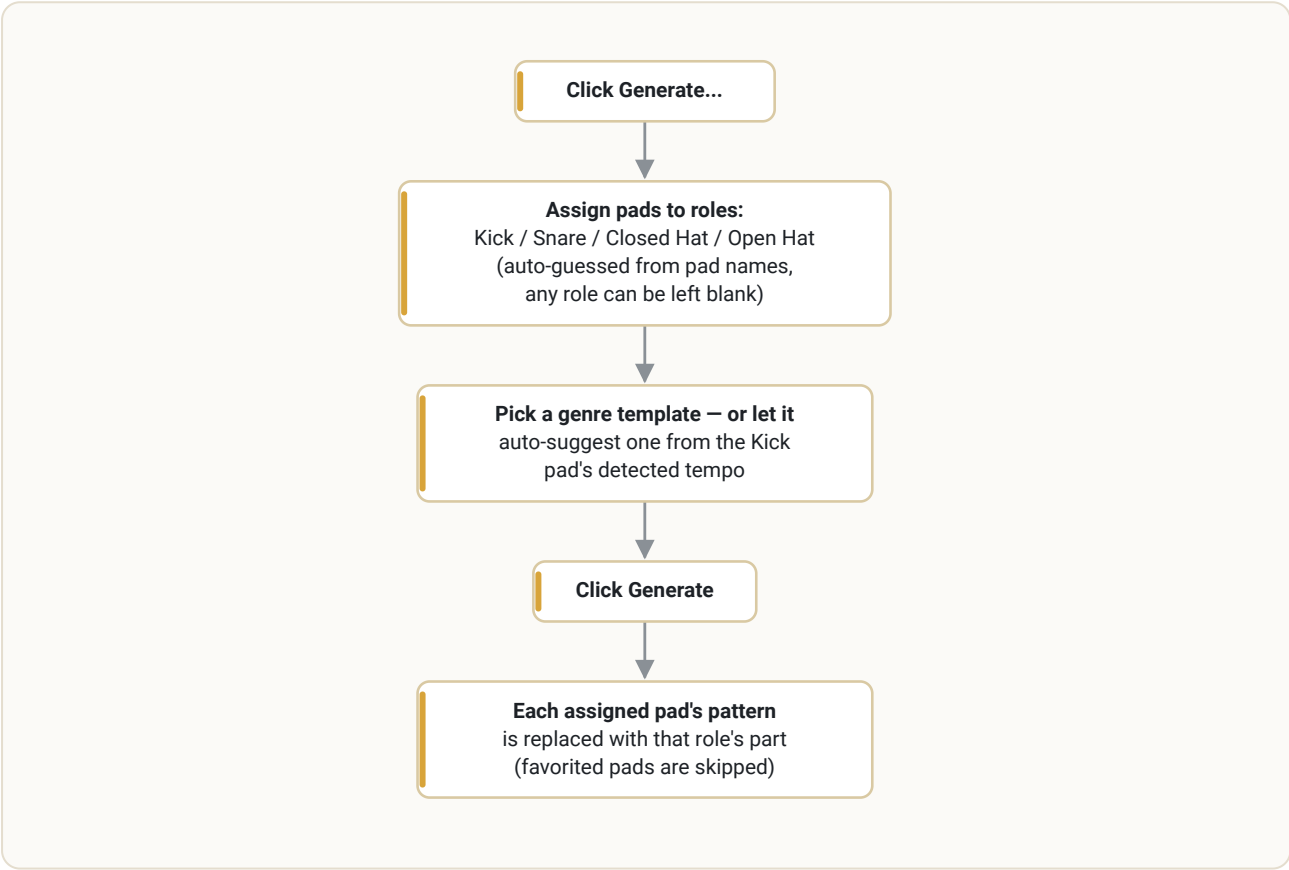
Step sequencer

SEQ tab. The whole beat in one grid: a row for every pad in the active bank (its number and name on the left), across the 16 steps of one bar of 4/4, grouped by beat in shaded blocks of 4.

- **Click a pad's name** to select that pad. The tools that work on one pad (Copy / Paste, Humanize, Randomize, Clear Pattern, Nudge) use the selected pad; it's the same selection as on the PADS tab.
- **Click a step** on any row to toggle it on/off.
- **Scroll on a lit step** to adjust its velocity; **Shift+scroll** to adjust its trigger probability (shown as a % on the step once it's below 100 — a step at less than 100% has a chance of skipping each time round, for evolving/generative patterns).
- **Copy / Paste** — copy the selected pad's whole pattern (steps, pitch, length, velocity, probability) and paste it onto another pad.
- **Humanize** nudges the velocity and timing of already-lit steps a little, for a less mechanical feel. **Randomize** regenerates which steps are on/off at roughly the same density as before — the two compose (Randomize first, then Humanize the result, or either alone). **Flip** generates a fresh pattern across every pad that's a slice of the same chop as the selected pad — a quick way to hear a chopped loop rearranged. Right-click a pad and

choose **Favorite** to protect its pattern from both Randomize and Flip — a favorited pad (shown with a small gold star) keeps its pattern exactly while everything else changes around it, so you can lock in a part you like and keep regenerating the rest.

- **Shift-click Randomize** to keep the steps on the beat (1, 2, 3, 4) and redraw only the ones between.
- **Generate...** builds a whole drum pattern across several pads at once from a genre template — different from Randomize/Flip, which each work on one pad's existing pattern. 19 styles, from Boom Bap and Golden Era to Amapiano, Gqom, Kwaito and Baile Funk; besides Kick, Snare and the hats it can write a **Clap / Rim** and a **Perc** part. **Density** thins the pattern out (under 50) or fills it in (over 50) -- the kick on 1 and the backbeat always stay; **Complexity** adds ghost notes, a syncopated kick and hat rolls; **Variation** changes the later bars from the first; **Fill in the last bar** ends on a snare run. **Continue bar 1** keeps the first bar you made and carries it through the pattern's other bars, varied by Variation. Opens a small popup:



  - Roles left as **(none)** are skipped — you don't have to fill in all four, e.g. assign just Kick + Snare and leave the hats alone.
  - A pad named something like "Kick 1.wav" gets auto-selected for the Kick role (matched by name, e.g. "kick"/"bd", "snare"/"sd", and similar hints for the hats) — always changeable, never required.
  - 8 genre templates: Boom Bap, Trap, House, Dembow, Halftime, Afrobeats, Drill, Lo-Fi — each a different kick/snare/hat feel.
  - The Template dropdown auto-suggests whichever template best matches the Kick pad's detected tempo (closest match if none fits exactly) — a starting point, always changeable.
  - Generate **replaces** each assigned pad's whole pattern (not a merge) — same one-undo-snapshot safety net as Randomize/Flip, so Undo gets you back if you don't like the result.
  - Favorited pads are protected here too, same as Randomize/Flip.

- **Export WAV...** renders the current pattern (1-8 bars, the length beside it) to a WAV file, and **Export Full Song** the whole arrangement. Both are the whole mix as you hear it: the pads, the bass and the KEYS synth part, with their effects, sends, Duck and buses, and the master's effects, fader and limiter. A pattern's reverb and delay tails wrap round to its start, so the file loops cleanly; a song's ring out at the end.
- **Drag straight into your DAW.** The small dotted grips skip the save dialog: drag the one beside **Export WAV** for the beat as audio (as many bars as chosen), beside **Export Full Song** for the song (it renders first, so give it a moment), and beside **Export MIDI** for the pattern as MIDI. Drop it on a track. The file is named with the tempo and key ("Boom Bap Beat - 4 bars - 90 BPM - A Minor.wav"), so a DAW that reads tempos from names lines it up, and it's kept in **Music > Boom Bap Drag Out**, by date -- your DAW keeps pointing at it, so don't empty that folder while a song still uses something in it.
- **Export SFZ...** exports the current kit (every loaded pad's trimmed sample + volume/pan/tuning/ADSR/reverse, Play To End or Loop, and choke groups) as an SFZ instrument you can load in any SFZ-compatible sampler outside this plugin. More formats (SoundFont, DecentSampler, sliced WAV, an FL Studio pack) and kit import are on the PADS tab's **Kit...** button -- see Kit import & export.
- **More...** (beside **Export Kit**):
  - **Stems**: one WAV per sound -- each loaded pad, the bass, the synth -- with its own effects, sends and bus, for this pattern (the bars chosen) or the whole song, into a new folder. The master's effects and limiter are left off, so the stems add up to the mix in your DAW.
  - **Each song section as its own loop**, one WAV each, named by section.
  - **MIDI**: one file with a track per pad, the bass and the synth, for the pattern or the song.
  - **Import drum MIDI** replaces the pattern: **by sound** puts each note on the pad that sounds like it (a General MIDI kick on your kick, a hat on your hat), **by note** puts note 36 on pad 1, 37 on pad 2... Undo puts the pattern back.
  - **Export options**: 16, 24 or 32-bit; the project's sample rate or 44.1 / 48 / 88.2 / 96 kHz; **Normalise** (the loudest peak at -0.3 dB); and **Tails** -- loops wrap round and songs ring out (as before), cut where the music ends, or ring out after. Saved with the project, used by every audio export.
- **Export Full Song** puts a marker at each song section in the WAV, named by it -- your DAW shows them.
- **Clear Pattern** wipes the selected pad's steps.
- **Right-click a row's name** to **mute** or **solo** that pad (the same as its M / S on the pad grid); the name shows [M] or [S].
- **Swing** delays every second step slightly, for a less mechanical feel. It reads as a percent: 50% is straight, 67% a triplet feel. **Groove...** picks a named feel (Straight, Nudge, Lean, Shuffle, Triplet, Lazy, Drunk).
- **Volume** — persistent per-pad level, audible immediately even on a currently-playing pad (separate from the transient per-hit velocity).
- **Pad Quantize** (toolbar) — a *live-triggered* hit (real MIDI or a click) waits for the next beat instead of firing instantly; sequencer steps are unaffected, they're already on the grid. See The toolbar.
- The sequencer follows your DAW's transport and tempo — press play in your DAW, not in the plugin. (Running the plugin standalone, without a DAW, is different — see Saving your work & running standalone.)
- Whatever's actually playing here (and on the BASS lane) can also be sent out live as real MIDI, in addition to triggering this plugin's own audio — see the toolbar's MIDI Out toggle.

Each pad has its own row and its own pattern, so you build the whole beat in one view.

Bank A / B / C / D buttons switch between 4 complete kits — see The pad grid & sample browser for what a bank switch carries with it.

Drum groove: accent, ratchet, flam, nudge and locks

The row under the SEQ buttons sets what a click on a step does -- **Edit**:

- **Steps** (the usual): a click turns a step on or off.
- **Accent**: a click makes the step louder (marked **A**), or back.
- **Ratchet**: each click plays the step 2 to 8 times inside its own 16th (marked **x2**-**x8**), then back to once -- rolls and stutters on hats and snares.
- **Flam**: a click adds a soft grace hit just before the step's hit (marked **f**), the way a drummer flams a snare.
- **Locks**: a click picks the step; then give that one step its own **Pan**, **Filter** cutoff and **Start** (how far into the slice it begins) -- marked *****. A hi-hat that opens up on the last step, a snare that hits from the middle of its sample, a kick panned for one hit. **Clear locks** puts the step back to the pad's own settings. A lock lasts until that pad's next hit; playing the pad live always uses its knobs. The same panel also has:
  - **Play** -- when the step plays (marked **?**). **1:2** is the 1st of every 2 times round its row, **2:4** the 2nd of every 4, and so on; **Fill** only while Fill is held; **First** only the first time round; **Prev** only when the row's last step with a Play setting played. A **!** means the opposite. A kick on 2:2 plays every other bar.
  - **Time** -- the step early (-) or late (+), up to half a step (marked **~**). Double-click it for on time.
  - The ratchet's **Even** / **Ramp up** / **Ramp down**: its repeats get louder or softer.
  - **Rev** -- the step plays its sample backwards (marked **R**).
  - **Sticky** -- the step's locks stay on for the pad's next steps that have no locks of their own, until a step with its own locks.

The lane, painting and the keys

The picker at the right of the Edit row opens a **lane** under the grid for the selected pad: **Velocity**, **Chance**, or one lock -- **Pan**, **Filter**, **Start**, **Time**, **Reso** (filter resonance), **Send** (its reverb send, live) or **Flam gap** (how far ahead a flam's grace hit is). Drag across it to draw a bar per lit step; right-click (or Alt-drag) clears a lock. A faint line means the step has no lock there.

- **Paint:** in Steps, press on a step and drag along the row -- every step you cross turns on (or off, if the first one was on).
- **Range:** the step you last clicked is outlined. Shift-click another on the same row (or shift + arrow keys) to pick the steps between. **Copy** then copies just those, and **Paste** puts them at the outlined step.
- **Keys** (after clicking a step): **Left / Right** move along the row, **Up / Down** to the next pad, **Space** or **Enter** clicks the step the way Edit says, **+ / -** changes its velocity, **Delete** turns it (or the range) off.

In any mode but Steps, clicking a step that's off turns it on with that change. The marks show on the steps (and the chance, if it's under 100%).

Nudge pad N (on the right when Edit isn't Locks): plays the selected pad a little late, up to half a 16th -- the laid-back snare that drags behind the beat. Each pad has its own; it's saved with the kit.

Groove... (beside Ghost...) takes the feel of a loop and puts it on your pattern. Take it from **this pattern**, from **the selected pad's sample** (a drum loop chopped onto a pad -- its hits are found for you, as 1 to 4 bars), or from **a MIDI file**; then **Apply** it to the selected pad or every pad at 25%, 50%, 75% or 100%. Each lit step moves that much toward the loop's timing and accent on that step (steps where the loop had no hit stay as they are). Undo takes it back. The groove is kept with the project.

Next to Nudge, **Swing: kit** gives the selected pad its own swing (kit = the PADS Swing) -- a straight hat over swung drums -- and **Swing steps / 8ths / 16ths / 32nds** picks what the swing moves: every second step, or the off 8ths, 16ths or 32nds.

Beside it, **< Shift** and **Shift >** move the selected pad's row a step earlier or later (what falls off one end comes round to the other), **Mirror** plays it last step first, and **Invert** turns its on steps off and its off steps on. They work over the row's own length; Undo takes them back.

All of it is saved with the pattern and its bank, copied by **Copy** / **Paste**, taken back by Undo, and heard the same in **Export WAV**.

Longer patterns

A bank's pattern can be **1 to 4 bars**. The row above the grid:

- **Bars** sets the length. Going longer starts the new bars as copies of the ones you have; going shorter keeps the bars past the end (unheard), so going back brings them back.
- **Bar 1-4** shows that bar in the grid (the grid is always one bar). **Follow** shows whichever bar is playing.
- **Speed**: **Half time** plays every step as an 8th (the pattern lasts twice as long), **Double time** as a 32nd -- the same steps, nothing rewritten.
- **Hide empty** takes the pads with no steps out of the grid, so a big kit is easy to read. The selected pad always stays.
- **Meter**: 4/4, 3/4, 5/4, 6/8 or 7/8 -- how long a bar of this pattern is. Bank and pattern switches, ARRANGE sections, the count-in and Export WAV all count its bars (a 3/4 pattern switches every 3 beats). A pattern can be up to 64 steps, so 5/4 goes to 3 bars. The bass line and synth part stay in 4/4. In 4/4 the page buttons are bars; in other meters they show the steps (1-16, 17-32...).
- **Heat** colours the steps by velocity -- teal for soft, red for hard.
- **Row** gives the selected pad's row its own length (1-64 steps; **Pattern** = the pattern's). It loops on its own: a 3-step hat row against a 16-step pattern drifts and lines up again every 48 steps -- the classic polymeter. Each row shows its own playhead, and steps past a row's end are drawn dark (they don't play).
- The picker beside **Row** puts the row on its own grid: **1/16T** or **1/8T** (triplets) or **1/32**. Its steps then play on that grid -- the columns are its own steps, not the 16ths above them (a 1/16T row has 24 steps in a 16-step bar). Accent, ratchet, flam and locks work as usual.

Each bank keeps its own length and speed. A bank switch still happens on the next bar of the song, and the new bank starts at its own bar 1. Generate, Euclid, Chords and Melody write a bar and repeat it over the pattern; Randomize, Humanize and Flip work over the whole length.

Pattern slots

Each bank holds **16 patterns** that share its kit -- the **Pattern** row:

- **1-16**: click one to play it. While the beat plays it waits for the next bar (**On beat**: the next beat) and shows a ***** until then; when stopped it switches at once. Lit buttons hold steps; gold is the one playing. Each pattern keeps its own Bars and Speed.
- **Duplicate** copies this pattern to the next empty slot and switches to it -- the quick way to start a variation.
- **Fill**: hold it and pattern 16 plays from the next bar; let go and the pattern you came from comes back. Keep your fill in 16.

- **Save... / Load...**: your pattern library (Documents > Boom Bap Producer Pads > Patterns). Load replaces the pattern you're on; Undo brings it back.
- **Bass** and **Synth part**: turn one on and the bass line (or the KEYS synth part) switches with the patterns too -- pattern 3 plays bass line 3. Turning it on copies the line you have into all 16, so nothing goes quiet; then change each one on its own. The numbers are shared by every bank.

Banks (**A**-**D**) are still whole kits; to chain patterns into a song, use the **ARRANGE** tab (right-click a section > **Pattern**).

Roll / note-repeat

Roll toggle + rate dropdown (1/8, 1/16, 1/32), next to Clear Pattern. Turn it on, then press and hold any loaded pad — it auto-retriggers at the selected subdivision for as long as you hold it, free-running from your DAW's current tempo. Release to stop. While Roll is on, dragging a pad (the sample-export gesture on the pad grid) is disabled to avoid the two gestures colliding.

With Roll on, a pad held on a MIDI controller rolls too, at the velocity you hit it with -- and pressing harder (aftertouch) plays it louder.

Roll... (beside the rate):

- **Latch** -- the roll keeps going after you let go; press the same pad again to stop it (another pad takes it over).
- **Triplets** -- the rate in triplets (1/16 becomes 1/16T).
- **Swing** -- every second hit late by the Swing amount.
- **Velocity** -- even, or ramping up or down over a bar.

With Record armed, a roll is recorded into the pattern as ratchets on the steps it crosses (a 1/32 roll over 16th steps = x2 on each).

CHAPTER 10

KEYS

KEYS has two editors -- pick one with the switch at the top left:

- **Synth part**: a part for the SYNTH tab's instrument -- chords and melodies, 1 to 8, 16 or 32 bars, looping with the transport. See the next section.
- **Pad melody**: the selected pad's own melody, one note per step (the rest of this page after the next section).

Synth part



KEYS: the synth part

A piano roll for the synth. It plays whenever the transport plays (the Play button at the top, or your DAW), looping every 1-8, 16 or 32 bars.

Note names (beside Move parts) names the notes and keys in both piano rolls as letters (C4), as **Do re mi** counted from the key (in A minor, A is do and C is me), or not at all. With the rows tall enough (Zoom), every row is named. It's remembered for every project.

- **Bars** sets how long it is; **Grid** is where notes land and how far they move: 1/4, 1/8, 1/16, 1/32, triplets (1/8T, 1/16T) or Off. **Width** spreads the beats out (then scroll sideways); **Zoom** (top right) makes the rows taller.

- **Draw**: click to add a note, the same length as the last one you made or set; drag right as you click to set its length. Stack notes on the same beat for **chords**. Drag a note to move it (hold **Alt** to copy), or drag its right edge to resize it. The same **Select** and **Erase** tools, **Key**, **Scale** and **Snap to scale** work here too.
- **Double-click** or **right-click** a note to remove it.
- **Keys**: **Delete**, **Up/Down** (transpose; **Shift**: an octave), **Left/Right** (a grid step), **Ctrl+A**, **Ctrl+D** (a copy of the selection straight after it), **Ctrl+Z** / **Ctrl+Y** (undo, redo -- or the **Undo** and **Redo** buttons), **Esc**.
- **Quantize** moves the selected notes (or all of them) onto the grid. **Clear** removes every note.
- **Chords...** writes a chord progression into the part: root, scale, progression, rhythm (one chord a bar, two a bar, or stabs on every beat), octave, and sevenths if you want them. It opens in the KEYS key.
- The keyboard on the left plays the synth; the lane underneath sets velocity (drag the bars); the bottom row shows and changes the selected notes' note, length (in 16th steps), velocity and chance -- and when they play: **Always**, **1:2** (the 1st of every 2 times round the part), **2:4**, **Fill**, **First**, **Prev** and their opposites (**!**).
- **Recording:** with **Record** armed and the beat playing, notes you play on the synth's MIDI channel go into the part where you played them (pulled to the 16ths by **Rec... > Quantize strength**).
- **Humanize** moves the selected notes (or all of them) a little early or late and a little louder or softer.
- **Notes...** works on the selected notes (or all of them): **Legato** holds each note until the next one starts; **Fix overlaps** cuts a note where the next one of its pitch begins (and removes doubles); **Length** sets them to one grid step, or doubles or halves them; **Velocity** makes them rise or fall from soft to loud, evens them out, varies them a little, or sets them all to 100.
- **Follow** keeps the playhead in view while it plays. Hold **Ctrl** and turn the mouse wheel to zoom in and out.
- **Notes... > Arpeggiate chords** turns each selected chord into an arpeggio over its length -- **up**, **down**, or **up and down** -- a note per grid step (set Grid to 1/8 or 1/16 first).
- **Play from the computer keyboard**: click the roll, then **A W S E D F T G Y H U J K O L** play the synth like a keyboard from C; **Z** and **X** move an octave down and up.
- **Ghosts** (in the tools) shows the other part faintly behind the notes -- the selected pad's melody here, the synth part in the pad melody -- to write one against the other. **Colour** colours the notes by loudness or by pitch (every C one colour, every D another). The **bass line** shows too, in faint gold, in both rolls.
- The **CC lane** (under the velocity lane): drag to draw the **Mod** wheel or **Slide** over time -- each value holds until the next; Alt or right-click removes a point. Send it somewhere in SYNTH's mod matrix (source Mod Wheel or Slide). **Notes... > Clear the CC lane** empties it.
- **Glide** (with notes selected): those notes slide into their pitch from the note before -- a slide mark shows on them.
- **Chord** (with pad melody notes selected): each plays a **triad**, a **seventh** or a **power** chord on its note, built in the key. The chord's other notes show lighter above it.
- **Free voicing**: in the pad melody, **Alt+click** above or below a note (in its columns) to add that note to the step -- any note from an octave below to two octaves above; Alt+click it again to take it away. The Chord menu then shows **Free**; picking a chord type there replaces it.
- **Phrase** (a keyboard mode, under the keys): hold a key and the synth part plays from its start, moved to that key (C4 plays it as written) -- a riff on every key, with or without the transport.
- **Chords from a sample**: on the PADS tab, in Note Map, **Chords** reads the sample's chords (one a beat) into the synth part.
- **Arp...** (in the pad melody) arpeggiates the synth part's chords onto the selected pad -- up, down or up and down, in 16ths or 8ths.

- **To pad melody...** (by the PIANO ROLL title) puts the synth part on the selected pad, one note a 16th (a chord keeps its top note); in the pad melody, **To synth part** does the reverse.
- **Stamp...** shows your key's seven chords as buttons (named as they'll sound). Pick a size and a length, then click chords in order: each one goes in after the last, so a few clicks write a progression. When the part is full the next one goes back to the start.
- The **chord lane**, under the bar numbers, shows each chord of the synth part as a named block lasting until the next one. Click a block to select that chord's notes (shift-click adds), then use Voicing... on it.
- Each chord in the synth part is **named** just above its top note -- C, Am7, Gsus4, and inversions as C/E (the bottom note after the slash).
- **Voicing...** works on the selected chords (notes that start together): **Invert up** / **Invert down** move the lowest note up an octave or the highest down one, **Spread** opens the chord out, **Close** packs it into an octave, and **Strum up** / **Strum down** start its notes a 32nd apart like a guitar (**Unstrum** lines them up again). Under **Reharmonise**: swap each for the chord **a third down** (C becomes Am) or **a third up** (C becomes Em) in your key, **simplify to triads**, or **make them sevenths**.
- **Step in:** turn it on and each note you play goes in at the cursor, a 16th long; let go and the cursor moves on a 16th (play a chord and it goes in together). The button shows the step it writes next.
- **Import MIDI...** (or drop a .mid file) brings every note of a MIDI file in, up to 8 bars; **Export MIDI...** and the drag handle next to it take the part out to your DAW.
- The sound is the SYNTH tab's: pick a preset there. **Export WAV** includes the part.

Pad melody



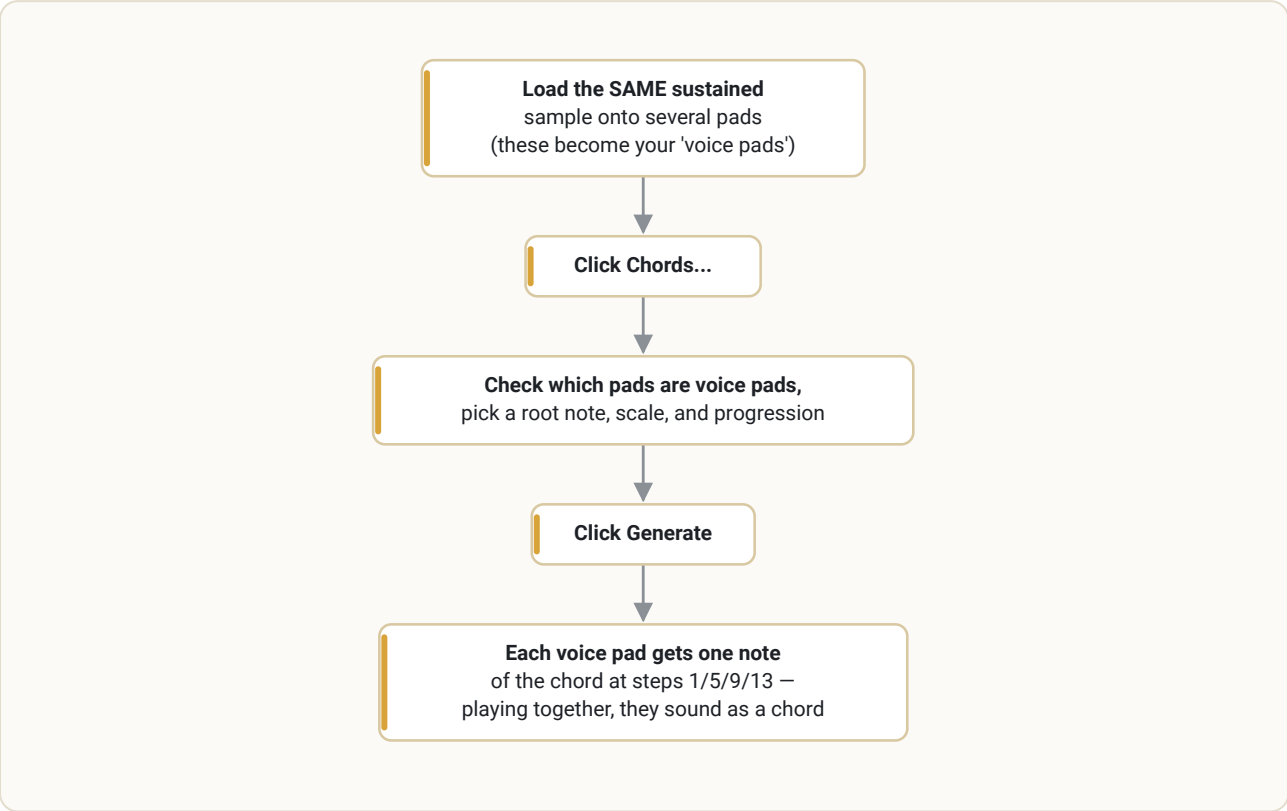
KEYS: a pad's melody

A vertical piano keyboard on the left plus a step x pitch grid (as long as the pattern -- up to 4 bars, see SEQ's **Bars**) (the piano roll) on the right, both scrolling together. This always plays and edits whichever pad is currently selected on the PADS tab — it's not a separate instrument, it's a melodic/chromatic way to play and sequence that same pad's sample.

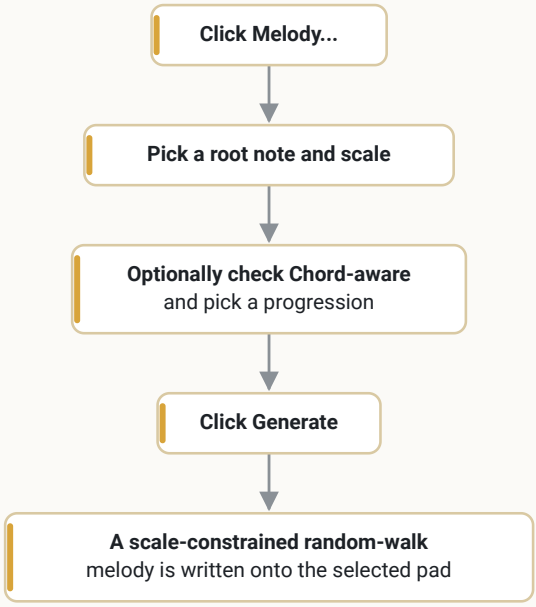
The grid fills the whole panel: the steps stretch to its width, and at the left end of **Zoom** every row fits its height (slide right for taller rows, then scroll). The beat numbers run along the top, and each note's velocity shows in the lane underneath.

- **Click or drag along the keyboard strip** to audition the selected pad at different pitches. The highlighted row is MIDI note 60 (middle C) — that's the pad's own natural pitch (its Tune/Fine settings), with every row above or below shifting by a semitone (±36 semitones total -- three octaves each way).
- **Three tools** (top left):
  - **Draw**: click an empty spot to place a note (drag right in the same move to make it longer). Drag a note to move it, drag its right edge to resize it. Hold **Alt** (or **Ctrl**) while dragging to copy instead.
  - **Select**: drag a box around notes to select them (**Shift** adds to the selection), then drag them to move or copy them together.
  - **Erase**: click or drag over notes to remove them.
- **Removing a note**: double-click it or right-click it, with any tool.
- **Keys** (after clicking the grid): **Delete** removes the selected notes, **Up/Down** transposes them (**Shift**: an octave), **Left/Right** moves them a step, **Ctrl+A** selects every note, **Esc** clears the selection.
- **Key** and **Scale**: the scale's rows are shaded lighter and the key's own note is tinted, so it's easy to stay in key. Scales: Major, Minor, Harmonic Minor, Dorian, Phrygian, Lydian, Mixolydian, Pentatonic Major and Minor, Blues, Melodic Minor, Locrian, Phrygian Dominant, Double Harmonic, Hungarian Minor, Whole Tone, Diminished, Hirajoshi, Major Blues, and **Custom** -- **Notes...** picks its notes; **None** shows the piano's white and black keys. **Snap to scale** makes new and moved notes land on the scale's notes (on BASS too). Your project remembers all of it. This key is the whole project's: BASS shades it, and a pad's **Key Snap** rounds its Tune onto it.
- **Fold** shows only the scale's notes in the synth part's roll -- every row a note that fits.
- **Move parts**: with it on, changing the key or scale moves the synth part and the bass line into the new key (Undo takes it back).
- **Velocity lane**: drag up or down over a note's bar to set how hard it plays; drag across several bars to draw a shape.
- **Selected note** (bottom row): the note, its length, velocity and chance (how likely it is to play each time round) of the selected note. Change one and every selected note gets it.
- Notes stop (with a short fade, not a hard click) once they reach their length, rather than always playing the pad's full sample. Louder notes look brighter, less likely ones fainter.
- One note per step: this edits the pad's own step pattern. For chords and longer parts, use the **Synth part** (above).
- The playhead highlights the current step in sync with the regular step sequencer — this is the *same* pattern data, just a pitch-and-length-aware view of it. Anything you place here also shows as "on" in the plain step sequencer, and vice versa.
- **Export MIDI...** — saves the selected pad's pattern as a standard `.mid` file, so you can drag it into your DAW's own piano roll/playlist.

- **Import MIDI...** (button, or just drag a `.mid` file onto the panel) — loads a `.mid` file's notes onto the selected pad's pattern, replacing whatever was there. Reads the file's own tempo resolution, so patterns exported from other software should import correctly too.
- **Chords...** generates a chord progression across several pads at once — different from everything else on this tab, which edits one pad's pattern at a time.

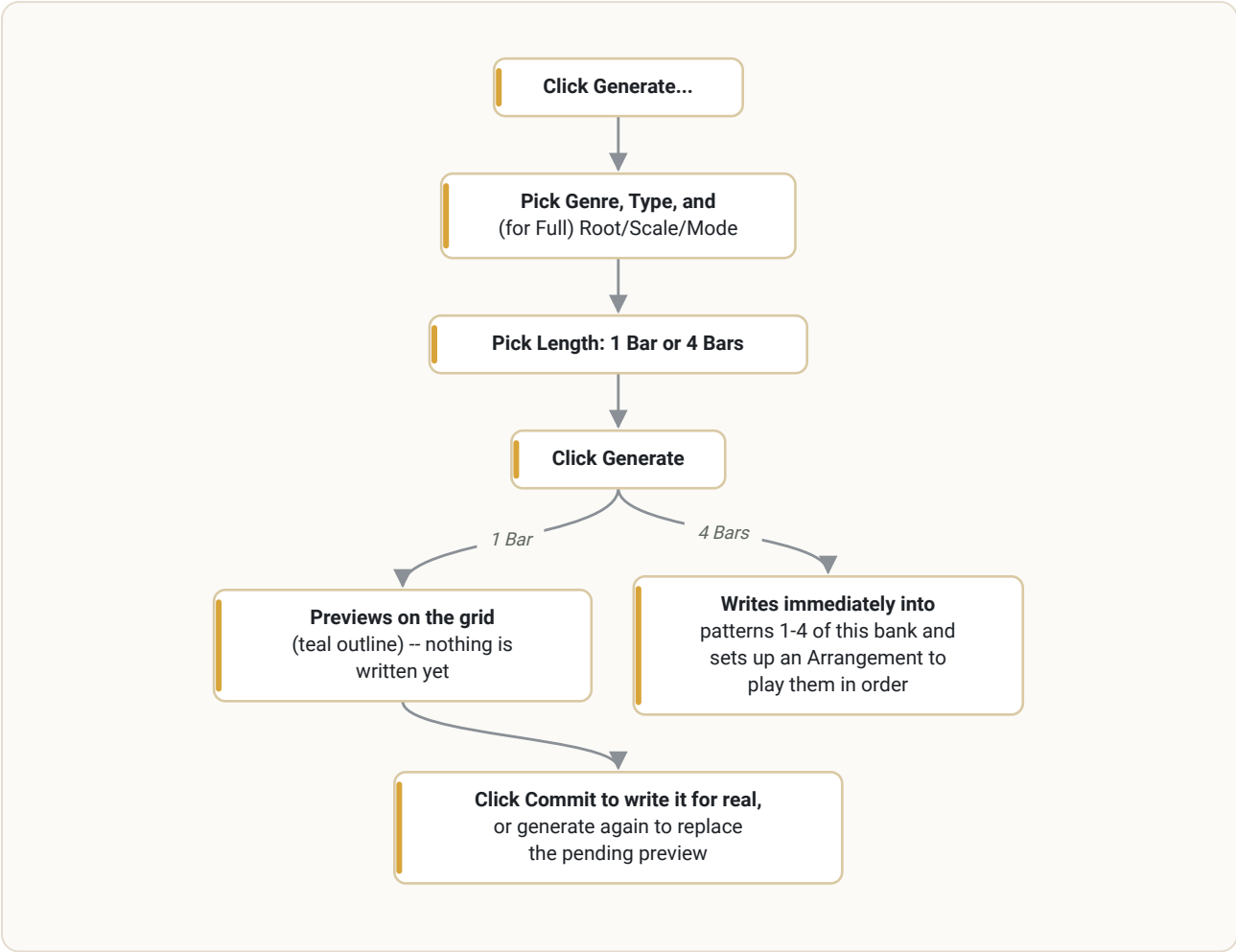


  - This needs pads with the **same sustained sample already loaded** — the button doesn't load anything for you, it only writes pitch/timing into pads you've already prepared.
  - More voice pads than a triad has notes (3) doesn't repeat a pitch — extra voices add the same notes an octave higher instead, so a 5-voice chord still sounds like real harmony, not doubling.
  - Generate **replaces** each checked pad's whole pattern (not a merge) — same one-undo-snapshot safety net as every other generator here.
  - Favorited pads are protected here too.
  - **Use Pad's Key** reads the lowest-numbered checked pad's detected key (shown on its SEQ/PADS tab sample info once loaded) and fills in Root/Scale for you — only works if that pad's sample has a detectable tonal key; the button is just a shortcut, still editable afterward.
  - **Export as .mid...** (inside the same popup) saves whichever checked pads' patterns as one MIDI file — works even before you've clicked Generate, so you can export a chord you built by hand too.
- **Melody...** generates a melody on the currently selected pad — unlike Chords, this only ever touches one pad. **Shape** sets how the line moves over the bar: **Wander**, **Rise**, **Fall**, **Arch**, **Dip**, or **Motif** (a one-beat idea moved up the bar). **Range** keeps it within one or two octaves, **Leaps** limits how far it jumps, **Notes** how many 16ths play; **End on the root** lands the last note home, and **Answer the first half** makes the bar a call and its response. **Write a counter-melody** puts a second line on this pad under the pad you pick in **Counter to**: in harmony on its beats, moving the other way, with passing notes where it rests.



- The melody's pitch drifts up and down within your chosen scale rather than jumping randomly — a more musical result than pure randomization.
- **Use Pad's Key** fills in Root/Scale from this pad's own detected key, same as the Chords popup's button above — disabled if this pad has no detected key.
- **Chord-aware** (checkbox) makes the melody outline a chord progression's notes on each beat instead of freely wandering the scale — pick the same progression you used in the Chords popup to get a melody that fits the chords you already wrote. Off by default.
- Generate **replaces** the pad's whole pattern, same one-undo- snapshot safety net as every other generator here. Favorited pads are protected.

- **Generate...** is the one-stop version of Chords/Melody above — pick a **Genre**, then a **Type**: **Full** (Root/Scale/Mode/Length, writing chords, melody, or both together) or **Rhythm** (Kick/Snare/Closed Hat/Open Hat role assignment, the same drum-generation idea as the SEQ tab's own Generate..., reached from here too). **Mode** (Full type only) picks Chords, Melody, or Both — Chords needs voice pads checked the same way the Chords popup does, Melody needs a target pad picked from a dropdown.



- **1 Bar** only previews — nothing is written to the live pattern until you click the **Commit** button (next to Generate...) that lights up once a preview is ready. Generating again before committing just replaces the pending preview, no separate discard step needed.
- **4 Bars** writes right away into **patterns 1-4** of the bank you're on (see SEQ's pattern slots -- your kit and the other banks stay as they are) and arranges them in sequence. There's no preview step for this size, since it's already a whole song section. If any of patterns 1-4 already has steps, you'll get a one-time confirm prompt first.
- Favorited pads are protected here too, same as every other generator.

Slices of one pad

Set the keyboard strip's **Mode** to **Slices: 16 equal** or **Slices: at transients** to play the selected pad in pieces: each key plays one slice, from note 60 (C3) up, and stops at the slice's end. The slices stay on the one pad -- nothing is cut onto other pads -- so its sound, effects and choke group all apply. Pick another pad and the keys follow it.

From your DAW, send the notes on **MIDI channel 6**: note 60 is the first slice. In FL Studio, give the plugin's piano roll channel 6. If you haven't switched the keyboard to a Slices mode since opening the project, those notes play 16 equal slices of the selected pad.

CHAPTER 11

SYNTH



The SYNTH tab

A synthesiser, played like an instrument. The tab is an **instrument slot**: the **Instrument** menu at the top holds the Synth now, and more instruments will join it there.

Playing it

- **The keyboard at the bottom** plays the synth while the SYNTH tab is open (its Mode switches to **Synth**, and back when you leave the tab). You can also pick **Synth** in that Mode menu on any tab.
- **MIDI channel 5** plays it from your DAW, a controller or a MIDI keyboard: notes, pitch bend, the mod wheel, the sustain pedal, aftertouch (pressure) and CC 74 (slide). Nothing else in the plugin listens to channel 5.
- **MPE controllers** (each note on its own channel, so every finger bends, presses and slides on its own): see **MPE** below.
- **Voices playing** (top) shows how many notes are sounding.

Presets

Pick one from the **Preset** menu, or step through with **<** and **>**. The menu is in folders by type, then character: **Bass** (Sub & 808, Synth Bass, Reese & Wobble, Acid, Plucked), **Keys** (Electric Piano, Lo-Fi Keys, Synth Keys, Organ, Clav), **Bells & Mallets**, **Pads** (Warm, Dark, Airy, Evolving, Vinyl & Lo-Fi), **Strings**, **Brass & Winds**, **Leads**, **Plucks & Stabs**, **Vocal & Choir**, **Arps & Sequences**, **FX & Textures** and **World** (sitar-, koto- and steel-drum-like) -- with **Init** at the top and your own presets under **My Presets**. **<** and **>** step through every sound in the menu's order, on into the next folder when one ends.

Browse... opens the sound browser: a **folder tree** on the left (All sounds, then Built-in, My Presets and Kits, each folder with how many sounds are in it), the list on the right, and search, categories, tags and favourites -- see the toolbar. Click a folder to see everything in it (that clears the search). Typing in the search box looks in every folder. **A** / **B** compares two versions of the sound: switch to B, change it, and flip back to A to hear the difference; **A > B** copies the side you're on to the other. Turning knobs, loading a preset or a wavetable are all **Undo** steps.

Save... keeps your sound as a preset file (in Documents / Boom Bap Producer Pads / Synth Presets). It asks where: **My Presets** itself, a folder you already have, one of the type folders (Bass > Sub & 808...), or **New folder...** (type `Leads/Mine` for a folder inside a folder); **Choose a place...** saves anywhere. It asks before replacing a preset of the same name. Folders you make in Explorer show up too; **Load...** opens one from anywhere. You can also drop a `.bbsynth` file on the tab. Your project saves the synth's whole sound either way.

Double-click any knob to put it back to where Init has it.

To Pads: resample the sound

To Pads... (beside Load...) plays the sound you've made into empty pads, so you can chop it, sequence it on SEQ, put it through the pad effects or export it in a kit like any sample:

Choose	What you get
One note	One pad: the note held 1.5 seconds, with its release.
Three velocities, as a velocity group	Three pads, soft, medium and hard, joined as a velocity group: hit the first pad softly and it plays the soft take, harder and it plays the harder ones (see Layers).
One per octave, over four octaves	Four pads, an octave apart, from an octave below the note to two above -- set up as **key zones**: select the first pad and play the keyboard, and every note plays the nearest take (see Key zones).
A chord stab of the key	One pad: a chord of your key -- pick its degree (I to vii) and a triad or a seventh -- played on this sound with its release, named after the chord.

- The note is your project's key (the KEYS / BASS **Key**), around middle C -- or C if no key is set. The BASS tab's **Edit sound...** has the same button, two octaves lower.
- The pads used are the first empty ones side by side in this bank. If there aren't enough, it says so: clear some or pick a bigger Grid Size.
- The sounds are saved as WAV files in **Music > Boom Bap Captures > Resampled**, by date, so your project keeps them. **Undo** takes the pads back.
- It's the sound itself: the synth's own effects are in it; its mixer strip (volume, pan, bus) isn't.

The sections

- **OSC 1, OSC 2, OSC 3**: each has a **Wave** (Sine, Triangle, Saw, Pulse, Noise, Wavetable), a **Level** (0 = off), **Octave**, **Semi** and **Fine** (cents). **Shape** is the pulse width for Pulse and the position in the table for Wavetable.
- **Wavetables**: with Wave set to Wavetable, the built-in table sweeps sine to triangle to saw to square as you turn Shape. Click **Table...** > **Load a sound or a wavetable...** (or drop an audio file on the oscillator) to make one from any sound: its first few seconds are cut into single cycles, and Shape moves through them. Try a vocal, a drum loop or a chord -- each gives its own colour. Wavetable files from other synths load as they are (their own frame size, up to 256 frames).
- **Draw a wavetable**: **Table...** > **Draw a wavetable...** opens an editor with two waves, **A** and **B**. **Draw** one with the mouse, or switch to **Harmonics** and build it from 32 bars (bar 1 is the note, bar 2 an octave up, and so on); **Sine**, **Saw** and **Square** give a starting point. **Frames** sets how many steps the table takes from A to B -- Shape sweeps through them. **Apply** saves it in Documents/Boom Bap Producer Pads/Wavetables and plays it on that oscillator.
- **STACK**: **Voices** stacks up to 16 copies of each oscillator, **Detune** spreads their pitch (cents) and **Width** spreads them across left and right -- for thick leads and wide pads. **Sub** adds a sine an octave under OSC 1. **FM 2>1** lets OSC 2 bend OSC 1's wave for bells and metallic tones. **Sync** makes OSC 2 start its wave again every time OSC 1 does -- tune OSC 2 up and sweep it for the classic tearing sync lead. **Ring** multiplies OSC 1 by OSC 2: metallic, bell-like tones (turn the two oscillators' own levels down to hear only the ring). **Spread** places the stacked voices: **Even**, or **Supersaw** -- close round the note and wide at the edges, the classic big saw (try 7 voices, Detune about 30). **Noise** colours the Noise wave: **White** (bright hiss), **Pink** (fuller, like rain) or **Brown** (a deep rumble).
- **FILTER**: **Type**, **Cutoff**, **Reso**, **Drive** (grit before the filter), **Key** (higher notes open the filter more), **Env** (how far the filter envelope moves the cutoff, up or down) and **FM** (OSC 1 sweeps the cutoff at audio rate -- growls, buzz, metallic edges; put a sine on OSC 1 with its level at 0 and play a saw on OSC 2). The types:

Type	What it does
Low 12, Low 24	Low-pass, gentle or steep.
High 12, High 24	High-pass.
Band, Notch	Keeps or cuts a band around the cutoff.
Ladder 24	A steep low-pass with a round, driven sound; its resonance thins the bass a little, the classic way.
Diode 18	A gentler, squelchier low-pass that clips one way sooner -- the acid-line sound.
Comb +	Rings at the pitch the **Cutoff** sets (and its harmonics); **Reso** makes it ring longer. Metallic, plucked, flanged.
Comb -	The same, ringing between those harmonics -- hollow, an octave down.
Vowel	Three formants that say A, E, I, O, U as **Cutoff** moves left to right; **Reso** sharpens them. Put the filter envelope or an LFO on it and the synth talks.

- **FILTER ENV, AMP ENV, MOD ENV**: attack, decay, sustain, release. The amp envelope shapes the volume, the filter envelope the cutoff, and the mod envelope whatever the mod matrix sends it to. Each has a **Curve**: right

of centre a punchy attack and a snappy fall, left of centre a slow swell and long tails; centre is a straight attack, as before.

- **LFO 1, LFO 2, LFO 3** (the bottom row): a **Shape** (sine, triangle, saw, square, random) at a **Rate**; **Restart** starts it from the top on every note. **Delay** keeps it still for that long after each note and **Fade** brings it in gradually -- LFO 1 to Pitch with a Delay of 0.5 s is vibrato that arrives on a held note, like a singer's.
- **MOD MATRIX**: four rows of *source*, *where it goes* and *how much* (drag left for the other way); **MOD MATRIX 5-8** (bottom row) has four more. Sources: the LFOs, the mod, amp and filter envelopes, velocity, the mod wheel, the bend wheel, a random value per note, the key, **Pressure** (aftertouch) and **Slide** (CC 74) -- per note with MPE -- and the eight macros. Destinations: pitch, each oscillator's level and shape, cutoff, filter 2's cutoff, resonance, drive, FM, filter FM, detune, sub, ring, each LFO's rate, volume, pan. Example: LFO 1 to Pitch at a small amount is vibrato; Mod Env to Pitch with a short decay is an 808's punch.
- **Drag to modulate**: drag the title of **LFO 1-3**, **MOD ENV**, **AMP ENV** or **FILTER ENV**, or a macro's name, onto a knob (cutoff, an oscillator's level or shape, drive, detune, an LFO's rate...). It fills the next empty matrix row at a small amount; a ring round the knob shows how far it moves (teal up, red down). Change the amount in the matrix.
- **WARP** (bottom row): each oscillator's wavetable read a different way -- **Bend** leans the wave, **Sync** runs through it several times a cycle (a sync sound from any table), **Mirror** plays it forward then back, **Squeeze** is a pulse width for any wave, **Quantize** steps it (lo-fi). **Amount** at 0 is off. Wavetable oscillators only.
- **MACROS**: eight knobs that do nothing on their own -- choose **Macro 1-8** as a source in the matrix, send it to several places (cutoff up, drive up, detune up), and one knob turns a sound from soft to wild. Saved with the sound.
- **Steps...** (on the MOD MATRIX title): two step sources you draw, **Steps** and **Steps 2** -- 16 bars, how many steps go round, each step's length (1/32 to a bar, at the song's tempo) and **Smooth**, the same editor as a pad's Matrix. They start with every note; turn on **Song** and they follow the song's beats instead. Pick **Steps** or **Steps 2** as a source: Volume for a gate, Cutoff for a rhythmic filter, Pitch for an arpeggio-like line on one held note.



The SYNTH tab's Steps editor

- **SHAPE**: a movement you draw (**Draw...**), its **Length** (1/16 to 4 bars) and **Loop**. It starts with every note. Pick **Shape** as a source in the MOD MATRIX to send it anywhere -- a stepped pitch line, a rhythmic filter, an envelope with as many stages as you like.
- **PLAY**: **Mode** Poly (chords), Mono (one note, restarts the envelopes) or Legato (one note, glides without restarting); **Glide** time; **Bend** range in semitones; **Velocity** (how much playing harder makes it louder); **Voices** (the most notes at once); **Priority** -- in Mono and Legato, which held key sounds: the **last** pressed, the **lowest** (a bass that stays down) or the **highest** (a lead on top).
- **ARP** (the arpeggiator): switch it **On** and held notes play one at a time, in time with the song -- **Mode** Up, Down, Up-Down, Random or As played; **Rate** 1/4 to 1/32 or triplets; **Octaves** 1-4 (the pattern climbs through them); **Gate** (short and plucky to smooth). When the song is stopped it keeps time at the tempo.
- **CHORD**: every note you play becomes a chord -- Major, Minor, 7th, Minor 7th, Major 7th, Sus2, Sus4, Power, Octave. With the ARP on too, the chord's notes are what it arpeggiates.

- **INSERT 1-5**: effects in order, the same ones the pads have (chorus, flanger, phaser, reverb, delay, saturation and more) with their own knobs and a **Mix**.
- **OUTPUT**: **Volume** and **Pan** (the same as the synth's mixer strip).

The bottom row: character

- **DRIFT**: an old analogue polysynth never plays two notes quite alike -- each voice is a little off in pitch, filter and envelope, and wanders. **Drift** is how much (0 is steady, as before); **Pitch**, **Filter**, **Env** and **Glide** share it out (Pitch moves up to 20 cents, Filter up to about half an octave). Each voice keeps its own character, so a chord sounds wider and alive rather than random. **Voice Pan** spreads the voices across left and right, each voice in its own place.
- **FILTER 2**: a second filter, any of the types above (**Off** is one filter, as before), with its own **Cutoff**, **Reso** and **Env**. Key and Drive are the first filter's. **Routing**: **Serial** runs filter 2 after filter 1 (a low-pass and a high-pass in series make a band); **Parallel** runs them side by side, and **1 / 2** mixes them (left: filter 1 only, right: filter 2 only, the middle both). Cutoff in the mod matrix moves both.
- **RESPONSE**: what the filters do to the sound, drawn from 20 Hz to 20 kHz (the lines are 100 Hz, 1 kHz and 10 kHz; the flat line is 0 dB). It follows the knobs: the envelopes and the mod matrix move it further as you play.
- **LAYERS**: two sounds in one synth. **One** is a single sound. **Split** plays layer **B** below the **Split** key and layer A from it up -- a bass under your left hand, keys under the right. **Stack** plays both on every note -- a pad under a pluck. **A** and **B** choose which layer the knobs edit (the section titles say **- B** while you edit B); each layer has its own oscillators, filters, envelopes, LFOs, mod matrix, play mode, inserts and output. **B Sound...** gives layer B a built-in sound or a copy of layer A. The ARP and CHORD play both layers. Presets: a built-in preset is one sound and goes to layer A, leaving B and the layer settings as they are; a saved preset (**Save...**) and your project keep both layers.
- **MPE**: for controllers that put each note on its own channel (an MPE keyboard or pad surface, or a DAW's MPE clip). **Off**: the synth plays channel 5 only, as always. **Ch 5-16**: it takes channels 5 to 16 (set your controller to an MPE *upper zone*, channel 16 the master) and channels 1-4 still play the pads and bass. **All ch**: it takes every channel (a *lower zone*, channel 1 the master) and nothing else in the plugin hears MIDI. Each note's own bend moves it by **Note Bend** semitones (48 is the MPE standard); the master channel's bend moves all notes by the PLAY **Bend** range. Pressure and slide come into the mod matrix as **Pressure** and **Slide** -- for example Pressure to Volume or Cutoff, Slide to Cutoff.
- **FM**: any oscillator bending another's wave -- **FM 1>2**, **1>3**, **2>3**, **3>1** and **3>2** (STACK has **FM 2>1**). The first named is the source: turn its level down to hear only what it does to the other. A sine into a sine gives bells and electric pianos; more amount, more bite.

The BASS tab's **Edit sound...** has DRIFT, FILTER 2 and RESPONSE too, but no layers or MPE: the bass is one sound, played as a line.

The sixth row, and the header's tools

- **ARP STEPS**: the arp's 16 steps. Click a step to make it a rest (it plays nothing), click again to bring it back; right-click for an accent (louder). **Steps** sets how many it uses before going round. The step playing is outlined. In the ARP section, **Swing** makes every second step late, and **Latch** keeps the arp playing after you let go (the next chord replaces it). **Mode** adds **Chord** (all the notes at once) and **Converge** (outside notes first).

- **CHORD** has more types (9ths, 11th, 13th, 6/9, diminished, augmented) and **Custom...**: tick the notes above the one you play. **STRUM** plays them one after another, up, down or alternating.
- **LO-FI**: an old sampler's converter over the whole synth, the same modes as the pads' crush; **Mix** blends it.
- **VOCODER**: the synth shaped by the plugin's sidechain input. Route a vocal or a beat to the plugin's sidechain in your DAW (in FL Studio, from the mixer), hold a chord, turn up **Mix**. Live only -- exports have no input to shape it with.
- **ENGINE**: **Quality** -- Eco is lighter on the CPU (two unison voices, eight notes), High and Ultra run the voices at 2x and 4x for less harshness on bright, driven sounds. **Steal** picks which note gives way when every voice is busy: the oldest, the quietest, or one that isn't the lowest or highest. **A4** tunes the whole sound: 440 Hz is standard; 432 Hz, or 415 Hz for baroque pitch. The button on the ENGINE title (**12-TET**) picks the tuning: **Load a Scala scale (.scl)...** plays the synth in any of thousands of tunings -- just intonation, maqam, gamelan, historical temperaments (free .scl files are easy to find online). Middle C stays put; the other keys follow the scale. It's saved with the sound, and KEYS clips play in it too.
- **GLIDE**: how **Glide** (PLAY) moves between notes. **Mode**: Smooth (eases in, as before), **Time** (a straight slide that arrives in exactly the Glide time, however far) or **Rate** (an octave per Glide time, so bigger jumps take longer). **Always**: in Mono and Legato, every note glides from the last one, not only when a key is still held.
- **SCOPE**: the sound as a wave, and the amp envelope with a dot where the newest note is.
- **Random** (header): a new sound, or just its oscillators, filter, envelopes, modulation or effects, kept to useful ranges. Undo goes back.
- **Morph...**: pick a built-in sound and slide from yours toward it.
- **Auto-sample...**: play this sound across the keys into a folder -- an SFZ and a DecentSampler instrument with the WAVs -- for any sampler.
- There are now 553 built-in sounds, named by family ("Pad - Warm Saw", "Bass - Punchy Square"), so the sound browser files them by category.

In the mixer

The MIXER tab has a **Synth** strip after the Bass: volume, pan, **Duck** (dips it when a duck-source pad plays), mute, solo and its meter. **Routing...** there can send it to Bus A or Bus B. See the MIXER tab.

CHAPTER 12

BASS



The BASS tab

One bass line, one note at a time -- and notes can slide into each other. Pick a sound at the top, then draw the line in the grid. The bass has its own strip in the MIXER and doesn't use a pad.

1. Pick a sound

Sound: Synth or Sample.

- **Synth** (the default) is a bass synth, ready to play. Pick a preset -- **808**, **Distorted 808**, **Sub Bass**, **Reese**, **Acid**, **Fat Bass**, **Pluck Bass**, **Wobble Bass** -- or one of the 70-odd more in the menu's folders (Sub & 808, Synth Bass, Reese & Wobble, Acid, Plucked, Electric & Upright), or step through them all with **<** and **>**. Then shape it with six knobs:
  - **Tone**: darker to brighter.
  - **Punch**: the 808 knock -- each note starts higher and drops onto its pitch.
  - **Drive**: grit and weight, so it cuts through small speakers.
  - **Length**: how long each note rings.
  - **Glide**: how long a slide takes to reach the next note.

- **Sub**: a pure low sine an octave under, for weight.

Edit sound... opens every control of the synth (the same ones as the SYNTH tab: oscillators, filter, envelopes, LFOs, mod matrix, effects) and every preset, and you can save your own. **< Back to the bass line** returns.



Edit sound: the bass synth's every control

- **Sample** plays your own bass sound (an 808 you like, a sub, a bass note): **Load...** it or drop it on the tab. Its knobs are the sample's filter (**Cutoff**, **Res**), envelope (**Attack**, **Decay**, **Sustain**, **Release**), **Tune**, **Fine**, **Glide** time and **Loop**. Right-click a knob to MIDI-learn it. Beside them:
  - **Slide: smooth / even / late** -- how a slide moves to its note over the Glide time. Smooth (as it always was) moves fast and settles; even moves at one speed and lands exactly on time; late holds back and drops in at the end, the classic 808 fall.
  - **Octave: off / up / down** and its **Octave** level -- the same sample played again an octave up (bite on phone and laptop speakers) or down (more weight) under every note.
  - **Start**: where in the sample each note begins -- skip a slow attack, a click or the silence at the front.
  - **Drive**: saturation after the filter, for a thicker bass that holds up on small speakers.
  - **Punch**: a louder first 25 ms or so of each note, so its attack cuts through the kick.
  - **Sub**: a clean sine at each note's real pitch under the sample, for weight that stays in tune. It needs a sample with a clear note (its **Root** shows top right).
  - **Tune...** (next to Load): 808 auto-tune. It tells you the note the sample sounds as and tunes it (Tune and Fine) so that note sounds as **C**, as the **key's root** (when KEYS has a key), or as **the kick's note** -- the kick is your kit's kick pad (by name or the Kick colour), measured as it plays. Tuned to C, a note on the C row plays a C. One Undo step.
  - **Zones...** (next to Tune...): a multi-sample bass. **Add samples...** adds up to 7 more recordings of the same bass at other notes (pick several at once). Each note of the line plays the sample recorded nearest to it, so

a low E isn't a stretched-down A. Each zone's note comes from its file name (**Bass_E1**) or is heard; change it in the list if it's wrong, or remove a zone with **x**. Zones need the main sample's note to be heard (its **Root**); until then the main sample plays everything. A slide stays on the sample it started on.

Volume and **Pan** (right) work for both, and are the same as the BASS strip in the MIXER.

2. Draw the line

The grid is a small piano roll: notes up the side, steps across (16 per bar).

- **Click** to add a note. It holds until the next note starts, which is what most 808 lines want (the rest of a held note shows lighter). Drag right as you click to give it a set length instead.
- **Drag a note** to move it (up and down changes its note), or drag its right edge to change its length.
- **Right-click** or **double-click** a note to remove it.
- **Selecting several: Shift-click** notes to add or take them away, or **Shift-drag** a box over them. Drag any selected note and they all move together. The selected notes are outlined in gold.
- **Keys** (click the grid first):
  - **Delete** or **Backspace** erases the selected notes.
  - **Ctrl+C**, **Ctrl+X**, **Ctrl+V**: copy, cut, and paste at the last place you clicked (the faint line).
  - **Ctrl+D** duplicates the selection straight after itself.
  - **Up / Down** move the selection a semitone (**Shift**: an octave); **Left / Right** a step (**Shift**: a beat).
  - **Ctrl+A** selects every note; **Esc** none.
  - Everything but dragging is one Undo step, and so is **Clear**.
- Moving, copying and pasting a note keeps its Roll, Bend, Timing and when it plays.
- One note at a time: a new note inside a held one ends it there.
- **Top right of the grid**: **Now** -- the note sounding and its frequency, as you hear it (the sound's own octave and Tune included, so it can differ from the grid's row name), its key lit orange on the keyboard; and the line's **Range**, lowest to highest note. Rows too low for most speakers (under 30 Hz) are tinted red, and the Range says so when the line goes there. A sample bass needs a clear note (its Root) for these.
- **Scroll** over the grid to see higher or lower notes. The keyboard on the left plays a note when you click it.
- If you've set a **Key** and **Scale** on the KEYS tab, the grid shades the notes in that key (and says so, top right) -- stay on the lighter rows and the bass fits the rest of the beat.

Under the grid, one cell per note:

- **Slide**: glide into this note from the one before instead of starting it fresh (the note before has to run up to it). The Glide knob sets how long the slide takes.
- **Accent**: play this note harder.
- **Right-click** a note's Slide or Accent cell to pick when it plays: **1:2** (the 1st of every 2 times round the line), **2:4**, **Fill** (only while Fill is held), **First**, **Prev** (when the line's last such note played) and their opposites (**!**). The choice shows on the Accent cell.
- **Roll**: play the note again, evenly across its length -- **x2**, **x3**, **x4**, **x6** or **x8** -- for 808 rolls. A note that holds to the next one rolls across one step. Click the cell to step through the rolls, right-click it to pick one. The first hit can still slide in; the rest start fresh, at the same pitch, and the next note ends a roll.

- **Effect**: **Rev** plays a sample-bass note backwards (from the end of the sample, or from Start counted from the end); **Stop** winds the note down like a tape machine stopping -- the speed (and pitch) falls to nothing over its length, fading at the very end. Tape stop works on the synth bass too; Reverse is for samples. Click the cell to step through, right-click to pick.
- **Velocity**: drag the bars up or down.
- **Bend**: click **Velocity** (the lane's name, on the left) to switch the lane to **Bend**, and back. Drag a note's bar up or down from the middle line to bend it up to 12 semitones up or down: the pitch moves there evenly over the note's length (or until the next note, when it holds) -- an 808 drop or rise. Double-click a bar to straighten the note. A roll's hits follow the bend. MIDI out sends the notes without the bend.
- **Slide ms**: the next view of that lane -- a slide's own time, 10 to 1000 ms, for this note only (drag up; double-click puts it back to the Glide knob). It matters on notes with Slide on (bright bars), and on the synth when Legato ties notes.
- **Timing**: click the lane's name again (Velocity, Bend, Timing, Slide ms, round and back) and drag a note's bar up to play it late or down to play it early, by up to half a step, in steps of 5%. Double-click puts it back on the step. Lay a few notes late for a lazy, behind-the-beat bass line.
- **Recording:** with **Record** armed and the beat playing, notes you play on MIDI channel 4 go into the line on their nearest step, as long as you held them.
- **Step in:** turn it on and each note you play on channel 4 goes in at the cursor (one step long); let go and the cursor moves on a step.
- **Humanize** moves each note's velocity a little at random; **Swing: kit** gives the line its own swing (kit = the PADS Swing).
- **Chords** (on by default) shades, under each step, the notes of the KEYS synth part's chord playing there -- land the bass on them to follow the harmony. **Follow chords** does it for you: every note moves to the nearest of those chord notes (notes already on one stay; Undo takes it back).
- **MIDI ch** picks the channel the bass plays and records on from a MIDI keyboard: 4 (as it always was) or 6-16.
- **Feel...** sets how the line plays, and its tone:
  - **Legato**: a note that starts while the last one is still sounding carries on from it at its own pitch, without starting again -- no new attack, one long phrase. With the synth bass the notes are tied, and the sound's own Mode and Glide decide how they join.
  - **Length lock**: with a sample bass, every note keeps the sample's own length. Normally a high note plays the sample faster (and runs out sooner) and a low one slower; locked, a long 808 rings just as long on every note. Best with samples that have a clear note (their **Root**).
  - **Accent**: how much harder an Accent plays a note (0-80 velocity; 40 is how it always was).
  - **Vel > Tone**: with a sample bass, softer notes play darker -- the filter closes by up to four octaves on the softest. 0 is off.
  - **Layer**: plays the other sound under the one you picked -- the synth under your sample bass, or the sample you loaded under the synth. Both play every note of the line, each with its own knobs. A favourite: a sampled 808's knock over the clean **Sub Bass** synth. 0 is off. The two only line up if the sample is in tune -- use **Tune...** first.
  - **Harmonics**: adds the bass's own harmonics, so it's still heard on phones and laptops where the lowest notes don't come through (your ear fills in the note). Works for the synth and a sample. 0 is off.

- **Drive type**: how the sample bass's **Drive** knob saturates -- **Tape** (smooth, as before), **Tube** (warmer, lopsided) or **Fuzz** (hard and buzzy). The synth bass has its own Drive.

3. Tools

- **Bars 1, 2, 3, 4, 8 or 16**: how long the line is. Going longer starts the new bars as copies of the ones you have.
- **Octave - / Octave +**: every note down or up an octave.
- **Clear**: remove every note.
- **Ref tone** (beside Edit sound...): a steady tone at the song's key (the KEYS tab's), low where an 808 sits -- the button names the note. Play your bass sample against it and tune it until the two stop beating. Switch it off when you're done; it isn't saved or exported.
- **Generate...**: write a line from a chord progression's root notes, across every bar. It opens on the KEYS tab's key; use the same Root, Scale and Progression as your chord pads to keep them together. Then:
  - **Genre**: a starting point for a style -- Boom Bap, Lo-Fi, Trap, Drill, G-Funk, House, Afrobeats, Dub, Acid, Halftime. It fills in everything below (you can still change any of it). With **Its sound and feel too** ticked, **Generate** also sets its synth sound (switching the bass to Synth), its **Feel** settings, and slides into the notes for the sliding styles (trap, drill, acid). Untick it to keep your own sound. Undo takes back the line, not the sound.
  - **Style**: **Chord roots** (a note on each chord) or **Follow the kick** (a note on every kick in the pattern, on the chord's root there). Kick pads are found by name ("Kick", "BD") or by the Kick colour Name and Colour by Sound gives them; with none, nothing changes and it says so.
  - **Notes**: **Hold** (each lasts until the next) or **Short** (one step each, bouncier).
  - **Octave**: **Low** (as before) or **Mid**, an octave up.
  - **Ghost notes**: quiet one-step pickups in the longer gaps, two steps before the next note.
  - **Approach notes**: a note a semitone below (or above) on the step before each chord change, leading into it.

MIDI and audio

MIDI / audio... (right of the sound row): export the line as MIDI (the line's 0 is C2, note 36) or as audio -- its bars through the bass strip, ready to loop -- or **Import MIDI** onto it (it replaces the line; Undo puts it back; where notes overlap, the lowest wins). The two handles beside the button drag the MIDI or the audio straight into your DAW.

Playing it live

The keyboard at the bottom plays the bass while the BASS tab is open (its Mode switches to **Bass**, and back when you leave). From a DAW or controller, **MIDI channel 4** plays it (or set **MIDI in** to **Bass**). A sustain pedal holds its notes.

Your older projects

A project saved before this version keeps its bass line (moved into the grid as it was) and, if it had a bass sample, plays that sample. Projects without one get the Synth's 808.

CHAPTER 13

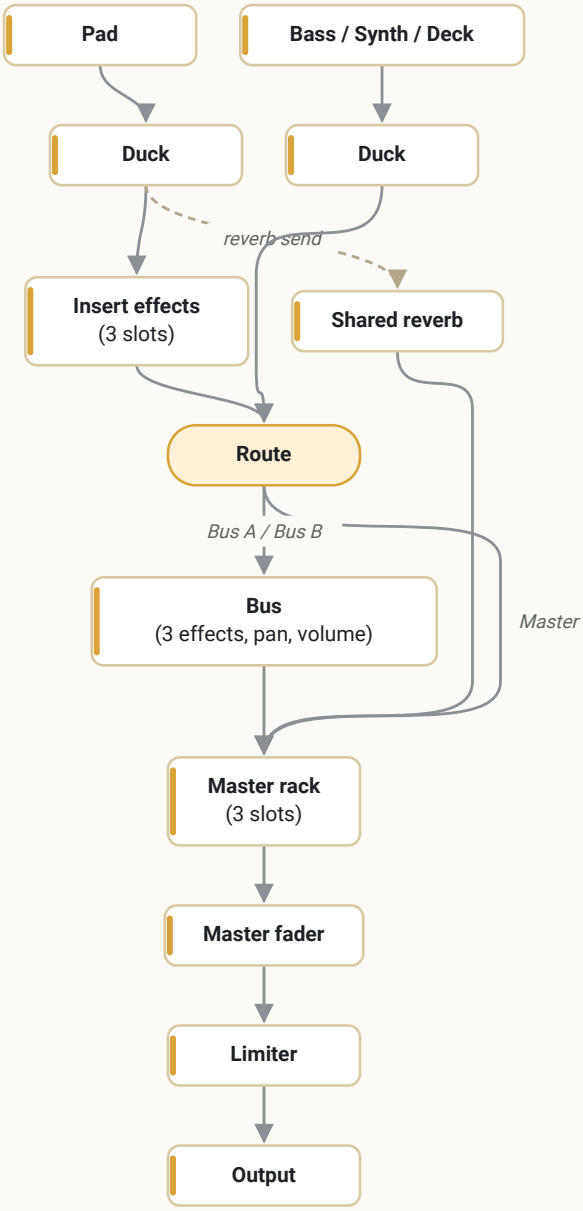
MIXER



The MIXER tab

A console-style row of vertical channel strips — one per sound source in the plugin (every pad in the active bank's grid size, the bass, the synth and both turntable decks), plus the **Master** strip pinned first on the left and **Bus A** and **Bus B** beside it. Everything here is the same underlying parameters the PADS/BASS/TURNTABLE tabs already control, just laid out side by side for a whole-kit view instead of one source at a time.

- **Solo-active banner** (above the strips) — shown whenever anything in the kit is soloed, so silence elsewhere in the mix reads as "something's soloed," not "broken."
- **Signal Flow** — opens a plain explainer of the signal path below.



The **Duck** dips a source each time a pad marked **D** plays (the sidechain pump); the duck-source pad itself never ducks.

Every strip

The MIXER looks like a hardware console: each strip is a module, with wood at the ends. From the top of a strip:

- A **colour bar**, the strip's **name**, and under it what it is -- the pad's sample, the synth's sound, the deck's record, Bus, Reverb return.
- **Out** -- where it goes: the master or a bus (a bus goes to the master or a later bus). The same routes as **Routing...**.
- A **VU meter** -- the needle shows the strip's level the way an analogue console's does; keep the loud parts around 0.

- Its **effects**, by name, each with a dot: lit is on, hollow is bypassed -- click the dot to switch it. Click a name for its menu (the type, **Knobs...**, Bypass, chains). **+ FX** adds an effect in the next free slot.
- **HPF**, **EQ** and **COMP** -- the strip's own channel strip, lit when on; the thin red bar under them shows how much the compressor is taking off. Their knobs are in the dock (select the strip).
- **Pan** -- a knob (double-click: the centre).
- Two **readouts**: the fader's level in dB (double-click it to type one) and the loudest peak just now (red if it hit 0 dB; click it to reset).
- The **Volume** fader -- from silence to **+6 dB**, with 0 dB three quarters of the way up (the white mark) -- and an **LED meter** for the left and right channels (green, amber near the top, red close to 0 dB), with its dB scale beside it.
- **M** / **S** — round Mute and Solo buttons that light up, same cross-source behavior as elsewhere in the plugin (soloing one source silences every other un-soloed source).

Pad strips only

- **5 insert-effect slots** at the top — the same per-pad Insert FX chain from the pad grid's right-click menu, shown here as clickable slots instead. Picking a type here and via right-click on the pad are the same setting either way.
- The strip's top accent bar reflects the pad's own color tag (set via right-click → Color on the PADS tab) instead of a fixed color, so a glance across the console tells you which strips belong to which color-grouped pads.
- A reverb send level and a **D** (duck source) control with its own amount — sends this pad into a shared reverb bus, or marks it as a sidechain-ducking source for other pads to duck against.

The dock and the effects browser



A bus selected: its effects in the dock, the effects browser on the right

- **Click a strip** (on its face, away from the controls) to select it -- it's outlined. The master is selected at first.
- The dock starts with the strip's **channel strip**: **PRE** -- a high-pass filter (HPF, 20-400 Hz) to clear rumble; **EQ** -- Low (a shelf at 100 Hz), Mid (a bell you sweep with Freq, 200 Hz-6 kHz) and High (a shelf at 10 kHz), +-12 dB each; **COMP** -- **Peak red** (turn it up for more squeeze), **Gain** (make the level back up) and **FAST** for drums, with the gain reduction beside them. **IN** switches each section on. They come after the strip's effects.
- The **dock** under the strips shows the selected strip's effects side by side: each with its on / off switch, **<** and **>** to move it earlier or later in the chain, **x** to remove it, its knobs and **Mix** (an EQ has **Edit EQ...**). **+ FX** adds one. With the master selected the dock shows its **loudness**: the integrated LUFS, how it sits against the target, short-term, momentary and true peak; **More...** opens the full readings, and **Master FX...** the master rack.
- The **effects browser** on the right lists every effect with what it does. **Drag** one onto a strip to add it there, or **double-click** it to add it to the selected strip. Type in the search box to find one.
- **Effects** and **Dock** (at the top) show or hide them.

The Bass strip's effects

The **Bass** strip has **five effect slots** too, for whichever bass is playing, sample or synth. Click one to pick an effect (everything but EQ: the bass has its own filter on the BASS tab), then **Knobs...** and **Mix** as on a bus. They come after the strip's fader, so the fader also sets how hard they're pushed. For a dusty bass, try **Lo-fi Box**, then a **Saturation**.

Moving an effect

Effects run top slot first. In a strip's slot menu, **Move earlier in the chain** / **Move later in the chain** swaps that effect with its neighbour, settings and all.

Bypass

Click a strip's effect slot and pick **Bypass** to switch that effect off without losing its settings: the slot greys out, and exports leave it out too. Pick it again to switch it back on. The master rack has an **Off** button beside each slot.

Insert chains

Click any strip's effect slot (a pad, a bus or the bass) for **Insert chains**: **Save this strip's chain...** keeps its three effects, their knobs and Mix under a name, and every saved chain is listed to load onto that strip. They're the same chains as a pad's own menu, so a chain made on a pad loads on a bus or the master rack, and the other way round. What a strip can't hold is left out: an EQ loads onto the bass as an empty slot, and the master rack plays every slot fully wet.

A strip's channel panel

Click a strip's **name** for its channel panel. On pads and buses:

- **Trim** (+-18 dB) turns the sound up or down *before* its effects. A quiet sample can come up here, where the fader only turns down.
- **Drive** pushes it into a console's soft saturation *after* its effects: the peaks round off and the body comes up. A little warms it; a lot is grit.

On the **Bass** strip it holds **Offset**: the bass up to 10 ms later. When a kick and a bass that hit together sound thin, slide this until they sound fullest.

Gain staging

Gain staging... (top of the Mixer, beside Routing) sets every loaded pad's volume so its loudest hit lands on the same level: **-6 dB** for a loud kit, **-12 dB** (a good start) or **-18 dB** for lots of headroom. A pad that's already quieter than that stays at full. It's a level start, not a mix: balance by ear from there. **Undo** puts every fader back.

Master strip

Leftmost, always visible, outlined in the accent colour. Volume fader, the level and peak readouts, and a left / right meter — no pan/mute/solo, since there's nothing to pan or solo on an already-summed master bus. Its **FX** button opens the same 3-slot master rack popup as the toolbar's own Master FX button — one rack, two entry points. At its top, **Preset: Tape & glue** puts a little console warmth in slot 1 and a gentle compressor in slot 2 (slot 3 stays as it is). Beside it, **Chains** saves the rack's three slots under a name or loads a saved chain (see below). Under its three slots is **Dust**: one slider of old record for the whole mix (the top rolls off, it saturates a little, crackle and hiss come in). Double-click it for off. Under it, **Speed** is varispeed: everything faster and higher, or slower and lower, together, like running a tape fast or slow. In the Standalone the tempo follows; in a DAW it pitches the plugin's sounds (the decks and tape loops keep their own speed). Double-click for normal.

Under FX:

- **Mono** plays the whole mix in mono while it's lit -- check how it holds up on a phone or one club speaker, where anything that cancels out in mono disappears. It's only for listening: it isn't saved and never reaches an export.
- **Low** makes everything under **80**, **120** or **160 Hz** mono (pick from its menu; off to start), so kicks and bass sit dead centre however wide the rest is. It's saved with the project and applies to exports too, after the master fader and before the limiter.

LUFS (beside "Master") opens the loudness readings -- how loud the mix is the way streaming services and broadcasters measure it:

- **Momentary**: the last 0.4 seconds.
- **Short-term**: the last 3 seconds.
- **Integrated**: the whole song since you pressed **Reset** (quiet gaps and silence don't drag it down). Press Reset, play the song through once, and read it at the end. Streaming services play most music at around -14 LUFS, turning louder songs down.
- **Target**: pick what to measure against (streaming -14, Apple Music -16, broadcast -23) and the line under Integrated says how many LU over or under it you are. Over isn't wrong, it just means the service will turn your song down to match.
- **The vectorscope** (the square on the right) draws the stereo picture. A thin vertical line is mono; a cloud is wide; something leaning toward L or R is louder on that side; a shape lying flat means parts are out of phase and will thin out in mono.
- **The spectrum** (along the bottom) shows the whole mix from 20 Hz on the left to 20 kHz on the right: a big hump in the lows, a hole in the mids, too much fizz up top.
- **Reference track** (the bottom row): **Load...** a finished song you want yours to sit next to, then **Hear reference** to switch to it -- click again for your mix. With **Match level** on, it's turned to your mix's loudness when you switch, so you're comparing the sound, not the volume. While it plays, every meter here reads the reference. It's never in an export, and closing the readings switches back to your mix.
- **True peak (dBTP)**: the highest the waveform has gone since Reset, counting between the samples, where a plain peak meter can't see. It turns **red above -1**: a song that peaks there can clip once it's made into an MP3 or AAC. Lower the master (or the Limiter's ceiling) until it stays under.

The thinner, lighter bar inside the master's meter is the **average level** (RMS), which moves slower and is closer to how loud the mix feels. The gap between it and the peaks is your punch: a big gap is dynamic, a small one is squashed.

The master's meter holds its **peak** for a moment (a line that falls after a second and a half). The small light on top turns **red** if the output ever reaches 0 dB -- it stays lit until you click the meter, so you'll see it even if you were looking away. With the toolbar's Limiter on it shouldn't light at all. Every other strip's meter has the same peak hold and clip light; a red light there means that one sound is clipping before it even reaches the master.

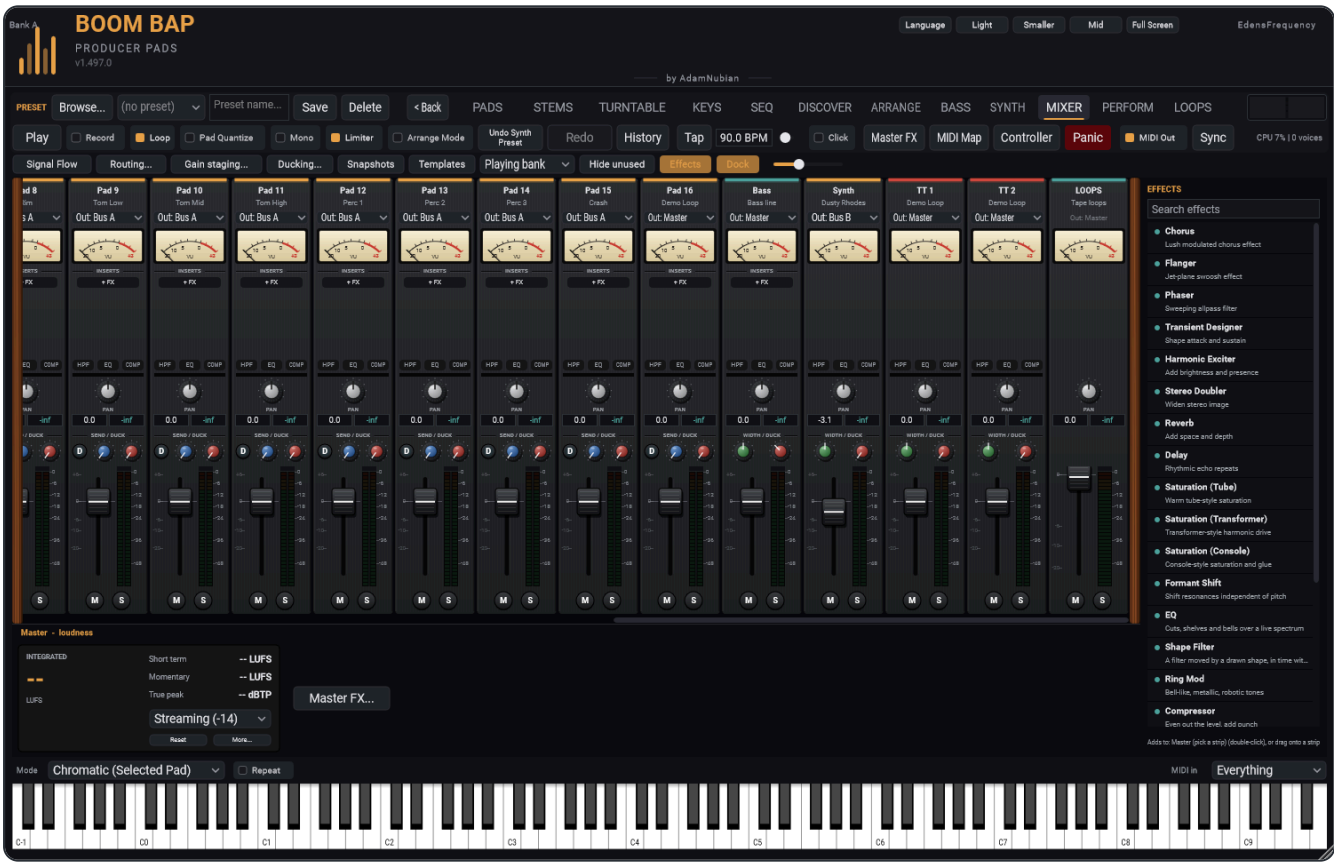
Under the fader, **Corr** is the correlation meter: how alike the left and right sides are. **+1** is mono; most mixes sit between **0** and **+1**. When the marker goes **red** -- below 0 -- something in the mix is out of phase and will get quieter or vanish in mono: check it with **Mono**, then try a pad's **Phase** button or less width.



The master rack

Bass, Synth and Turntable strips

At the right end of the console (scroll across to them):



The last strips: bass, synth and the decks

Same Pan/Volume/Mute/Solo/meter as any strip, plus a **Duck** knob: how far it dips each time a pad marked **D** (duck source, on the pad strips) plays. Turn it up on the Bass so the 808 makes room for the kick -- the classic sidechain pump -- or on the Synth for a pumping pad.

LOOPS

The last strip is the **LOOPS** tab's four Tape Loops together. Its fader, pan, Mute, Solo and meter come after each track's own Vol and Pan:

- Pull it down to sit the loops under the beat.
- Mute it to drop them all at once.
- Solo it to hear just the loops.

It fades rather than clicks, it follows Solo like any strip (right-click for **Solo-safe**, a name or a colour), and exports follow it too. It has no Duck, sends or inserts; each track has its own Reverb and Delay sends on the LOOPS tab.

Buses A-D

Next to the master: four group buses. Send several sources to one bus and treat them together -- all the drums through one saturation for glue, the synth and a deck through a shared delay.



Routing: every source against Master, Bus A and Bus B

- **Routing...** (top left of the MIXER) shows every source -- each pad, the Bass, the Synth, both decks -- against **Master**, **Bus A** and **Bus B**. Click a cell to send that source there. Everything starts on the Master.
- A bus has three **insert slots** (click one for an effect; **Knobs...** in the same menu sets its controls), **Pan**, **Volume**, **Mute** and a meter. It joins the master before the master rack. Each bus effect's **Knobs...** panel ends with a **Mix** slider (how much of the effect you hear), and its menu has **Parallel compression (preset)**: send your drums to a bus and pick it for thicker drums that still hit. A bus also has a **Width** knob (under its name): 0% squeezes everything on it to mono, 200% spreads it twice as wide; double-click for 100%. The Bass, Synth and turntable strips have the same Width knob, beside Duck. **Right-click** a Width knob for **Mono below** (off, 80, 120 or 160 Hz): the low end of that strip goes mono, so its bass stays centred however wide it is.

- A bus can go into a **later** bus: in **Routing...**, the bus rows at the bottom pick the master or a bus after it (Bus A into Bus B, never back). Send drums to A and the bass to B, both into C, for one glue compressor over the lot.
- A pad sent to its own output in your DAW (multi-out) still goes there, not to a bus.

Returns, sends and ducking

- **Reverb 1, Reverb 2, Delay**: the return strips after the buses. Each has a level, a mute and a meter. Click a reverb's name for its **Size**, **Damping** and **Width**. Reverb 2 starts as a bigger hall.
- **Right-click** any pad, the Bass, Synth or a deck, then **Sends...**: how much goes to each return. A pad's sends can be **before the fader**, so its reverb stays when you pull the pad down.
- **Ducking...** (top row): two duck groups. Each **follows** the pads marked **D**, the bass, the synth, a deck, or the **sidechain input** (route a track to the plugin's sidechain -- in FL Studio, from the mixer). **Attack** and **Release** shape the dip; the bar shows how far it's dipping now. Right-click a strip, **Ducks with**, to pick its group; its **Duck** knob sets how far.

Snapshots, links and the view

- **Snapshots**: store the mixer in one of eight, and recall it whole or **just the mutes** (a mute scene: drop the drums for a bar). The **Mixer Snapshot** parameter recalls them from your DAW -- automate it in FL Studio to switch mixes through a song. The LOOPS strip is part of a snapshot too, so a scene can bring the loops in or drop them out. A snapshot keeps faders pushed above 0 dB too (the master's as well).
- Right-click a strip: **Link** (faders in a group move together, mutes follow), **Solo-safe** (still heard when you solo something else), **Rename...** and **Colour**.
- **Zoom** (the slider at the top) makes the strips narrower -- more of the mix on screen -- or wider; **Hide unused** hides pads with no sample; the bank picker shows another bank's pads (their level, pan and mute).
- Every strip shows its level in dB above the fader -- double-click it and type one. **Undo** takes back a burst of mixer moves in one go.
- **Templates** saves the buses (with their effects), returns and ducking, to load into the next beat.
- **Signal Flow** shows how a sound reaches the output; click a step to go to it.
- **Export WAV** and **Export Full Song** render the mix the way you hear it: each pad's effects, sends and Duck, the buses, the master rack, the Master fader and the safety limiter.

The EQ

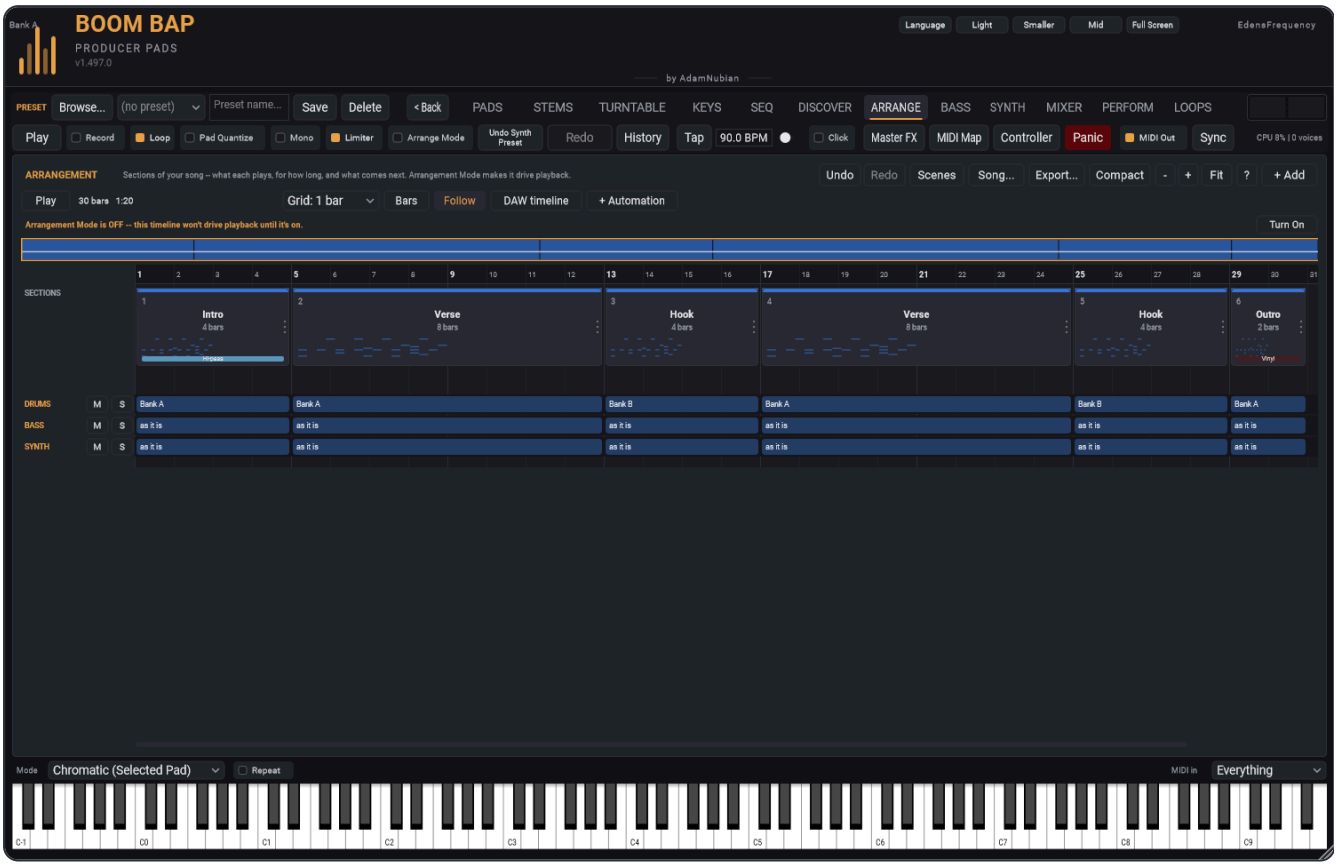
Any insert slot -- a pad's, Bus A / B's or the master rack's -- can hold an **EQ**. Pick it, then choose **Edit EQ...** from the same menu (or click the slot's button):

- The curve sits over a live picture of the sound. **Before / After** shows what goes in or what comes out.
- **Bands 1-4**: drag a round handle sideways for its frequency, up or down for boost or cut; the mouse wheel over it makes it wider or narrower; double-click puts it back. Each band is a **Bell**, **Low Shelf** or **High Shelf**.
- **Low cut / High cut**: the square handles. Turn one on and pick how steep it is (12 to 48 dB/oct).
- **Dynamic** on a band: it only moves while its own frequencies are louder than the **Threshold**, by up to the **Range** (negative cuts, positive boosts) -- tame a boomy note only when it booms.
- **Output** sets the level after the EQ; **Flat** resets everything.

A slot switched to EQ starts flat. The EQ is saved with your project, the bank, kits and the pad's A/B, and sounds the same in Export WAV.

CHAPTER 14

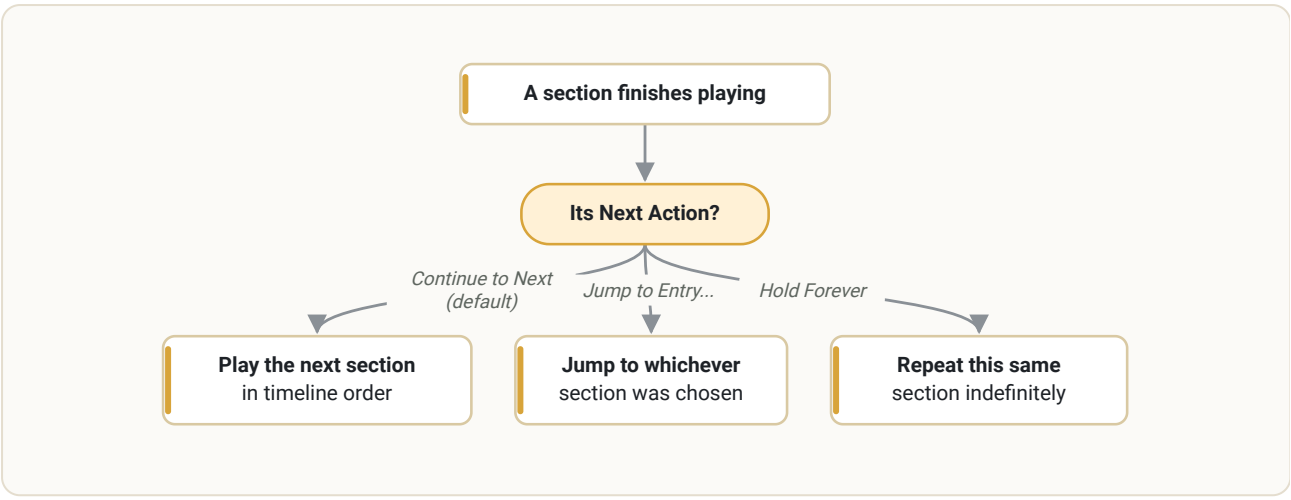
ARRANGE



The ARRANGE tab

Build a timeline out of your 4 kit banks (see The pad grid & sample browser for what a bank is) — sections you can arrange into a song structure instead of switching banks by hand.

- **+ Add** — adds a new section (4 bars, same bank as the last section you added).
- **Double-click** a section, or right-click and choose **Rename...**, to give it a name (e.g. "Intro", "Chorus") — leave it blank to fall back to showing which bank it plays.
- **Drag a section's right edge** to resize it, in whole bars.
- **Drag a section** to reorder it within the timeline — a gold line shows where it'll land.
- **Right-click → Edit Next Action...** to choose what happens when a section finishes:
  - **Continue to Next** (default) — play the next section in order.
  - **Jump to Entry...** — jump to a specific other section instead. The section shows a curved arrow to its target, plus a small "-> N" label.
  - **Hold Forever** — repeat this section indefinitely. Shown with a small looping-arrow icon on the section itself.



- **Right-click → Pattern** picks which of the bank's 16 patterns the section plays ("Whatever the bank is on" leaves it to the bank). The block shows it as "/ P3".
- **Right-click → Key** plays the section's synth part and bass line a few semitones up or down (-5 to +6) -- a key change for a chorus or a last section. The block shows it as "/ +2".
- **Right-click → Duplicate** adds a copy of a section right after it; **Right-click → Delete** removes it.
- Each section's block width is proportional to its length in bars, so longer sections are visibly wider.
- **The view:** a ruler with bar numbers along the top and **Zoom in**, **Zoom out** and **Fit** (the whole song in view). The button beside them switches to the **Compact** view (plain blocks, no ruler) and back. **?** explains the tab in a few lines.

Arrangement Mode (toolbar toggle, next to Limiter, or the **Turn On** button in the bar that says it's off) turns the timeline on: while it's on, playback advances through your sections automatically following each section's Next Action — a gold highlight shows which one is currently playing — and the SEQ tab's A/B/C/D bank buttons are disabled, since the timeline is what's driving bank switches now. Turn it back off to return to switching banks by hand.

Sections, round two

- **Select** a section by clicking it; **Ctrl**-click adds or removes one, **Shift**-click selects a run. Right-click settings apply to all of them where that makes sense.
- **Right-click** a section for more:
  - **Type** (Intro, Verse, Hook, Bridge, Break, Drop, Outro) and **Colour** -- a type brings its own colour.
  - **Repeats** -- play it up to 8 times before moving on.
  - **Bass** and **Synth** -- leave them as they are, make them silent in this section, or switch to one of their 16 slots.
  - **Mute pads...** -- tick the pads that sit this section out.
  - **Tempo...** -- the section's own BPM, and a ramp from the section before. The standalone plays it; in a DAW the DAW's tempo leads.
  - **Pickup** -- the drums come in only for the last 1-3 beats of its first bar.
  - **Drop-out** -- the drums stop for the last 1-4 beats of its last bar: the gap before a hook. An orange mark runs along the end of the block.

- **Comment...** -- a note on the section (a lyric idea, "add a riser here"). The block shows a small note; hover over it to read it.
- **Mixer snapshot as it starts** -- recall one of the MIXER's snapshots.
- **FX preset** -- treat the whole mix while this section plays, and switch it back off when the section ends. The change fades over a few milliseconds, so it never clicks. Section FX are in your exports too, but not in stem exports, which leave them out the same way they leave out the master rack.

Preset	Sound
Hi-pass	The low end drops away, which suits a build into a hook
Lo-fi	Dull and crunchy: a lower sample rate and fewer bits
Telephone	Thin, mid-only and a little driven
Room, **Hall**, **Stadium**	Reverb on everything, from small to huge
Vinyl	A narrower, softer record with hiss and the odd crackle

- **Play from here**, **Loop this section** (or the selection), **Split** before one of its bars, **Merge with the next section**.

- **Edit Next Action...** also has **Random section** and **Stop (the end of the song)**, and a Jump can be taken a set number of times, then the song goes on.
- Each block shows a small picture of its pattern; the strip above the timeline is the whole song in its colours with an energy line -- click it to scroll there.

The whole song: Song...

Song... in the header has everything about the song as a whole.

Start from a template

Start from a template lays out a whole song, with drop-outs leading into the hooks:

Template	What you get
Boom bap song	Intro, verse (16), hook (8), verse (16), hook (8), bridge, hook, outro -- 72 bars
Short song	Intro, two 8-bar verses and hooks, outro
Hook first	Opens on the hook: intro, hook, verse, hook, verse, hook, outro
Beat tape	A short instrumental: intro, the beat, a break, the drop, outro
Long loop	32 bars of one beat to write or freestyle over

Verses play bank A and hooks bank B, so make your hook beat on bank B. A template replaces the sections there (it asks first); **Undo** brings them back.

Build a song from my patterns

Build a song from my patterns writes the song for you from what's already in your project. Pick a length:

Choice	What you get
Classic (about 3 minutes)	Intro, verse, hook, verse, hook, bridge, hook, outro
Short loop (about a minute)	Intro, verse, hook, outro
Extended (about 4 minutes)	Classic with a longer intro, outro and last hook

It looks at your patterns rather than filling in a fixed shape:

- **Banks.** Verses use the first bank that has a beat, and hooks use the next one. A third bank becomes the bridge. If you've only made one beat, the whole song uses it.
- **Intro and outro.** The kick and snare wait while the hats and anything else play. If there's a bass line or synth part, the drums wait instead and the music plays. If there's both a bass and a synth, the bass comes in with the first verse.
- **Into each hook.** If the verse's bank has a second pattern slot with hits, a bar of it plays as a fill. Otherwise the drums drop out for the last 2 beats.
- **Bridge.** With no third bank, the hook's beat plays with the kick and the bass taken out, as a breakdown.
- **One beat only.** The extra percussion (claps, open hats, shakers) waits until the hook, so the hook lifts.
- **Section FX.** The intro gets **Lo-fi** when only the music plays, or **Hi-pass** when the drums do. Each fill bar gets **Hi-pass**, so it builds into the hook. A breakdown gets **Room** and the outro **Vinyl**. Untick **With Section FX** at the bottom of the menu for a song without them.

If your beat is only a kick and a snare, and there's no bass or synth, the intro plays them rather than going silent.

It knows which pad is the kick, snare or hat from the pad's name (`Kick`, `BD`, `hh`...) or from the colour **Name and Colour by Sound** gave it. Run that first if your pads have plain file names. Every section it makes has a comment saying what it did, so you can see it and change it. It replaces the sections there (it asks first); **Undo** brings them back.

Fit to a length

Fit to a length... asks how long the song should be -- type `3:00`. Sections of 4 bars or more stretch or shrink, 4 bars at a time, until the song is as close as it gets; short ones (a 2-bar intro) stay as they are. **Undo** puts it back.

Beat tape mode

Beat tape mode puts a bar (or two) of silence after each section, so the arrangement plays as a run of separate beats -- the way a beat tape plays -- instead of one song. Not while **DAW timeline** is on.

Save and load an arrangement

Save arrangement... keeps the sections, the loop, lanes, automation and markers in a `.bbarrange` file; **Load arrangement...** brings them into another project, replacing its sections (**Undo** brings them back). Sections play banks and patterns by number, so they play whatever that project's banks hold.

Lanes and automation

Under the sections are three lanes: **DRUMS**, **BASS** and **SYNTH**. Each cell shows what that section does with the part; click it to change it. **M** mutes a part for the whole song, **S** solos it.

+ Automation adds a lane for a mixer level (the master, bass, synth, a deck, a bus, a return, or a pad). Click to add a point, drag to move it, right-click to delete it. Or turn on **R**, play the song and move that level in the MIXER -- the lane records it.

Playing it

- **Play** / **Stop** (the standalone; Space too). In a DAW, its own transport plays it.
- **Click the ruler** to play from that bar; **right-click** it to add a **marker**, or to rename or delete one.
- **Follow** keeps the playhead in view. **DAW timeline** makes the DAW's bar the song's bar, so moving the DAW's playhead (FL Studio's song position too) moves the song.
- **Scenes** shows the sections as big buttons: while playing, click one and the song goes there at the next bar.
- The time label shows where the song is and how long it is; **Bars** / **Time** switches the ruler. **Grid** sets what sections and points snap to.
- **Tails ring on** when a section changes bank: a long sound still playing as the new section starts (a crash, a held chord, an 808) carries on with its own sample instead of being cut, for up to 8 seconds, while the new bank's pads play over it. It goes through the pad's current settings (filter, envelope, effects), which now belong to the new bank. A time-stretched pad still stops at the change.

Shortcuts: Space play / stop, Delete, Ctrl+D duplicate, Ctrl+Z / Ctrl+Y undo / redo, Ctrl+A select all, Left / Right select (Shift extends, Ctrl moves the section), Enter play from the selection, L loop, F2 rename, + / - zoom.

Exporting the song

Export... in the header: the song as one WAV with a marker at each section (named by the section), its **stems** (one WAV per sound across the song), **each section as its own loop**, or **MIDI** with a track per pad, the bass and the synth, or a **beat tape**: each section, all its repeats, as a numbered track ("01 Intro.wav") with a playlist. The SEQ tab's **More... > Export options** set the bit depth, sample rate, normalise and tails for all of them.

CHAPTER 15

TURNTABLE



The TURNTABLE tab, two decks and the crossfader

One dedicated deck with its own sample slot, not tied to pad selection. The platter sits on the left at a much larger size than other tabs' controls; everything else lives in a scrollable strip on the right, and the **Chop** lane runs along the bottom (see Deck Chop below): it chops whatever's on the deck straight onto your pads.

- **Full Screen** (or press **E**) — hides everything but the platter and a small Play/Cue strip, for an uncluttered view while performing. Press **Escape**, or click the button again, to return.
- **Load...** or **drag an audio file** onto the panel to load a sample onto the deck. A **video** works too: its soundtrack loads like any record (also from DISCOVER's **Send to Deck**).
- **Click-drag the platter** to scratch: dragging clockwise plays forward, counter-clockwise reverses, and the pitch follows how fast you drag. Release to resume normal playback from wherever the scratch left off.
- **Play / Pause** — normal playback at the current pitch.
- **Cue** — stops and jumps back to the start.
- **Pitch** fader — 0.5x–2.0x playback speed when not scratching.
- **Volume** fader — deck output level.
- **EQ / Filter / Reverb** — Low/Mid/High EQ (±24dB), a Filter knob (sweeps low-pass to high-pass through a neutral centre), and a Reverb send, all real host-automatable parameters.

- **Loop** — repeats a beat-length region from wherever playback currently is (needs a detected BPM, see the BPM/Key readout).
- **Stutter** — hold-to-engage rapid retrigger of a short slice, rate selectable (1/4 to 1/32).
- **Scratch Patterns** — Baby/Scribble/Chirp presets replay a canned scratch gesture; **Record** captures your own platter moves as a named custom pattern to replay later. The fader-work patterns are under Perform below.
- **Vinyl Sim** — Wow/Flutter, Vinyl Noise, and Saturation knobs (0–100%, no effect at 0) plus a **Motor Ramp** toggle (spins up to speed from a stop instead of starting instantly). All off by default.

Perform

DJ moves, in the control column under Scratch Patterns (scroll down to see them). They run inside the audio engine, so a fader click lands exactly where it should, and the patterns follow your **project tempo**.

Scratch patterns with the fader — each plays one bar: a hand rocking the record plus a crossfader cutting the sound, then the record is back exactly where it started.

Pattern	What it sounds like
Transform	The sound chopped into four quick stabs every beat while the record rocks back and forth
Crab	Four very fast taps on each push of the record
Flare	Each push starts open and is clicked off once in the middle — two sounds per push
Orbit	A flare on the push *and* on the pull
Tear	The push stalls halfway, splitting it into two sounds, with the fader open
Stab	A quick hard push with the sound cut right after the attack

The running pattern's button lights orange.

- **Brake** — the record slows to a stop, like hitting stop on a real turntable. (Scratch Speed sets how long it takes.)
- **Spinback** — throws the record backwards, then it stops.
- **Censor** — hold it and the playing record runs backwards; let go and it plays forward again.
- **Slip** — turn it on and scratching, Censor, Stutter, Loop and the patterns all happen *over* the record without moving it: a silent playhead keeps going underneath, and when you let go the record jumps to where it would have been. Your scratch stays in time with the beat.
- **Hot cues 1–8** — click an empty one to mark the playhead there (it turns gold); click a gold one to jump to it and play. Shift+click clears it. Jumps fade out and in over about a millisecond, so they don't click. Cues are saved with your project, and cleared when you load a different record.
- **Combos 1-10** (under the hot cues) — right-click one (or Shift+click) to set it: this record, from one of its hot cues, with a performance (Transform, Crab, Flare...) -- or just the cue. Then click it to fire the whole thing in time. A combo for another record loads that record first. Combos are saved with your project.
- **< and >** — jump back or forward by the amount in the box (1 beat up to 4 bars), using the record's tempo.

Stems

Stems splits the record on the deck into its **vocals**, **drums**, **bass** and **other** (everything else: keys, guitars, strings), so you can drop the vocal out, play only the drums, or bring the bass back in on the one.

1. The first time, press **Get separator**. It downloads the separator (about 250 MB, once): HT-Demucs, the separation model, and ONNX Runtime, which runs it. Everything then runs on your computer: there's no account or key, and your music is never uploaded.
2. Load a record and press **Separate**. Splitting takes a while -- on a typical laptop about twice as long as the record lasts, so a 4-minute record takes about 8 minutes. The deck keeps playing while it works, and **Cancel** stops it. The result is kept, so the next time you load that record its stems are ready straight away.
3. Click **Vocals**, **Drums**, **Bass** or **Other** to switch that part on or off (gold = on). Shift+click plays that part on its own; Shift+click it again to bring them all back. Parts fade in and out over a few milliseconds, so they don't click.
4. The fader under each part sets how loud it is while it's on -- pull the vocals down under the beat instead of cutting them. Double-click a fader for full. A new record starts with every part on and full. Below them, **Stem sends to** picks the mixer's **Reverb 1**, **Reverb 2** or **Delay**, and the teal fader under each part sends that part there -- reverb on just the vocals, a delay throw on just the drums. They follow the part's switch and fader, and the deck's volume, crossfader and mute.
5. **Export...** (beside Separate) saves them as WAV files in `Music\Boom Bap Stems\<the record's name>`: **the four stems** as they are, or **one mix as you hear it** -- your switches and faders, so switching off the vocals and exporting gives you the instrumental. The folder opens when it's done.

With every part on, you hear the original record, untouched. Scratching, loops, hot cues, Slip and the patterns all work on the stems too. Stems are separated per deck, and a record's stems are kept in `%APPDATA%\Boom Bap Producer Pads\Stems` -- STEMS > **Stem cache...** shows how much space they take and frees it (the plugin makes them again when asked), and its **Separate records...** splits a crate of records ahead of time, so they load here with their stems ready.

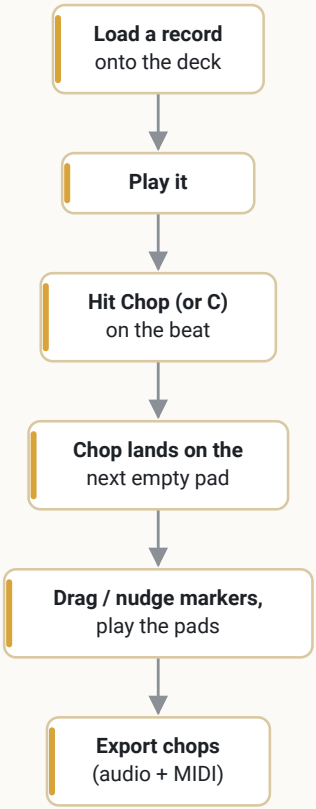
On a DJ controller, the **STEMS** pads work the stems once a record has them: see Pad modes.

The platter also responds to an external MIDI jog-wheel controller, not just mouse drag: Note On/Off at note 20 touches/releases the platter, and CC 20 carries relative jog motion (the standard sign-magnitude relative-encoder convention most DJ jog wheels already speak) — both on MIDI channel 1. Useful if you're driving this from a hardware controller or a DAW MIDI script rather than the mouse.

A real DJ controller: press **DJ Controller** (top left of this tab), switch your controller on, and teach the plugin its jog wheels, Play/Cue/Sync, faders, EQ, crossfader and pads: both decks follow it. See Using a DJ controller.

Deck Chop

Chop a record the way you'd do it on hardware: put it on the deck, play it, and hit a button on the beat. Every chop lands on a pad straight away and is playable immediately. There's no separate "commit" step, because the markers on the waveform **are** the pads: move a marker and its pad changes with it.



- **Chop** (or press **C**) — drops a chop at the playhead onto the lowest-numbered empty pad of the current bank. Works while the record plays (chop by ear) or while it's paused (chop exactly where you stopped). **Double-click** anywhere on the waveform to chop at that point instead.
- **Auto...** — chops the whole record for you, onto the empty pads: *Vocal phrases (from the stems)* -- once the record is separated into stems, a chop at the start of every vocal line (short breaths don't split a line; blips are skipped), so each phrase is a pad -- *Every hit*, *Only the big hits*, *Every bar*, *Every beat* (from the record's tempo, settling onto its real hits), *8* or *16 chops* (strong phrase starts, on the beat first), or *8 random chops*. One Ctrl+Z takes the whole lot back. These are the same slice modes as the Sample Editor's Slice row. **Into the next empty bank** has the same choices, but chops into a fresh bank (one with no sounds and no pattern) and switches to it, so your current kit stays as it is. **Chops to the stem mix (as you hear it)**, at the bottom: once the record is separated, set its stems the way you want (vocals off, the drums down...) and every chop pad takes exactly that slice of the mix. The pads become ordinary pads with their own files (in `Music\Boom Bap Stems\<record>\Chops`), so they keep that sound whatever the deck does next. One Ctrl+Z puts the chops back.
- **The mini pad grid** (right of the waveform) shows the current bank: gold pads are chops of this record, grey pads hold other samples, and grey pads **outlined in gold** are chops of the record you had on before. Tap an **empty** pad to chop onto that exact pad at the playhead; tap any other pad to play it. **Right-click** a pad to clear it (or to clear all the old chops at once).
- **Markers** — each chop is a numbered flag; the number is its pad. Click a flag to hear the pad, **drag** it to move the chop, or select it and press **Left/Right** to nudge it 10ms (hold **Shift** for 1ms). **Delete** (or right-click → Delete) removes the chop and empties its pad. Every one of these is undoable with **Ctrl+Z**.
- **Gate** (on by default) — each chop stops where the next one starts, so every pad is its own clean slice. Turn it off to let chops play on to the end of the record, e.g. to start a longer phrase from a pad.

- **Snap** — Off, 1/4, 1/8 or 1/16: chops land on the record's beat grid, worked out from its detected BPM and lined up on your first chop (so the first chop is where you say the beat is; the grid follows from there). The grid shows on the waveform while Snap is on (zoom in if it's too dense to draw).
- **1/2** and **x2** — tempo detectors often read a record at double or half its real tempo (a 94 BPM record read as 188). Fix it here; the deck's Sync and Loop use the corrected tempo too, and it's saved with your project.
- **Offset** — if your by-ear chops keep landing a hair late (you hear the beat a few milliseconds after it plays), pull this negative. It only applies to chops taken while the record is playing. Double-click to reset to 0.
- **Tap Pads** (off by default) — when on, hitting an **empty** pad on the PADS tab or on your MIDI controller while the deck plays lays a chop on that pad, right then. Leave it off if a DAW MIDI clip plays the pads while the deck runs, or that clip would lay chops too.
- **Stretch** — time-stretches every chop pad to your project's tempo without changing its pitch, and follows the tempo if it changes. Needs the record's BPM (fix it with 1/2 / x2 first if it's wrong).
- **-**, **Fit**, **+** zoom the waveform (or **Ctrl + mouse wheel**; the wheel on its own scrolls). While the record plays, the view follows the playhead.
- **Export...** — pick a folder and you get one audio file per chop, numbered in record order with its pad in the name, plus a MIDI file that plays the chops back on the pads in their original order and timing. Drag both into your DAW.
- **Clear** — removes chops from the current bank: this record's, or the old chops of the record you had on before (see below). If there are both, it asks which. Ctrl+Z brings them back.

Good to know:

- Chops go into the **current bank** only. A 4x4 bank holds 16 chops; pick a bigger grid on the PADS tab (up to 8x8) for up to 64, or switch bank and keep chopping.
- Chop never overwrites a pad that already holds a sample: when the bank is full it tells you instead.
- Loading a different record onto the deck leaves your existing chops on their pads, fully playable; they just stop being editable markers. The lane tells you how many are left, the mini grid outlines them in gold, and **Clear** removes them all in one go, making room for the new record's chops.
- Chops are ordinary pads everywhere else: sequence them on SEQ, shape them with DSP, mix them on MIXER. Your project saves them like any other pad, and they come back as editable chops.
- Only deck 1 has the Chop lane. In **Full Screen** the lane hides, but **C** still chops.

Chop a pad again

Any pad can go back onto the deck to be chopped further: a chop of a chop, a drum hit cut into smaller pieces, a phrase re-sliced on a different beat. Right-click the pad on the PADS tab and choose:

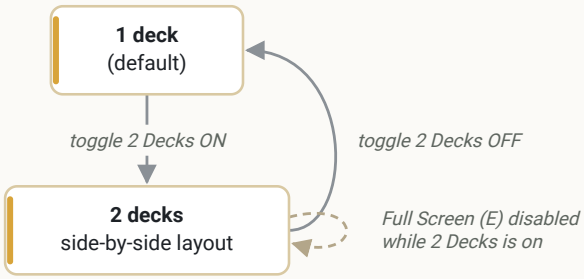
- **Send to deck 1, to chop again**: the pad's slice exactly as it is.
- **Send to deck 1 as it sounds**: the pad the way you hear it, with its Tune, Fine, Speed, Reverse, filter, envelope, LFO, bitcrush and insert effects (reverb, delay, saturation...) printed in, the reverb or delay's tail included. Use it to lock in a pitched-down chop before cutting it up again.
- **Send to deck 2** and **Send to deck 2 as it sounds**: the same, onto the second deck, to play or scratch against deck 1. The two-deck view comes up by itself.

The sound lands on the deck and the TURNTABLE tab opens, ready to play and chop (only deck 1 has the Chop lane). New chops only fill **empty** pads, so the pad you sent (and every other pad) stays as it is; when the bank is full, switch bank and keep going. You can go round as many times as you like: deck, pads, deck.

Each sent pad is saved as a WAV file in your Music folder, under **Boom Bap Pad Bounces**, named after the pad (like `Pad 3 - Break (as it sounds).wav`). Your project remembers the deck's record by that file, so keep the folder; delete files from it once you no longer need them.

2 Decks

Toggle **2 Decks** (top of the tab) to bring in a second, fully independent deck — its own sample slot, platter, EQ/Filter/Reverb, Vinyl Sim, scratch patterns, and MIDI Learn mappings, running alongside the first. Everything documented above applies identically to both; they don't share state or interact with each other beyond both being audible at once.



- With 2 Decks on, the platters split **left/right** instead of one platter filling the space — each half is a complete, independent deck.
- **Full Screen is unavailable while 2 Decks is on** (there'd be two platters competing for the same full-screen space) — turn 2 Decks off first if you want the single-deck Full Screen view.
- Turning 2 Decks off doesn't unload or reset deck 2 — it's just hidden; turn it back on and deck 2 is exactly where you left it.
- Deck 2's sample, settings, and MIDI Learn mappings are saved with your project/preset the same as deck 1's.

Crossfader

With 2 Decks on, a **crossfader** appears across the top: left plays deck 1, right plays deck 2. Double-click it to centre it, right-click it for **MIDI Learn** (put it on a real fader), and pick how it blends:

Curve	In the middle	Good for
Dipless (default)	Both decks at full level	Layering, and never changing a one-deck session
Smooth	Both a little quieter, an even blend	Mixing from one record into another
Scratch	Both full; each deck cuts out only at the very edge	Fader tricks — a tiny move cuts the sound

Reverse swaps the ends (sometimes called "hamster" style). The crossfader and its settings are saved with your project and can be automated from your DAW.

CHAPTER 16

STEMS



The STEMS tab

Splits a pad's sample three ways:

- **Bands** -- **Low / Mid / High**, by frequency, with filters. Quick.
- **Harmonic / percussive** -- the sustained part and the hits. Quick.
- **AI** -- **Vocals, Drums, Bass and Other**, by the same separator the decks' **Stems** use (see the TURNTABLE tab). It runs on your computer and takes a minute or more for a long sample, but it's the one that can really pull a vocal or a drum part out of a full mix. The first time, it offers to download the separator (about 250 MB, once, shared with the decks).

The bands and H/P split by sound, not by instrument: good for pulling apart a one-shot's range, not for remixing a song. That's what AI is for.

The tab has three columns.

Left -- the pads. Every pad, with its number and sound. Click one to work on it. Three small dots on the right show what it's been split into: gold for the bands, purple for harmonic / percussive, teal for AI. A pad whose stems are previewing has a teal number.

Middle -- the pad you picked. Its name, length, detected tempo and key, and whether it's been split; an overview of its sound with **Play** (the stems mixed together), the time and a position bar under it -- drag or click the bar to jump; then a card per stem, in its own colour, with its own waveform once it's been split:

- **Low** (below 250 Hz), **Mid** (250 Hz - 4 kHz), **High** (above 4 kHz) -- the band split. **Crossovers** (under Split into bands) moves those two points: Low / Mid from 40 Hz to 2 kHz, Mid / High at least an octave higher, up to 16 kHz -- say 120 Hz to keep just the sub, or 2.5 kHz to put a vocal's bite in High. Double-click one for its default. The cards show the points; split again to hear a change. **Slopes** picks how the bands meet: **24 dB/oct** (as before), or **linear phase** -- no phase shift, so each band lines up with the original -- **gentle** (a smooth octave-wide handover) or **steep**. **Presets...** beside it saves these settings (crossovers, slopes and H/P strength) by name and brings them back -- a few are built in (Sub only, Vocal bite, Clean bands, Hard drums) -- and lists **this pad's splits**, each with the settings it was made with: pick one to get those settings back. **A** and **B** (beside it) hold two sets of settings to flip between: set up A, split, switch to B, change things, split again, compare.
- **Harmonic** (the sustained, tonal part) and **Percussive** (the hits and transients) -- the H/P split. **H/P strength** (under its button) sets how hard it pulls them apart: low keeps them overlapping (smoother, a little of each in the other), high makes every sound all one or the other (cleaner, but it can turn grainy). 33 % is how it always was -- double-click for it. The two still add back up to the original at any strength. Split again to hear a change.
- **The view list** (by STEMS) switches the cards between those five, the AI split's four (**Vocals**, **Drums**, **Bass** and **Other**), and the parts of **More splits...**: **Transient + sustain** splits a pad into its attacks (hits, plucks, consonants) and what rings on after them -- turn the sustain down for a tighter loop, or the transients down to soften it. **Mid + side** splits it into what both speakers share (centre vocals, kick, bass) and what differs between them (width, the room, wide pads) -- turn Side up for a wider loop, or use Mid for a mono-safe one. **Tonal + noise** splits it into its notes (voices, keys, strings) and its breath, hiss and cymbal wash. **Clean + crackle** lifts the clicks and pops off a vinyl sample into Crackle, leaving Clean. **Cleaned + removed** is the sample cleaner: it takes out rumble, mains hum (when there is some) and steady hiss, and puts what it took in Removed. **2 bands**, **4 bands** and **5 bands** split into that many frequency bands (linear phase), spread between the two Crossovers -- the cards say where each sits. Like every split here, the parts add back up to the original.
- The AI view's fifth card, **Residual**, is what the four AI stems miss of the original (usually very quiet), so nothing is lost. Play and Export follow what's showing. The four AI cards share the first four cards' levels.
- Each card's round **icon** lights up once that stem exists. **Drag it** straight into your DAW, or **right-click** it for "Send to next empty pad" -- or **Tab** to it and press **Enter**. Stems you split earlier -- even in an earlier session -- show up as soon as the pad has that sample again.
- **S** solos a stem (only the soloed ones play) and **M** mutes it, in the preview -- not in Export. They're for listening; they aren't saved.
- The slim **meter** beside each level shows that stem while it plays.
- **Waveform / Spectrum** (by STEMS) switches the overview and every stem between waveforms and spectrograms -- time left to right, low frequencies at the bottom, louder brighter. Handy for seeing what each split really kept. In the Bands + H/P view, the pad's spectrogram shows the two crossovers as white lines -- drag one up or down to move it.
- Each card's **level** (shown in dB; double-click for 0 dB) is the stem's level in the Preview mix and in Export, its small **pan** knob (double-click: centre) places it left or right, and the teal **width** knob beside it narrows it to mono or widens it up to twice (double-click: as it is). Each pad keeps its own levels, pans and widths: pick another pad and its mix comes back (a pad not mixed yet starts full and centred). The levels are real, host-automatable parameters.

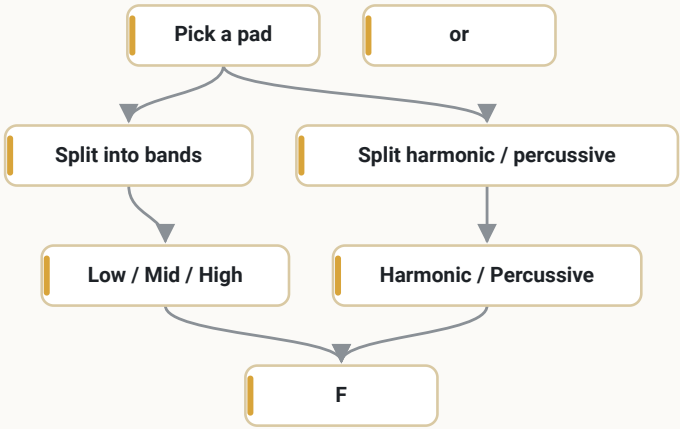
- **Right-click a card** for its **Low cut** (40-400 Hz: the rumble off a vocal, the sub off the drums) and **High cut** (3-16 kHz), in the preview and in Export.
- **Match** (by the view list) plays every stem at the loudness of all of them together, so a quiet stem soloed is heard fairly. Preview only.

Right -- what to do.

- **Split into bands** writes `..._Low`, `..._Mid` and `..._High` into a `Stems` folder next to the pad's sample file. **Split harmonic / percussive** writes `..._Harmonic` and `..._Percussive` there too -- a second, independent split; you can have both. It works best on material with real sustain (chords, vocal chops, pads); on pure drum breaks the two won't separate as cleanly.
- **Every pad: bands / H/P** run a split on every pad with a sample.
- **OUTPUTS** -- where each stem plays while you preview: **Main** (the plugin's main output, as always) or **Out 1-16**, the plugin's extra outputs ("Pad 1" to "Pad 16" in your DAW, the same ones pads can use). **Each to its own output** puts them on Out 1, 2, 3... so each stem gets its own mixer track: record them, or mix the preview there. Turn those outputs on in your DAW first (in FL Studio, the plugin's Processing settings; in Ableton, a track's "Audio From" the plugin's output). A stem on an output that's off plays on the main output. The choice is saved with the project.
- **Split with AI (4 stems)** writes `..._Vocals`, `..._Drums`, `..._Bass` and `..._Other` beside the others and switches to the AI view. The button shows how far it's got; click it again to cancel. A sample split before (on a pad or a deck) comes straight from the cache. One AI split runs at a time.
- **Drums to SEQ** (once a pad has its AI stems): reads the drum stem's hits into the pattern. Each hit is heard as a **kick** (low), a **hat** (very high) or a **snare** (the middle -- claps too) and goes on that step of your **Kick**, **Snare** or **Hat** pad -- the first pad named so (or coloured so by Name and Colour by Sound). It takes the pad's region as 1-4 bars at its tempo, replaces those three pads' steps, and is one Undo. The status line says how many of each it wrote, and which had no pad.
- **Quality** (beside them): in the other views, a **null check** -- whether the parts add back up to the original (the bands come very close but not exactly, from the crossovers' phase). In the AI view, how cleanly this pad's AI split came apart. Each stem gets a score out of 100 -- lower means its loudness moves with another stem's, a sign of bleed (drums ticking in the vocals) -- and how much of the mix it holds ("almost nothing" for an empty one, like the vocals of an instrumental). The **null** says whether the four add back up to the original (nothing lost). Parts of a song do move together, so use it to compare stems and splits rather than as an absolute grade.
- **Replace drums** (beside it): your kit instead of the loop's drums. The loop without its drums goes on the first empty pad (named "... (no drums)", same region and tuning) and takes over the loop pad's steps; then its drums go onto your Kick, Snare and Hat pads as Drums to SEQ does. The loop pad keeps its sample, so the original is still there. One Undo puts everything back.
- **To pads...**: **Each on its own pad** puts the stems the view shows on empty pads to build a new pattern from; **As one stack** puts them onto empty pads as one **stack** -- hit the first and they all play, lined up with the original (same region and tuning). Each stem is its own pad, so it has its own volume, pan, mute, sends and effects on the MIXER: turn the vocals down, put reverb on the drums only. The stack takes over this pad's steps (the pad keeps its sample). One Undo.
- **Chops play...**: every pad cut from this sample -- its slices and chops, in this bank -- plays one of its stems instead (**Its Vocals**, **Its Drums**, the bands...), or **The original** again. Each pad keeps its own cut, name and settings, so slice a loop once and flip all the slices to the acapella in one go. One Undo.
- **Fit...** (beside To pads): fit every stem shown to your beat -- a new **tempo** (from the sample's own) and a **key shift** for all of them together, time and pitch apart, and **Tune the bass to the key** to bring the bass stem onto

the key's root on its own. The fitted stems are new files (the originals stay) and go onto empty pads, as one stack or each on its own.

- **Stem cache...**: the AI split (here and on the decks) keeps every record's stems so it never splits the same file twice -- about 40 MB a minute of music. The list shows each record, when it was last used and its size, with the total above. **Remove selected**, **Remove missing** (records you've since moved or deleted -- their stems can't be used again), **Remove all**, or **Open folder**. Removed stems are simply made again the next time that record is split. **Separate records...** at the bottom prepares records ahead of time: pick files or whole folders and each is split in the background, one after another (a minute or more each; ones already split are skipped). The line beside it shows which one and how far; **Stop** ends the batch. Afterwards those records load on a deck with their stems at once, and **Split with AI** on a pad is instant for them.
- Splits run in the background: the pad shows **SPLITTING...** (and "..." in the pad list) until it's done, and you can keep working -- pick another pad, preview, export -- meanwhile. Several splits run one after another. While any run, **More splits...** reads **Stop splitting**. **More splits...** > **Split a folder** splits every audio file in a folder (and its subfolders) into bands and H/P, beside each file, so they arrive on pads already split.
- **Format: WAV / AIFF** -- what Split and Export write from now on; anything already split stays as it was written. **Export as** can write them instead as **24-bit WAV**, **32-bit float WAV**, **24-bit FLAC** or **24-bit AIFF** (with the levels, pans and widths). **Names** sets how the files are named, from {sample}, {stem}, {pad}, {bpm} and {key} -- say `{bpm} {key} {sample} - {stem}`. Export opens where you last exported to.
- **Tick the stems** Export should write (all are ticked to start with).
- **Drag all** (beside Export) drags every stem shown out at once -- into your DAW, one track each, or a folder.
- **Export this pad's stems...** copies the ticked ones to a folder you pick, with the levels baked in (a stem at 0 dB is copied untouched). **Export every pad's stems...** does it for every pad that has any, one subfolder per pad.



CHAPTER 17

PERFORM



The PERFORM tab

Macros

Four **MACRO** knobs, each moving up to eight controls at once.

To put a control on a macro:

1. Click that macro's **Assign** (it lights up).
2. Right-click any knob or switch anywhere in the plugin -- a pad's Cutoff, the reverb send, Swing, the bass volume -- and pick **MIDI Learn**. Instead of waiting for a MIDI knob, it joins the macro, and its name appears under the macro.
3. Add more the same way (up to eight), then click **Assign** again (or leave the tab) to finish.

How far each control moves: from where it is when you assign it, toward its far end -- a filter that's open closes, a quiet pad gets louder. So with the macro at 0 nothing changes; turning it up takes every control from where you set it to the other end. Set each control where you want the "start" of the move, then assign it.

Taking controls off: Remove... lists them; pick one, or **Clear this macro**.

Macros are saved with your project, and your DAW sees them as **Macro 1** to **Macro 4** -- automate them like any parameter.

The XY pads

Drag anywhere on the square: **across** moves **Macro 1**, **up** moves **Macro 2**. Put a filter on Macro 1 and a delay or reverb send on Macro 2 and you can sweep both with one finger.

There are four XY pads (two by two): **Macro 1 / 2** and **Macro 3 / 4** play those macros; **Filter** and **Repeat** play effects on the whole mix while you touch them. On **Filter**, left of centre closes a low-pass (darker), right of centre opens a high-pass (thinner), and up adds resonance. On **Repeat**, across picks how long a piece of the song loops -- 1/4, 1/8, 1/16 or 1/32 -- and up is how much of it you hear. Let go and the song comes back.

Hold-to-apply effects

Along the bottom: five buttons that act on the whole mix **only while you hold them**.

- **Stutter** catches the next 1/16 of the song and loops it until you let go.
- **Sweep** closes a filter down over a bar; let go and it springs open.
- **Tape stop** slows the whole mix to a halt, like switching off a tape machine.
- **Throw** sends the mix into a big reverb; let go and the tail rings on.
- **Glitch** chops the next 1/16 of the song into pieces and plays them back at random pitches, pans and directions, some gliding into their note.

They're for playing live: they aren't saved and aren't in exports.

Good to know

- A control can be on several macros; the last one moved wins.
- A knob on a macro still works by hand -- the macro sets it again only when the macro moves.
- MIDI Learn works as usual whenever no macro's Assign is on.

CHAPTER 18

LOOPS



The LOOPS tab

Open the **LOOPS** tab (top row, after PERFORM). There are four tracks side by side.

Loading a loop

Drop an audio file (WAV, AIFF, FLAC, OGG or MP3) onto a track's waveform box, or click the box or **Load...** and pick one. The track shows the file's name and length. **Clear** takes it off.

The sample browser sits on the left of the LOOPS tab too: drag a sound from it onto a track. Or right-click a file there and choose **Load into Tape Loop**, then pick the track.

Playing

Press **PLAY** on a track; press it again (**STOP**) to stop. Loops fade in and out over a few milliseconds, so they never click.

You can also start and stop each track from a MIDI controller. In the Controller setup, learn **Tape Loop 1 Play** (up to 4) on any button or pad. The button lights while its track plays. Learn **Tape Loop 1 volume** on a knob or fader to ride each track's level. See Using a DJ controller.

Sync

With **Sync** on (the default), the loop is fitted to the beat the way a tape machine would: sped up or slowed down so it lasts a whole number of beats at the tempo. That means the **pitch follows the tempo** -- a loop made at 90 BPM played at 96 BPM sounds a little higher. While your DAW (or the Standalone's own Play) is running, the loop stays in time with the song; start it at any moment and it comes in on the beat.

Turn Sync off and the loop plays at its own speed and pitch, free of the tempo.

Beats

When you load a loop, the plugin works out how many beats it lasts: from the tempo in the file name if it has one (`dusty-keys_90bpm.wav`), otherwise from its length at the current tempo. If a synced loop plays twice as fast or slow as it should, use **/2** or **x2** beside the beat count.

Half, Norm, Double

Half plays the loop an octave down over twice as many beats; **Double** an octave up over half. Both stay in sync.

Reverse

Plays the loop backwards (still in sync).

Seam

If a loop clicks or bumps where it goes round (it doesn't end where it starts), turn on **Seam**. The last 20 ms of the loop fade into the first, so the join is smooth. The loop gets that little bit shorter, which you won't hear in Sync. Leave it off for loops that were cut cleanly.

The knobs

Knob	What it does
Vol	The loop's volume (double-click: back to the default)
Pan	Left / right
Wear	Old tape: slow wow and fast flutter, a duller top end, saturation and a little hiss
Reverb	How much goes to the shared reverb (the same one the pads send to)
Delay	How much goes to the shared delay

Good to know

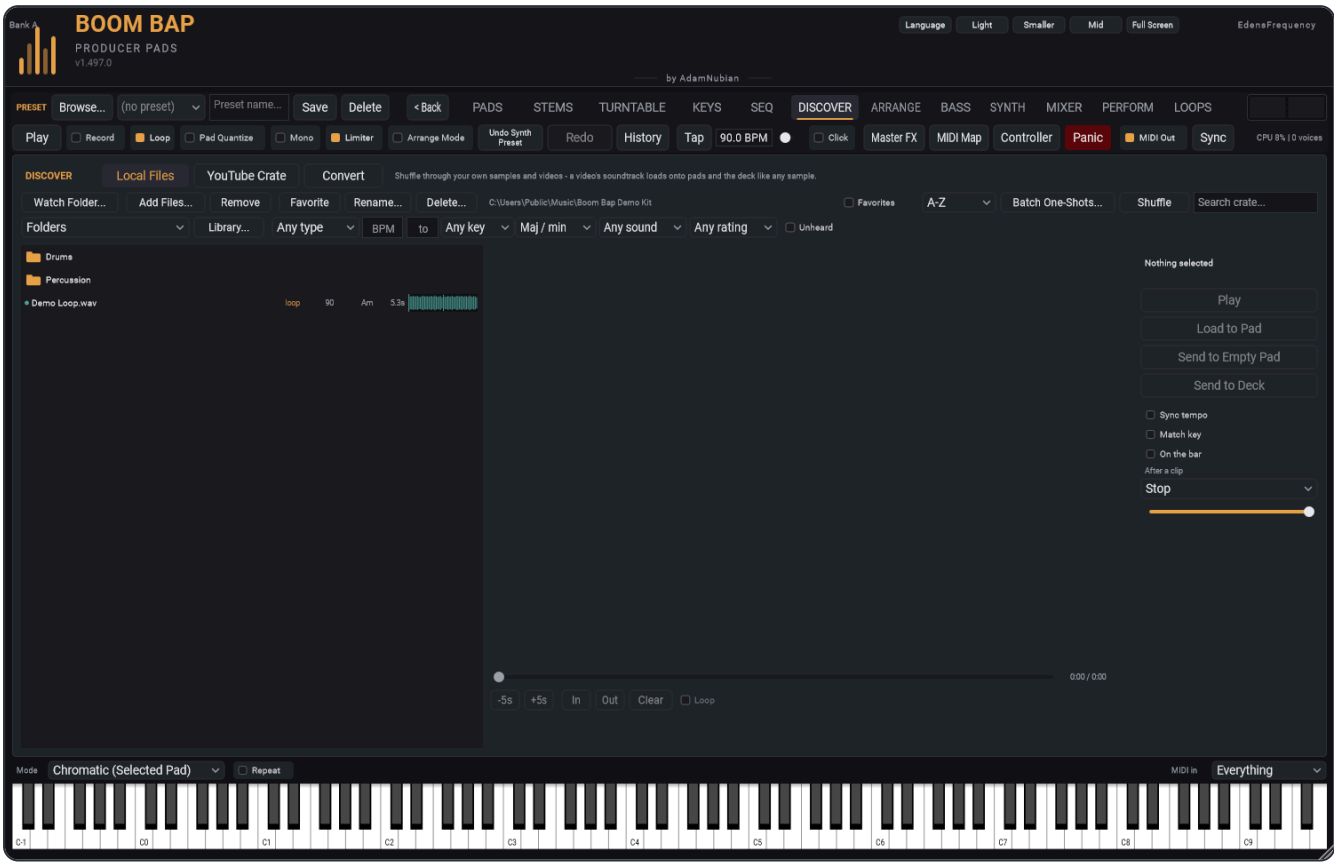
- Your loops, and every setting on the tracks, are saved with your project. The audio files themselves stay where they are -- if you move or delete a file, that track comes back empty.
- Tape Loops play on the main mix. Every audio export includes the tracks that are playing when you export. They start on the export's first beat at the export's tempo, so a synced loop lines up with the music. Stem

exports give them a file called **Tape Loops**. A track that's stopped isn't exported.

- On the MIXER, the **LOOPS** strip (the last one) is all four tracks together: a fader, pan, Mute, Solo and a meter after each track's own Vol.
- Loops up to 10 minutes long can be loaded.

CHAPTER 19

DISCOVER



The DISCOVER tab

Three sub-tabs: **Local Files** (a crate/shuffle audition browser over your own local sample/video library), **YouTube Crate** (filter, search, and get suggestions from YouTube videos you've tagged yourself) and **Convert** (turn audio files and videos into WAV, AIFF, FLAC or OGG, a whole folder at a time). Nothing here downloads or fetches anything automatically — Local Files and Convert only work with media you already have on disk, and YouTube Crate only embeds videos the same way any website embeds one.

Local Files

- **Watch Folder...** — point it at a folder (your downloads, a "new samples" folder, wherever you drop fresh material) and its audio and video files show up in the crate automatically, re-checked every couple of seconds.
- **Folders inside it.** Its subfolders are listed first, with a folder mark; **double-click** one (or select it and press **Return**) to open it. Inside a folder, the top row **..** goes back up. You can nest them as deep as you like, but never above the watch folder itself. The path above the list shows where you are.
- **Add Files...** — add specific files to the crate directly, without needing them all in one watched folder. You can also **drag files or a folder** from Explorer onto the crate: a folder adds its audio and video (not its subfolders).

Remove takes an added file back out (watch-folder files aren't removable this way — they just reflect whatever's actually in the folder).

- **Search** looks through the watch folder and every folder in it, and lists each match with its folder (*Drums\Kicks\kick 1.wav*), so typing a folder's name finds what's in it. **Favorites** shows only starred clips, from every folder. Clear the search (and Favorites) to go back to the folder you were in. The sort box lists the crate **A-Z** or **Newest** first (newest file on disk, handy for fresh downloads).
- **Managing clips.** Select a clip, then:
  - **Favorite** stars it (a gold star in the list); click again (**Unfavorite**) to unstar it.
  - **Rename...** renames the actual file on disk (the extension stays the same). If the clip is on pads or a deck in this project, they follow the new name, so the project saves it. Other projects that use the file will look for the old name.
  - **Delete...** moves the actual file to the Windows **Recycle Bin**, after asking. You can restore it from there. If it's on pads or a deck in this project, the message says where. They keep their sound until the project is closed, but won't find the file next time. The **Delete** key does the same from the list.
  - **Right-click** a clip for all of the above, plus Play, Load to Pad, Send to Empty Pad, Send to Deck, Show in Explorer and Remove from Crate. Right-click a folder to open it or show it in Explorer. Folders themselves are never renamed or deleted from here.
- Click a file to select and preview it. Audio plays through the built-in preview voice; video files marked `[VIDEO]` play with picture and sound in the central preview pane. **Shuffle** picks one at random from the folder you're in and every folder inside it, opens its folder and previews it. **Play** replays/toggles the current selection (for video). With the list focused, the **arrow keys** audition the next/previous file.
- **Load to Pad** (or double-click) puts the file on the selected pad; **Send to Empty Pad** puts it on the first empty pad of the current bank. The line under the buttons shows how it's going ("Loading ... onto pad 3", then "Loaded ..."): a long video takes a few seconds.
- **Send to Deck** loads it onto the turntable deck and switches to the TURNTABLE tab once it's there — ready to scratch, or to chop onto pads with Deck Chop.
- **Videos work too.** All three load a video's **soundtrack**, exactly like an audio file: chop a music video, a live clip or a phone recording the same way you'd chop a record. A few things to know:
  - What plays is whatever Windows itself can decode: MP4 and MOV (H.264/AAC — almost every phone, camera and download), WMV/ASF, 3GP and 3G2, camcorder files (MTS, M2TS, TS), MPG, and MKV/AVI with common codecs. WEBM needs the free Windows codec from the Microsoft Store. FLV isn't supported (Windows has no reader for it); convert it with another tool first. Audio: WAV, AIFF, FLAC, OGG, MP3, M4A/AAC, WMA, M4B (audiobooks), M4R (ringtones) and MKA. If Windows can't decode a video's sound (or it has none), it's marked *can't play* and loading it tells you so.
  - A video whose sound is completely silent — a screen or webcam recording made without a microphone — loads, but you get a message saying it's silent.
  - The picture and the sound are decoded separately. If Windows can't show a video's picture in the preview pane, Local Files plays just the sound (marked *sound only*), and it still loads onto pads and the deck.
  - Surround soundtracks are mixed down to stereo.
  - Pads and the deck take the **first 20 minutes** of a very long file (a whole film would need gigabytes of memory). To keep all of it, convert it first (below).
  - Saving a preset or sharing a kit stores the soundtrack as a WAV, not the whole video.
- M4A and AAC audio files load too (also through Windows' own decoder).

The library

DISCOVER keeps a library of your local files: it listens to each one in the background and remembers its **tempo**, **key**, **length**, whether it's a **loop** or a **one-shot**, what kind of **sound** it is (kick, snare, hat, perc, bass, vocal, keys, fx, loop) and its **waveform** -- plus everything you say about it. "Analysing... N to go" by the filters shows it working; **Library... > Analyse every file in the library** does them all at once.

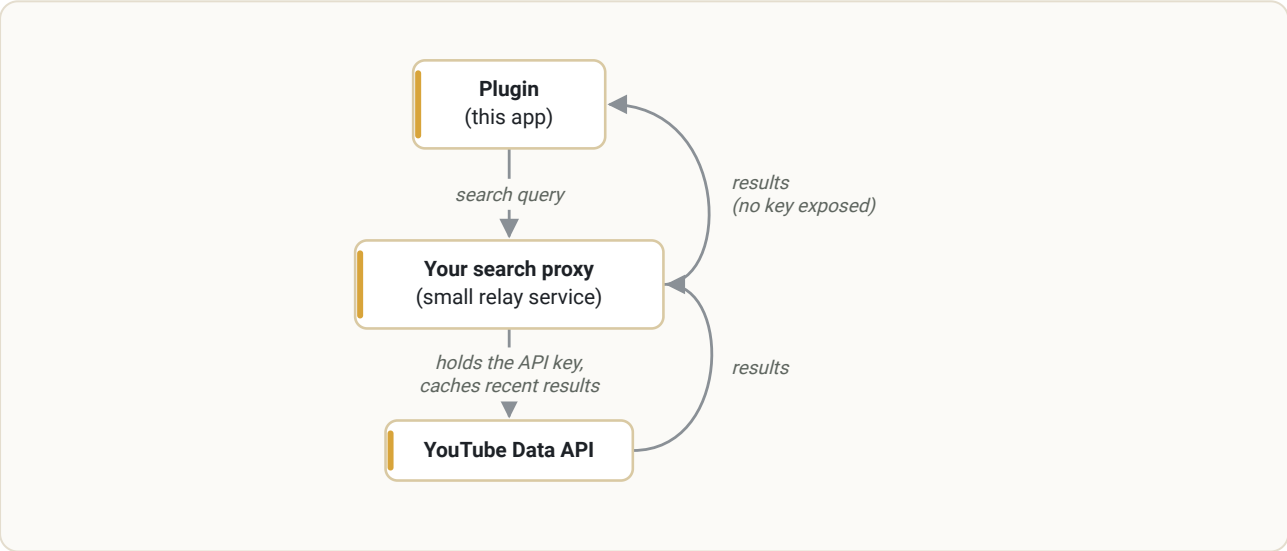
- **The list** shows, for each clip: a colour bar, a dot if you haven't heard it yet, its name and tags, its stars, how often you've used it (x3) and a * if it has a note, then its sound, BPM, key, length and waveform. Hover a clip for the details (and which pads use it in this project).
- **Click the waveform** to play from that point; drag along it to scrub.
- **The timeline** under the preview: drag the bar to move through the clip, **-5s** / **+5s** to skip, **Play** to pause and carry on. **In** and **Out** mark a region (shaded on the bar): previews start at In and stop at Out -- or go round with **Loop** on -- and **Load to Pad** / **Send to Empty Pad** load just that part (the pad's start and end set to it). **Clear** forgets the region. Each clip remembers its own.
- **Filters** (the row over the list): loops or one-shots, a BPM range, key and major / minor, sound, a minimum rating, **Unheard**. A filter looks through every folder at once, like search.
- **Sort** by name, newest, BPM, key, length, rating, recently added, sound or most used.
- **Preview options** (on the right): **Sync tempo** plays loops at your project's tempo; **Match key** plays clips in your key (the key set on the KEYS tab), moved the shortest way; **On the bar** waits for the next bar while your song plays, so a preview lands in time. **After a clip**: stop, play the next, or play the most similar one you haven't heard (Up Next). The slider is the preview volume.
- **Right-click a clip** to rate it (1-5 stars), give it a colour, add tags or a note, mark it heard or unheard, add it to a collection, find similar clips, or move / copy it. With several clips selected (Ctrl/Shift-click) it applies to all of them.
- **The view menu** (left of the filters) switches between your **Folders**, each **collection**, each **smart crate**, the clips most **like** one you picked, and **Duplicates** (the same file in more than one place).
- **Library...**: analyse everything, **save the filters as a smart crate** (it keeps filling itself as files arrive), delete crates or collections, switch **library folders** (every watch folder you choose is remembered; smart crates, similar and duplicates look in all of them), make a **new folder**, show duplicates, **relink missing files** (moved your samples? pick the folder they're in now and they're found by name) and **statistics** for the list on show.
- **Type** the start of a name with the list focused to jump to it.
- **Send to empty pad** with several clips selected loads each onto the next empty pad.

Drag and drop, folders

- **Drag a clip out** (or several selected) into your DAW, onto Explorer, or onto a pad.
- **Drag clips onto a folder** in the list -- or onto **..** to go up a level -- to move them into it. The folder lights up under the mouse. Hold **Ctrl** as you drop to copy instead. Files dragged in from Explorer onto a folder are copied there.
- **Right-click a folder** for **New folder inside...**, or to move or copy the selected clips into it. **Move to...** / **Copy to...** on a clip pick any folder.
- A name that's already taken gets a number. Pads and decks using a moved file keep finding it, and its stars, tags, notes and collections go with it.

YouTube Crate

- **Paste a YouTube URL** (or bare video ID) and hit **Watch** to embed that video using YouTube's own player, right in the tab. This is view-only — nothing is downloaded, extracted, or saved, the same as embedding a video on any webpage.
- **Search YouTube...** searches YouTube directly by keyword instead of requiring you to already have a URL — type a query, get a results list, double-click one to jump straight into the **Add to Crate** dialog with the title/channel already filled in.
  - This needs a **search proxy URL** configured first (a small relay service — see below for why). Paste it into the field at the top of the search popup; it's remembered for next time. Until one's configured, the button still works but shows "No search proxy configured" instead of results — Watch/paste-a-URL keeps working either way, this only affects the *search* convenience.
  - Why a proxy instead of searching YouTube directly: a real YouTube search needs a Google API key, and a key baked into a program that gets installed on other people's computers can be extracted and abused. The proxy holds that key server-side instead — this plugin only ever talks to the proxy, never to Google directly.



- **Add to Crate...** tags the pasted video (title, artist, channel, genre, style, year, key, BPM) and saves it to your own local crate — this is what the filter bar and Up Next queue search over. Nothing is auto-populated; you tag what you add.
- The **filter bar** (Search / Genre / Style / Decade / Key / Mode / BPM min-max / Starred only) narrows your crate down; **Apply** runs it, **Clear** resets everything.
- **UP NEXT** ranks the filtered results by similarity to whatever's currently playing (genre/style/artist/channel match, era proximity, key compatibility, BPM closeness) — click any entry to watch it. **Shuffle** picks something at random from the filtered set; **Next** advances to the top of the queue.
- **Favorite** stars the current video (usable with the Starred-only filter); **Remove** deletes it from your crate entirely.
- Load to Pad isn't available here — a YouTube embed was never a local file to begin with, so there's nothing to load onto a pad.

Convert

Batch-converts your own audio files and videos into audio files. The classic job: a folder of music videos or live footage in, a folder of WAVs out, ready to dig through.

1. **Add Files...**, **Add Folder...**, or drag files and folders onto the list. **Include subfolders** decides whether Add Folder (and dropping a folder) goes into subfolders too. Videos are marked `[VIDEO]` and convert to their soundtrack. **Remove** takes the selected rows out; **Clear** empties the list.
2. Pick what to convert to:
   - **Format** — **WAV** or **AIFF** (uncompressed, for any DAW or sampler), **FLAC** (lossless and about half the size), or **OGG Vorbis** (lossy, smallest; choose the **OGG quality**, 128-320 kbps).
   - **Sample rate** — keep the original, or 44.1 / 48 / 88.2 / 96 kHz.
   - **Bit depth** — 16-bit, 24-bit, or 32-bit float (WAV only; AIFF and FLAC go up to 24-bit, OGG has no bit depth).
   - **Channels** — keep, **Mono** (both sides averaged) or **Stereo**.
   - **Normalize peaks** — turns each file up or down so its loudest moment sits just under full scale (-0.3 dB; -1 dB for OGG).
   - **Trim silence at start and end** — cuts quiet lead-in and tail (anything below -60 dB), keeping a few milliseconds so soft attacks and decays aren't clipped.
3. Choose where files go: **Save next to each original**, or an **Output Folder...** (it starts as *Music\Boom Bap Converted*). With an output folder, **Keep folder structure** recreates each file's subfolders from the folder you added it from.
4. Press **Convert**. Files convert one at a time in the background — keep working in other tabs. Each row shows its progress and then *Done* (with the new file's name) or *Failed* and why. **Cancel** stops after the current block; nothing half-written is left behind.
5. Double-click a finished row to show the file in Explorer. **Watch Output Folder** opens the output folder in **Local Files**, so you can audition what you converted and load it straight onto pads or the deck.

Nothing is ever overwritten: if a file with the same name already exists (including the original itself), the new one gets a number, like `Break (2).wav`. Pressing **Convert** again converts only what hasn't succeeded yet; once everything has, it converts the whole list again (handy for making the same files in a second format). Your settings are remembered for next time.

CHAPTER 20

MIDI Learn

Right-click any DSP knob, toggle, the Volume slider, or the Swing slider, and choose **MIDI Learn** — then move a control on your MIDI hardware (a knob, fader, or similar) to bind it. Right-click the same control again and choose **Clear MIDI Learn** to unbind it. Bindings persist with your saved state (project/preset), same as everything else.

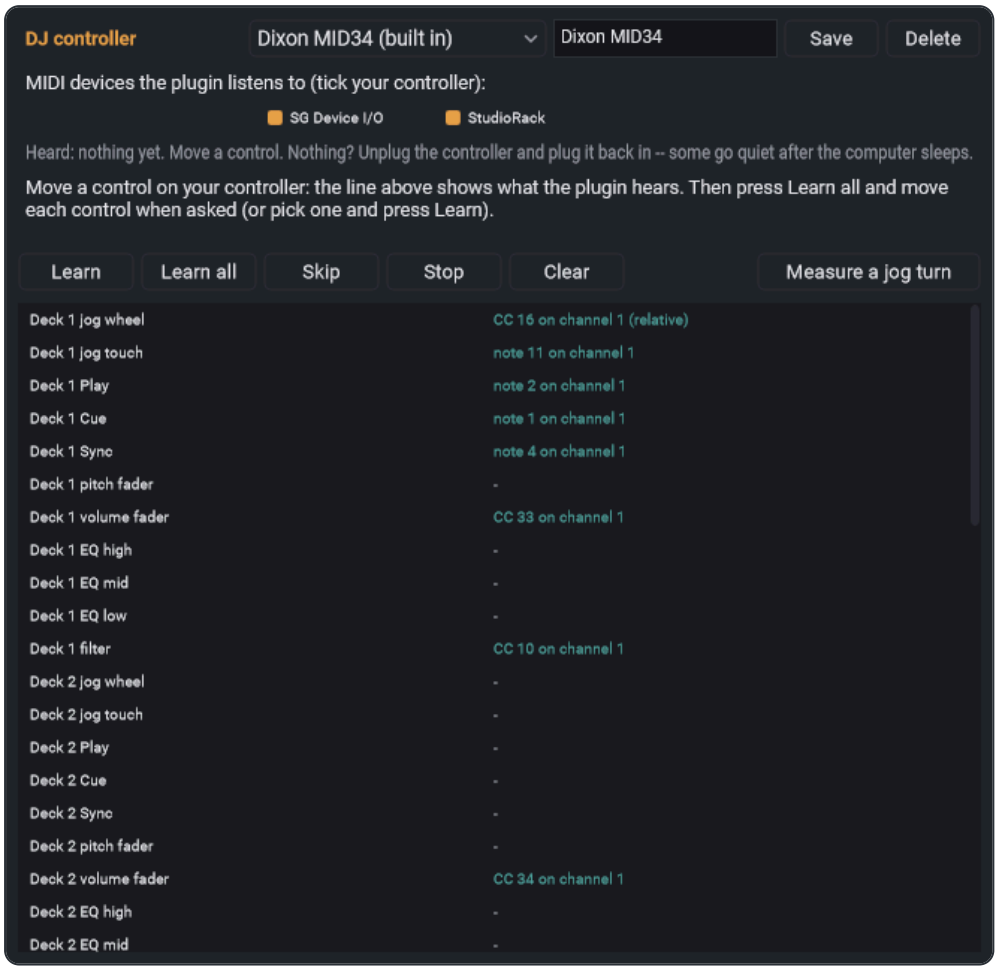


MIDI Map

MIDI Map (toolbar button) opens a single popup listing every active binding across the whole plugin — every learned knob and toggle, on every tab — so you can review or clear a mapping without hunting down whichever control it's attached to.

CHAPTER 21

Using a DJ controller



The Controller window

In one line: teach the plugin your DJ controller once, and its jog wheels, buttons, faders and pads play the two decks and the pads.

Most DJ controllers only work out of the box inside the DJ app they came with. The **DJ Controller** button on the TURNTABLE tab (and **Controller** in the toolbar, next to **MIDI Map**) opens a window where you switch your controller on and teach the plugin your controller yourself: it asks you to move each control, works out what that control sends, and remembers it. You can save the result as a named controller and it's there next time, in the Standalone and in your DAW.

Controllers that work straight away

Some controllers need no teaching at all. Plug one in and open the Standalone: the plugin switches it on by itself, its buttons light up briefly to say it's connected, and its jog, buttons, faders and pads work at once.

Controller	What drives what
Dixon MID34 (Windows calls it "MIDI34")	**DECK A/B** chooses which deck the jog, **CUE**, **PLAY**, **SYNC** and the pads play (it lights up for deck 2, and deck 2 comes on screen). The jog scratches that deck (touch the top to grab the record; hold **SHIFT** to search through it quickly); turning the jog's **edge** while it plays nudges it faster or slower for a moment, to line up a beat. The side faders are the two decks' volumes, the side knobs their filters; the crossfader; **BROWSE** and **LOAD** pick files (see Browsing from the controller); the side buttons choose what the 8 performance pads do -- see Pad modes.

Your own saved controller always wins: teach a built-in one differently and save it, or pick another from **Saved controllers**. In a DAW the built-in layout is picked for you too when the controller is plugged in (since version 1.154.0); the DAW, not the plugin, decides which devices it hears, so switch the controller on in your DAW's MIDI settings and send it to the plugin's track.

Pad modes (Dixon MID34)

The buttons down the sides of the MID34 choose what its 8 performance pads do. The button of the mode you're in lights up, and so do the pads that do something in it. Everything acts on the deck **DECK A/B** chose.

Button	The pads	With SHIFT held
SAMPLER (where it starts)	play pads 1-8	SHIFT + pad plays pads 9-16. SHIFT + SAMPLER switches **tempo adjust** on or off: while SAMPLER blinks, the jog changes the deck's tempo instead of moving the record
HOT CUE	jump to hot cues 1-8, or set one where the record is if it's empty (lit = set)	SHIFT + pad deletes that hot cue. SHIFT + HOT CUE = **saved loops**: an empty pad saves the running loop (or a loop from where the record is), a lit pad plays it again, SHIFT + pad clears it. They're saved with your project, like hot cues, and a new record on the deck starts with none
LOOP	top row: loop 4, 8, 16 or 32 beats (press the lit one again to stop looping); bottom row: jump back 4 / 1, forward 1 / 4 beats	SHIFT + LOOP = **roll**: hold a pad to stutter at 1/4, 1/8, 1/16 or 1/32
STEMS	with the record's stems (see below): top row vocals, drums, bass and other on or off, bottom row that part on its own (again: all back). Without them: the record's bass, mids and highs	SHIFT + STEMS = **beat jump**: top row back 1, 2, 4, 8 beats, bottom row forward 1, 2, 4, 8
FX ON	Brake, Spinback, Censor (while held), Stutter (while held)	SHIFT + FX ON = **FX page 2**: Reverb, Vinyl noise, Wow/flutter, Motor ramp (each on or off), low-pass and high-pass filter (while held), Slip, Censor (while held)
SCRATCH	the six scratch patterns: Transform, Crab, Flare, Orbit, Tear, Stab; pad 7 stops a pattern	SHIFT + SCRATCH switches **Slip** on or off
SYNC	matches the deck to the project tempo	SHIFT + SYNC puts the pitch back to normal

STEMS works with the record's real **stems** -- vocals, drums, bass and other -- once you've split the record: press **Separate** in the deck's Stems section on the TURNTABLE tab (see Stems). A pad is lit while its part plays. Until the record has stems, STEMS falls back to frequency bands, the same split the STEMS tab uses: "bass" is the low end (kick and bass line), "mids" the middle (most vocals and melody), "highs" the top (hi-hats and air); a switched-off band comes back at the level it had.

Browsing from the controller

Turn **BROWSE** to move through the sample browser (on the PADS tab). Press **LOAD** (or push BROWSE) on a folder to open or close it, and on a file to load it: onto the selected pad on the PADS tab, or onto the controller's deck on the TURNTABLE tab. On the TURNTABLE tab, where the browser isn't on screen, a small note under **DJ Controller** shows the file you're on.

The controller's lights

The plugin lights your controller's buttons for you, the way DJ apps do: **Play** is lit while its deck plays, **Cue** and **Sync** while the deck has a record, and a **pad** while it holds a sound. When you close the plugin, the lights go off. Only buttons you've taught (or a built-in controller's buttons) are ever lit.

In a DAW (since version 1.154.0) this works too, with two differences: the lights come on once you've opened the plugin's window (so a DAW scanning its plugins never lights up your controller), and with several copies of the plugin in a project, the first one you open drives them. If they stay dark, your DAW may be holding the controller's output for itself: in its MIDI settings, switch the controller's *output* off there (its input stays on).

Step 1: switch your controller on

- **If the plugin can't hear it, close your DJ app.** On some Windows setups only one program at a time can use a MIDI device.
- **Standalone:** open the controller window (**DJ Controller** on the TURNTABLE tab). The top of the window lists your MIDI devices: tick your controller. The Standalone remembers it for next time. If it won't stay ticked, another program is using it: close that program and tick it again.
- **In a DAW:** switch the controller on in your DAW's MIDI settings and send it to the track the plugin is on. The plugin only hears the MIDI its track gets.

Then move a knob or press a button on the controller. The **Heard** line under the devices shows the last thing the plugin received, for example "Heard just now from MIDI34: CC 16 on channel 1". If it still says "nothing yet", the controller's MIDI isn't reaching the plugin, and nothing below will work until it does. **Unplug the controller and plug it back in:** some controllers go quiet after the computer sleeps, even in their own DJ app, and a replug wakes them. The Standalone picks it up again by itself.

Step 2: teach it your controller

1. In the same window, the list shows every control the plugin can drive: for each deck the **jog wheel**, **jog touch**, **Play**, **Cue**, **Sync**, **pitch fader**, **volume fader**, **EQ high / mid / low** and **filter**; then the **Crossfader**; then **Pads 1-16**.
2. Press **Learn all**. The window tells you which control to move next:
    - **buttons** (Play, Cue, Sync, pads): press it once;

- **faders and knobs**: move it from one end to the other;
- **jog wheels**: turn it back and forth for a couple of seconds, turning the top of the platter;
- **jog touch**: touch the top of the platter, then let go.

3. When it has what it needs it says **Got it**, shows what it heard, and moves on by itself. A control your controller doesn't have? Press **Skip**. Want to stop part way? Press **Stop**. What you've learned so far already works.
4. Press **Save** to keep it. Give it a name first if you like (it starts as "My controller").

To teach or re-teach one control, select it and press **Learn** (or double-click it). **Clear** forgets the selected control.

While the window is listening, the controller only teaches: nothing plays. As soon as it stops listening, everything you've taught it works. The window can stay open while you try it out: the decks and pads still respond to your mouse and to the controller.

Measure a jog turn

After the jog wheel is learned, press **Measure a jog turn**, put a finger on a mark on Deck 1's platter, turn it exactly one full turn clockwise and press **Done**. From then on one turn of your jog wheel moves the record the same as one turn of a real 33 rpm record (1.8 seconds of audio). Without this, the plugin uses a fixed step per tick. That still works, but the record may move faster or slower than your hand.

No jog wheel? Use a knob, the mod wheel or the pitch wheel

Any MIDI keyboard or pad controller can scratch. Click **Learn** on a deck's jog and move a knob, your keyboard's **mod wheel** or its **pitch wheel**. The record follows how far you move it: one full sweep of a knob is half a turn of the platter. The pitch wheel springs back to the middle, so push it and let go for a quick back-and-forth. Then learn a pad or key as that deck's **jog touch**: holding it holds the record, like a hand on the vinyl. (There's no turn to measure for a knob.)

What each control does

Control	What it does
Jog wheel	Scratches the deck. With **jog touch** learned, only a touched platter scratches, and turning the edge of the wheel does nothing. Without jog touch, turning the wheel grabs the record, and it's let go a moment after you stop turning.
Jog touch	Grabs the record while your hand is on the platter, and lets it go when you lift off, like holding a real one.
Play	Starts or stops the deck.
Cue	Stops the deck and jumps back to the start.
Sync	Matches the deck's pitch to the project tempo (the same as the deck's own Sync).
Pitch, Volume, EQ high/mid/low, Filter	Move the deck's own controls.
Crossfader	Moves the crossfader between the two decks.

Control	What it does
Pads 1-16	Play the pads exactly as a pad controller would. Velocity-sensitive pads play soft and loud, and Live Pad Quantize, Tap Pads and recording into the pattern all apply.
Tape Loop 1-4 Play	Starts or stops that track on the LOOPS tab. Its light is on while the track plays.
Tape Loop 1-4 volume	That track's Vol, from silent to full on a knob or fader.

The on-screen controls follow your hardware, and your DAW can record the fader and knob moves as automation.

Good to know

- **Two decks that send the same numbers.** Many controllers send the same message for both decks and only change the MIDI channel. That's fine: the plugin tells them apart by channel.
- **Jog wheels count in different ways.** The plugin works out which way your jog wheel counts while you teach it, so you don't have to know.
- **Extra-smooth faders.** Some controllers send a fader in two halves for finer steps (14-bit). The plugin spots this and uses both.
- **Everything else still works.** Anything the controller sends that you haven't taught it goes on to the plugin as usual: pad notes, MIDI Learn links, and the older fixed jog input (note 20 and CC 20) all keep working.
- **Several controllers.** Save each under its own name and pick one from the **Saved controllers** list. Picking one makes it live. The plugin remembers which one you used last. **Delete** removes the one you've picked (it goes to the Recycle Bin); built-in controllers can't be deleted.
- **Where they're kept:** `%APPDATA%\Boom Bap Producer Pads\Controllers\`, one `.bbcontroller` file each. Copy the file to another computer to take your controller setup with you.

If something doesn't work

- **The controller did work, and now nothing reaches the plugin (or its own DJ app).** Unplug it and plug it back in; the Standalone picks it up again by itself. Some controllers (the MID34 among them) go quiet when Windows powers their USB port down to save energy. If it keeps happening, you can stop Windows doing that: in **Control Panel > Power Options > Change plan settings > Change advanced power settings**, set **USB settings > USB selective suspend setting** to **Disabled**. On a laptop that costs a little battery.
- **To see what the Standalone did with your controller**, open `%APPDATA%\Boom Bap Producer Pads\controller-log.txt`: it notes when the controller was switched on, when devices came and went, and (once a minute at most) whether anything arrived from it. If it says the controller was open but nothing arrived, the controller has gone quiet: unplug it and plug it back in.
- **"Nothing came through from the controller."** The controller's MIDI isn't reaching the plugin: the **Heard** line will say "nothing yet". Close your DJ app, then tick the controller at the top of the window (Standalone) or check the track's MIDI input (DAW).
- **"Couldn't switch on ..."** Another program has the controller open. Close VirtualDJ or any other DJ or MIDI app, then tick it again.
- **I can't find the Controller button.** It arrived in version 1.142.0. If your TURNTABLE tab has no **DJ Controller** button, install the latest version.

- **A jog wheel learned as a fader** (the record jumps instead of following your hand): select it, press **Learn** and turn the wheel back and forth, both ways, not in one long sweep.
- **The record moves too fast or too slow under your hand:** use **Measure a jog turn**.
- **A button does two things:** it was probably taught to two controls. Teaching it again to the one you want takes it off the other.

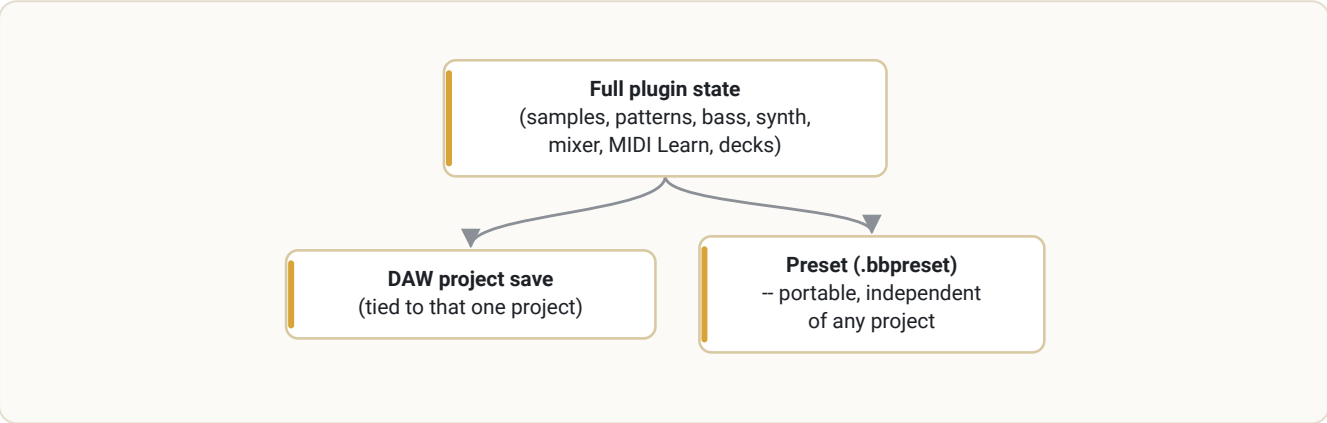
Related guides

- MIDI Learn: link any single knob to hardware
- Turntable: the decks themselves, scratching and the crossfader
- Pad grid: what the pads do

CHAPTER 22

Saving and the Standalone

Everything saves with your DAW project: every pad's sample and sound settings in all four banks, the patterns (with their pitch, length and groove), the bass line, the synth and its KEYS part, the arrangement, the mixer (levels, routing, buses and effects), MIDI Learn bindings and the decks' records — reopening the project restores everything, as long as the original sample files haven't moved or been deleted. Presets (Save/Load in the toolbar) capture the exact same full state as a portable, named file independent of any project — see The toolbar.



Running as a standalone app

The installer's Standalone component (or a manual build's `Boom Bap Producer Pads.exe`) is the same plugin, running on its own without a DAW. On first launch, use its audio settings to pick an output device (and a MIDI input if you want to play it from a controller).

There's no DAW transport to drive the step sequencer in standalone mode, so the toolbar's **Play** button (or the **Seq Play** toggle next to Roll) runs it, at the tempo in the toolbar's **BPM** box (type it, drag it, or **Tap** it). It has no effect when the plugin is loaded inside a real DAW (the DAW's own transport is always in control there); pads, Roll, the turntable, and the keyboard/piano-roll all trigger normally in standalone mode regardless of this toggle.

Your work is saved as you go

The Standalone keeps your session -- pads, decks, settings -- and brings it back next time you open it. Besides saving when you close it, it now saves every 30 seconds while you work (only when something changed), so a crash, a forced restart or a power cut costs you at most the last half-minute.

Open it once at a time. Two Standalone windows share one place to keep your work, so whichever you close last is what you'll find next time. If you open a second one, it tells you, and it doesn't save as you go -- close it and carry on in the first.

If the Standalone closes unexpectedly

If the Standalone ever crashes, it saves a short **crash report** on your computer first: which version it was, what went wrong ("access violation", say) and where in the program it stopped. It holds no recordings, none of your files and no settings, and your Windows user name is taken out.

Nothing is sent anywhere by itself. The next time you open the Standalone, it asks what to do with the report:

- **Send report...** opens a new issue on the plugin's GitHub page in your web browser, with the report already filled in. Read it, add what you were doing if you remember, and submit it (you need a free GitHub account). The report is also copied, so you can paste it into an email or a message instead.
- **Not now** leaves it.
- **Don't ask again** stops these questions.

To read a report without sending it, open the folder: reports are kept in `%APPDATA%\Boom Bap Producer Pads\CrashReports` (the last ten). Delete them whenever you like.

In a DAW, the plugin doesn't do this: a crash there is the DAW's to catch and report, and a plugin mustn't take that over.

CHAPTER 23

Languages

The plugin's text can be shown in another language. It ships in English; a language is a small text file that anyone can write, and the plugin picks it up by itself.

Choosing a language

Press **Language** in the banner (top right, next to Light / Dark) and pick one. The plugin uses it the next time you open its window: in the Standalone, close it and open it again; in a DAW, close the plugin window and open it again. Your choice is remembered.

The list shows **English**, **Pseudo** (see below) and every language file in your Languages folder.

Translating it into your language

1. Press **Language**, then **Make a template for translators**. A file called `template.txt` appears in your Languages folder (`%APPDATA%\Boom Bap Producer Pads\Languages`), and the folder opens.
2. Make a copy and name it after your language, for example `Francais.txt` (any name ending in `.txt`).
3. Open it in a text editor (Notepad is fine). Change the first line, `language: English (template)`, to your language's name, for example `language: Francais`. That's the name the Language menu shows.
4. Each line looks like this:

   ```
   "Load..." = "Load..."
   ```

   Replace the text on the **right** of the `=` with your translation, and leave the left side exactly as it is:

   ```
   "Load..." = "Charger..."
   ```

   A line you don't change stays in English, so you can translate a bit at a time. Inside the quotes, write `\"` for a quote mark and `\n` for a new line, as the template does.
5. Save the file as **UTF-8** (Notepad: *Save as* > *Encoding: UTF-8*), so accented letters come out right.
6. Choose your language from the **Language** menu and reopen the plugin window.

Almost everything can be translated. What stays as it is: the plugin's name in the banner, your own names (pads, presets, files), and a handful of labels built by joining pieces, where a piece like "Pad " (with its space) is translated on its own -- keep such spaces in your translation.

The Pseudo language

Pseudo is for testing, not reading: every translatable piece of text shows up in brackets and about a third longer, like `[Load...~~~]`. Text that shows *without* brackets isn't translatable yet, and text that gets cut off shows where a longer translation won't fit.

CHAPTER 24

License

Boom Bap Producer Pads is **free to use** — personal or commercial. You do not need to credit "Boom Bap Producer Pads" in tracks you make with it.

What you don't get

- The right to redistribute, sell, sublicense, or rehost the plugin binary yourself — get it from the official source.
- The right to reverse-engineer, decompile, or extract the source code from a compiled build, beyond what applicable law permits regardless of this license.
- Ownership of the "Boom Bap Producer Pads" name, its UI design, or its bundled assets — those remain the creator's ("AdamNubian", distributed by "EdensFrequency").
- Any warranty. The plugin is provided **as is**, without warranty of any kind, express or implied.

Credits

Built with JUCE.

The plugin's text is set in Roboto, built into the plugin under the Apache License 2.0. Roboto belongs to its authors, not to Boom Bap Producer Pads, and the terms above don't apply to it.

Questions

For anything not covered here — licensing for redistribution, commercial bundling, or anything else outside normal end-user use — reach out to the creator directly rather than assuming; nothing here is a substitute for that conversation.

Boom Bap Producer

Boom Bap Producer Pads 1.498.0

Free to use, personal or commercial.

Downloads, news and these guides: **edensfrequency.github.io/boom-bap-producer-pads**

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